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THE STANDARDS OF THE BRAZED ALUMINIUM PLATE-FIN HEAT EXCHANGER MANUFACTURERS' ASSOCIATION

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BRAZED ALUMINIUM PLATE-FIN HEAT EXCHANGER
MANUFACTURERS' ASSOCIATION

Chart Energy and Chemicals, Inc.

2191 Ward Avenue
La Crosse, Wisconsin 54601
USA

Tel: +1 608 787 3333
Email: bahx@chartindustries.com
http://www.chartindustries.com

Fives Cryo

25, Rue du Fort
BP 87
88194 Golbey Cedex
France

Tel: +33 (0)3 29 68 00 00
Email: bahx-france@fivesgroup.com
http://cryogenics-energy.fivesgroup.com/

Kobe Steel, Ltd.

Static Equipment Section
Machinery Business
2-3-1 Shinhama, Arai-cho
Takasago-Shi, Hyogo-Ken
676-8670, Japan

Tel: +81 794 45 7144
Email: alex@kobelco.com
https://www.kobelco-machinery-energy.com/en/energy/heat-exchanger/alex/

Linde GmbH

Linde Engineering
Schalchen Plant
83342 Tacherting
Germany

Tel: +49 8621 85 0
Email: plantcomponents@linde-le.com
http://www.linde-engineering.com

Sumitomo Precision Products Co., Ltd.

Thermal Energy Systems Engineering Department
1-10 Fuso-cho, Amagasaki
Hyogo Prefecture
660-0891, Japan

Tel: +81 6 6489 5867
Email: cryogen@spp.co.jp
http://www.spp.co.jp

PREFACE

This is the Fourth Edition of the Standards of the Brazed Aluminium Plate-Fin Heat Exchanger Manufacturers' Association (ALPEMA). It is the result of the work by a technical committee of all the Members to meet the objective of the Association to promote the quality and safe use of this type of heat exchanger. The Standards contain all relevant information for the specification, procurement, and use of Brazed Aluminium Plate-Fin Heat Exchangers.

The First Edition was published in 1994, the Second Edition in 2000, and the Third Edition in 2010. The ALPEMA Members review the Standards every year to consider whether updates are required and what these should be. Changes in the industry, experience with using the Standards, and feedback from Users indicated that the time was right for the Fourth Edition.

Comments by Users of the Standards are welcomed and should be sent to the ALPEMA Chair using the email address at www.alpema.org.

Additional information developed by the ALPEMA members, provided in the form of Frequently Asked Questions (FAQs), is available at www.alpema.org/faq.html. These FAQs are updated periodically as an Addendum to the current edition of the ALPEMA Standards.

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The Standards herein are recommended by The Brazed Aluminium Plate-Fin Heat Exchanger Manufacturers' Association to assist engineers, designers and operators who specify, design, install, and operate Brazed Aluminium Plate-Fin Heat Exchangers. These Standards are based upon sound engineering principles, research, and field experience in the manufacture, design, installation, and operation of these exchangers. These Standards may be subject to revision as further investigation or experience may show is necessary or desirable. Nothing herein shall constitute a warranty of any kind, expressed or implied, and warranty responsibility of any kind is expressly denied.

PLEDGE

ALPEMA Members will conduct themselves fairly and honestly, always practicing within legal and legislative boundaries.

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1 GENERAL DESCRIPTION AND NOMENCLATURE

1.1 General Descriptions

1.1.1 Background

Brazed aluminium plate-fin heat exchangers are the most compact and energy efficient heat exchangers for handling a wide range of services, noted particularly for their relative high thermal efficiency, compactness, low weight, and low maintenance. They provide low capital, installation, and operating costs over a wide range of cryogenic and non-cryogenic applications. Typically, these units have a total surface area of 1000-1500 m^2/m^3 of volume; this compares, for instance, with a shell-and-tube unit where the surface area per unit volume is of the order of 40 to 70 m^2/m^3. Brazed aluminium plate-fin heat exchangers with surface areas of 2000 m^2/m^3 are sometimes employed in the process industry. For these reasons brazed aluminium plate-fin heat exchangers find applications in aircraft, automobiles, rail transport, offshore platforms, etc. The main applications are in industrial gas processing, natural gas processing and liquefied natural gas (LNG), refining of petrochemicals and refrigeration services. Their ability to carry multiple streams, occasionally up to twelve or more, allows process integration in certain industrial processes, establishing them firmly in air separation processes and other cryogenic systems. The very large surface area per unit volume is particularly advantageous when low temperature differences apply. Such applications are typically found in cryogenic systems and hydrocarbon dewpoint control systems where temperature difference is linked to compressor power.

While plate-fin heat exchangers are available in various materials, this Standard refers solely to brazed aluminium plate-fin heat exchangers.

Where it is feasible to use a brazed aluminium plate-fin heat exchanger, it is nearly always the most cost-effective solution, often by a significant margin.

1.1.2 Definitions

Block, core

A brazed aluminium plate-fin heat exchanger consists of a block (core) of alternating layers (passages) of corrugated fins. The layers are separated from each other by parting sheets and sealed along the edges by means of side bars, and are provided with inlet and outlet ports for the streams. The block is bounded by cap sheets at the top and bottom.

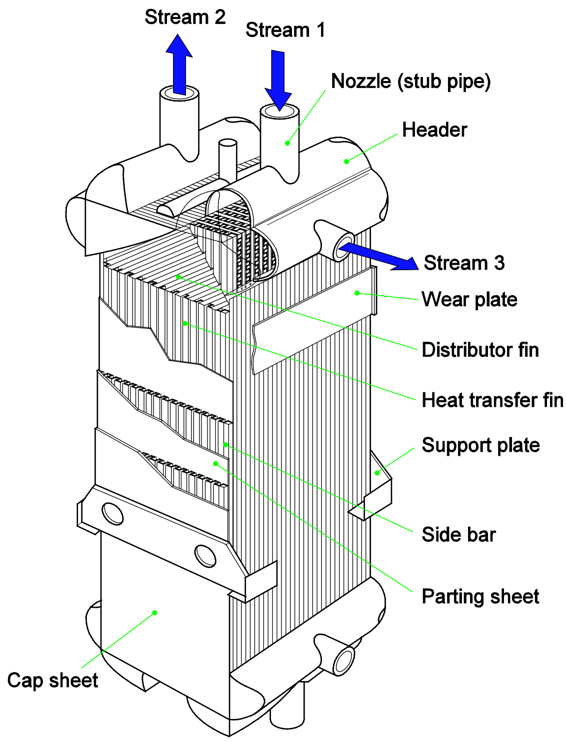


Figure 1-1: Illustration of a typical multi-stream brazed aluminium plate-fin heat exchanger

Block-in-shell heat exchanger

The term block-in-shell is used to describe a heat exchanger system of one or more brazed aluminium plate-fin heat exchanger blocks installed in a shell made either of aluminium or steel (see Figure 9-1). Enclosing the block in a larger shell is sometimes advantageous because it avoids the need to have a separate knock-out drum (see Section 9.1.2).

The arrangement is similar to a tubular exchanger known as a kettle reboiler. Hence these are known as block-in-shell exchangers. The block-in-shell exchanger offers several advantages over the tubular kettle reboiler.

Brazed aluminium plate-fin heat exchanger

A heat exchanger consisting of a block to which headers and nozzles have been attached. Also referred to as 'Plate-fin heat exchanger'.

An illustration of a multi-stream brazed aluminium plate-fin heat exchanger is shown in Figure 1-1.

The stacked assembly is brazed in a vacuum furnace to become a rigid core. To complete the heat exchanger, headers with nozzles are welded to the side bars and parting sheets adjacent to the ports.

The size of a brazed aluminium plate-fin heat exchanger shall be specified by width W, stacking height H and length L of the rectangular block. See Figure 1-2.

The three dimensions shall be given always in the same sequence as $W \times H \times L$, e.g., 900 × 1180 × 6100 mm.

Cold box

The term cold box is used to describe a casing typically of carbon steel which houses cryogenic equipment (see Section 9.2).

Cyclic service

A process operation with periodic variation in temperature, pressure, and/or flow rate.

Design temperature

Minimum and maximum metal temperatures.

Drain

A vent with the added characteristic that gravity assists liquid to exit an internal area.

Dry nitrogen

Oxygen-free nitrogen with dewpoint of -40 °C or better.

Dummy layer

An unpressurized, drainable, inactive layer that is typically located in the outermost layers of a block.

Heat transfer area

The sum of the primary and secondary heat transfer surface areas of all heat transfer passages in contact with a stream. See Section 7.2.1.

Hydrogen service

Services that contain hydrogen at a partial pressure exceeding 350 kPa (50 psi) absolute. Note: The source of this threshold is API-941, in which a phenomenon called "High Temperature Hydrogen Attack (HTHA)" is described, valid for temperatures above 200 °C (392 °F). This temperature is above the relevant temperature range of brazed aluminium plate-fin heat exchanger applications.

Minimum design metal temperature

The lowest metal temperature at which pressure-containing elements can be subjected to design pressure. Examples are ambient and process fluid temperatures.

Temperature

Temperature refers to stream temperature unless referred specifically to metal or other temperature.

Thermosyphon

The term thermosyphon refers to a specific effect in which natural convection mechanisms are used to circulate fluids.

Transition joint

A prefabricated, non-separable, metallurgically bonded joint used for the connection of dissimilar metal piping components which are not weldable to each other, e.g, between aluminium piping and other piping materials.

Vent

An opening to a core internal area to prevent it from retaining pressure. The opening is sufficiently large to prevent pressurization of the area.

Vent cover

A covering for an unpressurized area to reduce moisture and debris ingress between shipping and installation.

1.1.3 Applications for Brazed Aluminium Plate-Fin Heat Exchangers

1.1.3.1 Typical Services

Most brazed aluminium plate-fin heat exchangers have been installed in process plants used to separate a feed gas into its constituents, for example by the partial liquefaction of the feed and subsequent distillation and separation. The products and waste streams are then re-warmed against the feed streams. Condensers and reboilers are associated with distillation columns. Often chillers using standard refrigerants are used.

Brazed aluminium plate-fin heat exchangers are well suited for these and many other services, including those in Table 1-1.

Table 1-1: Typical services

Name	Service
Main exchanger	To cool inlet feed streams against return product and residue streams
Reversing exchanger	Air separation application to cool air and remove atmospheric water and CO_2 by reversing flow
Subcooler	To subcool liquid products or other liquid streams
Reboiler	To reboil column bottoms or vaporize tray liquids. Often this exchanger is installed inside a column
Overhead condenser	To condense column overheads, usually against a refrigerant stream
Chiller	To cool a process stream with a vaporizing refrigerant
Liquefiers	To liquefy the feed gas in a closed loop
Dephlegmators, reflux condensers	To condense overheads with vapour and liquid in counter-current flow and perform simultaneous heat and mass transfer
Aftercooler	To cool vapour coming from a compressor discharge
Block-in-shell exchangers	Type of reboiler with horizontal block inside a shell; operates as a kettle reboiler (see Chapter 9)

1.1.3.2 Plant Types

Brazed aluminium plate-fin heat exchangers have been successfully used in the above services in a variety of applications. The major applications have been in the cryogenic separation and liquefaction of air (ASU); Natural Gas Processing (NGP) and Liquefaction (LNG); the production of petrochemicals and treatment of off gases; and large refrigeration systems. Table 1-2 gives some typical applications where brazed aluminium plate-fin heat exchangers are used extensively and have proven reliable over many years of service.

Table 1-2: Plant types and applications

Plant types	Products and fluids	Typical temperature range, °C	Typical pressure range, bar.a
Industrial Gas Production – Air Separation – Liquefaction – Nitrogen Production Supporting Enhanced Oil Recovery (EOR)	Oxygen Nitrogen Argon Rare gases Carbon dioxide	+65 to -200	1 to 100
Natural Gas Processing (NGP) – Expander type – Nitrogen Rejection Unit (NRU) – Liquefied Petroleum Gas (LPG) – Helium recovery	Methane Ethane Propane Butane Pentane Nitrogen Helium Hydrogen Hexane Carbon dioxide	+100 to -130	15 to 100
Liquefied Natural Gas (LNG) – Base Load – Peakshaver – Floating LNG – Boil-off gas reliquefaction (BOG) – Small and midscale LNG	Liquefied Natural Gas Multi-component refrigerants Nitrogen refrigerant	+65 to -200	5 to 100
Petrochemical Production – Ethylene – Ammonia – Refinery off-gas purification – HYCO plant	Ethylene Propylene Ethane Propane Ammonia Carbon monoxide Hydrogen	+120 to -200	1 to 100
Refrigeration Systems – Cascade cooling – Liquefaction	Helium Freon Ethylene Propane Propylene Nitrogen Hydrogen Multi-component refrigerants	+100 to -269	15 to 45
Coal-to-Liquids (CTL) and Gas-to-Liquids (GTL)	Hydrocarbons Air / Oxygen	Same as for air separation (as Oxygen is necessary in GTL/CTL plants) and natural gas processing plants	

1.1.4 Limits of Use – Maximum Working Temperature and Pressure

The maximum working pressure for brazed aluminium plate-fin heat exchangers can be over 100 bar g while also having a natural ability to accommodate a full vacuum design condition. Manufacturer dimensional constraints and thicker components are required at high pressures, however. It is possible to have over twelve process streams, at various pressures from minimum to maximum, in a single heat exchanger.

The maximum temperature rating is typically specified at 65 °C. An upper limit of 65 °C is suitable for most applications, and it allows Manufacturers to use 5083 aluminium alloy piping which is more economical. However, designs are available for up to +204 °C at lower pressures. The minimum design temperature is -269 °C. See Chapter 6 for a full material listing.

1.1.5 Acceptable Fluids

Brazed aluminium plate-fin heat exchangers are capable of handling a wide variety of fluids in many different types of applications. In general, fluids need to be clean, dry, and non-corrosive to aluminium. Trace impurities of H_2S, NH_3, CO_2, SO_2, NO_2, CO, Cl and other acid-forming gases do not create a corrosion problem in streams with water dewpoint temperatures lower than the cold-end temperature of the brazed aluminium plate-fin heat exchanger.

Under specific conditions, mercury can corrode aluminium and therefore caution shall be used when handling mercury-containing fluids. However, there are many instances where brazed aluminium plate-fin heat exchangers have been successfully used with fluids containing mercury provided that the proper equipment design and operating procedures are implemented. In all cases, it is particularly recommended that the heat exchanger Manufacturer is consulted about specific conditions.

Further information on corrosion guidelines is given in Section 8.3.

Proper filters (usually mesh filters) shall be installed upstream of the heat exchanger system (see Section 4.10.2) and maintained according to the filter Manufacturers' recommendations to prevent plugging from particulates such as pipe scale or molecular sieve dust. If a brazed aluminium plate-fin heat exchanger is accidentally plugged, proven cleaning procedures are available. Further information on this subject is given in Section 8.2.

Fluids containing compressor lube oils, and other heavy hydrocarbons, are acceptable provided these contaminants do not precipitate out on the fin surface. In the event of heavy fouling, chemical solvent cleaning of the exchanger is a proven method of cleaning.

Hydrogen service fluids are acceptable for brazed aluminium plate-fin heat exchangers. There might be a concern, however, for pressure retaining parts made from materials other than aluminium and connected to the brazed aluminium plate-fin heat exchanger. For example, the shell material in block-in-shell units shall be reviewed by the Manufacturer for hydrogen service suitability.

1.2 Nomenclature of the Components

This section describes the components which are part of a brazed aluminium plate-fin heat exchanger. Typical geometrical dimensions are included in Chapter 5 of these Standards. Other items are described in the relevant chapters (e.g., supports are illustrated in Chapter 4 of these Standards).

1.2.1 Components of an Exchanger

For the purpose of establishing standard terminology, Figure 1-2 illustrates the components of an exchanger.

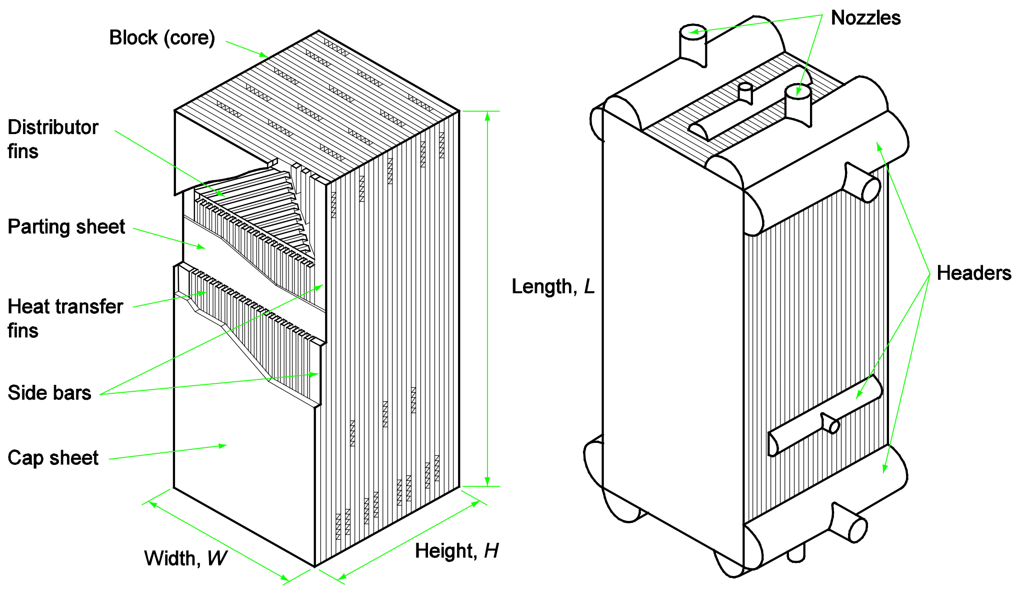


Figure 1-2: Components of a brazed aluminium plate-fin heat exchanger

1.2.2 Components of Manifolded Exchangers

Multiple brazed aluminium plate-fin exchangers may be connected in parallel, in series, or in parallel-series combination, to form one assembly.

Figure 1-3 illustrates an assembly of three brazed aluminium plate-fin heat exchangers connected in parallel. For this case, each individual stream enters the assembly through a manifold, is distributed to the inlet nozzles on each of the three heat exchangers, flows through the heat exchanger and leaves the assembly through the outlet manifold.

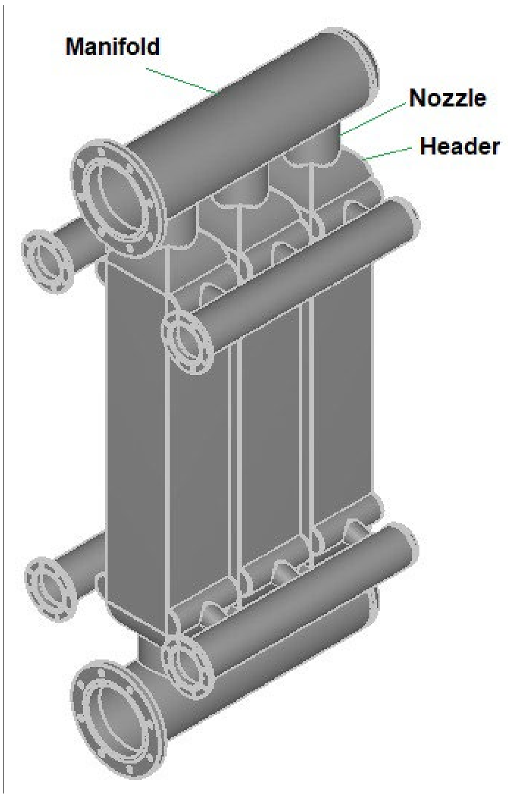


Figure 1-3: Typical assembly of three brazed aluminium plate-fin heat exchangers in parallel

1.2.3 Module Construction

Two or more individually brazed aluminium plate-fin exchanger blocks may be welded together in parallel to form one composite block. In this way the stacking height of the exchanger can be increased above a dimension limited by the manufacturing process or other restrictions.

1.2.4 Connection Options

Several options are available for connecting a brazed aluminium plate-fin heat exchanger to plant piping.

1.2.4.1 Aluminium Stub Ends

This option is selected if the exchanger nozzles are to be directly (butt-) welded to the connecting aluminium pipes.

1.2.4.2 Flanges

This option is available if the heat exchanger is to be connected to steel piping or if installation of the heat exchanger without welding is desired. Under this option, the Manufacturer normally provides the heat exchanger with aluminium flanges welded on the nozzles to fit with mating (normally steel) flanges on the piping. Details on the adequacy and design of such flanged connections are given in Section 5.12.3. As an option, the Manufacturer can weld transition joints on the aluminium nozzles and steel flanges on the end of the installed transition joints to provide steel flange connections to fit the mating flanges in the plant piping. Details of transition joints are described in Section 5.12.4.

1.2.4.3 Steel Stub Ends

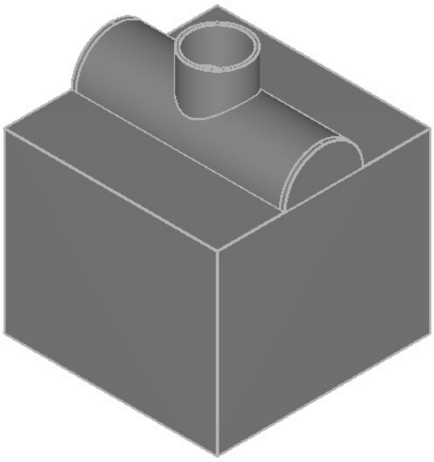
This alternative to flanges is available if the heat exchanger is to be directly (butt-) welded to steel piping. Under this option, the Manufacturer provides the heat exchanger with transition joints welded to the nozzles. The transition joints are to be directly (butt-) welded to the connecting steel piping.

1.2.5 Header/Nozzle Configurations

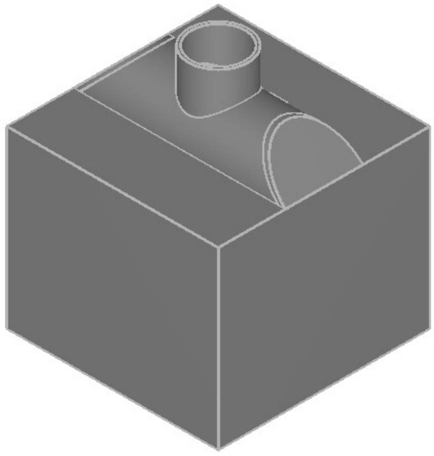
Streams to and from the heat exchanger enter and leave by means of various header/nozzle configurations.

The headers are normally made from half cylinders with flat and/or mitred ends or mitred dome headers. Typical variants are shown in Figure 1-4.

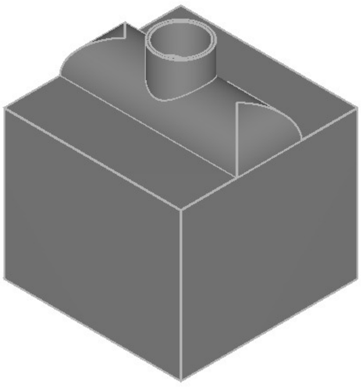
The nozzles may be radial, tangential, or inclined to the header and can also be equipped with flanges or transition joints. There are many different acceptable variations of header/nozzle configurations in use and some typical variations are shown in Figure 1-5.



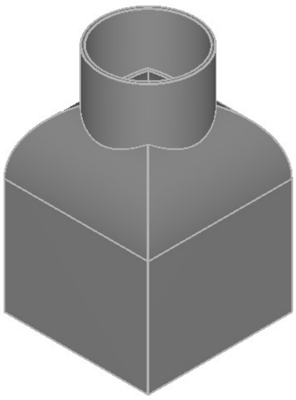
Standard header with flat ends



Header with inclined ends

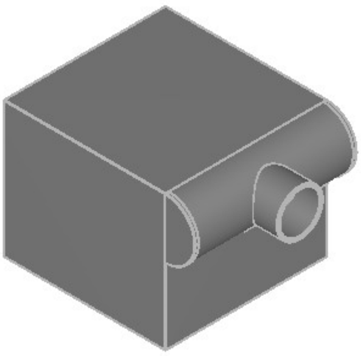


Header with mitred ends

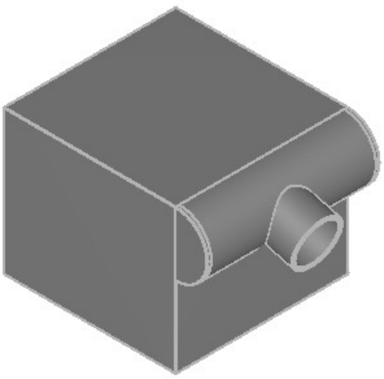


Dome header with mitred ends

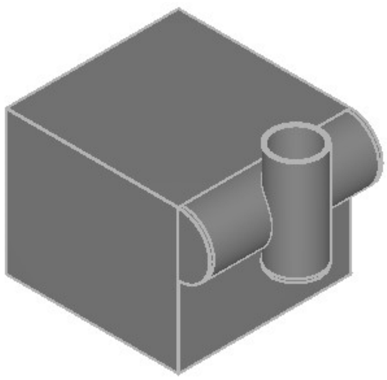
Figure 1-4: Typical header configurations



Radial nozzle



Inclined nozzle



Tangential nozzle

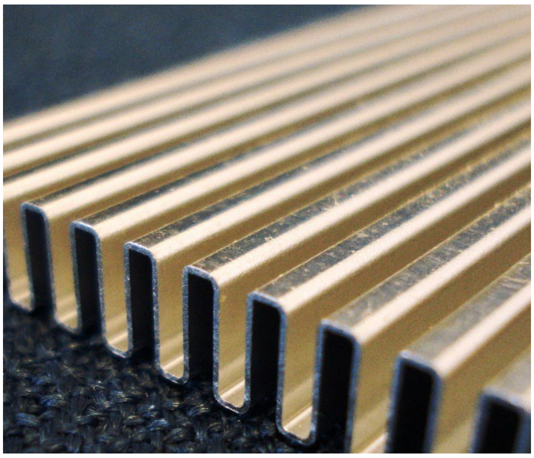
Figure 1-5: Typical header/nozzle configurations

1.2.6 Fin Corrugations

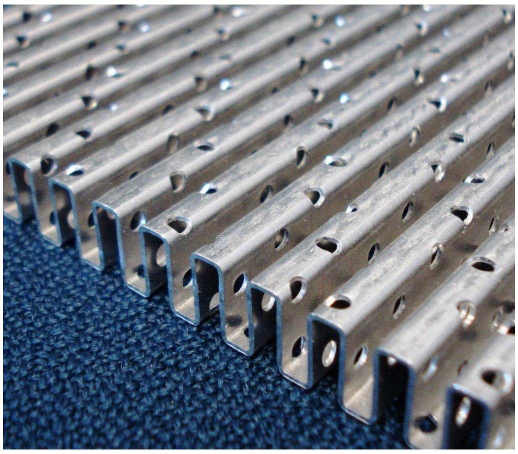
Various shapes of corrugated fins are available.

1.2.6.1 Principal Types of Fin

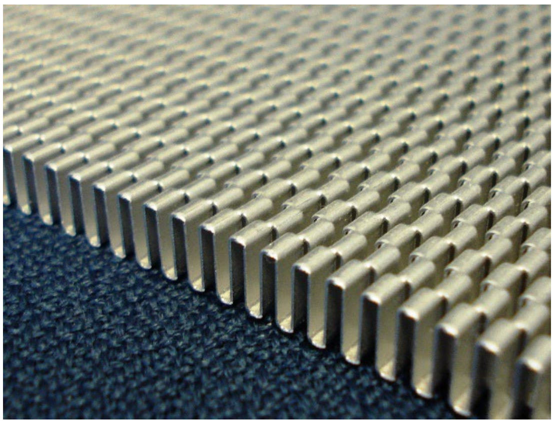
The principal types are plain, serrated, and herringbone as illustrated in Figure 1-6. Plain and herringbone fins may also be perforated as illustrated. For the thermal and hydraulic behaviour of each fin type, refer to Chapter 7 of these Standards.



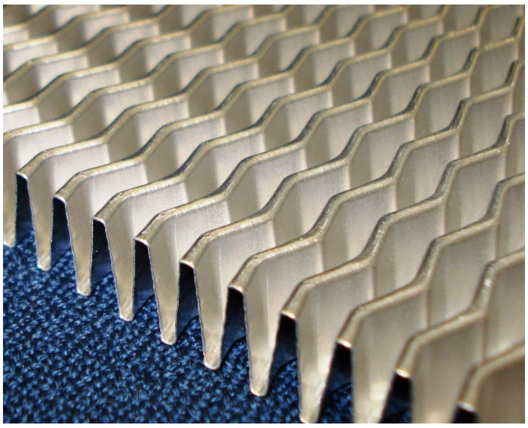
Plain fins



Plain-perforated fins



Serrated fins



Herringbone or wavy fins

Figure 1-6: Principal types of fin

1.2.6.2 Dimensions of Fins

Each type of fin is characterised by its height, *h*, thickness, *t*, and pitch, *p*, or by the number of fins per inch (FPI) or by the number of fins per meter as shown in Figure 1-7. Additionally, specification is required for the percentage perforation of perforated fins (e.g., 5% of corrugated area), for the length of the serration of serrated fins l_s, and for the distance between crests on herringbone fins.

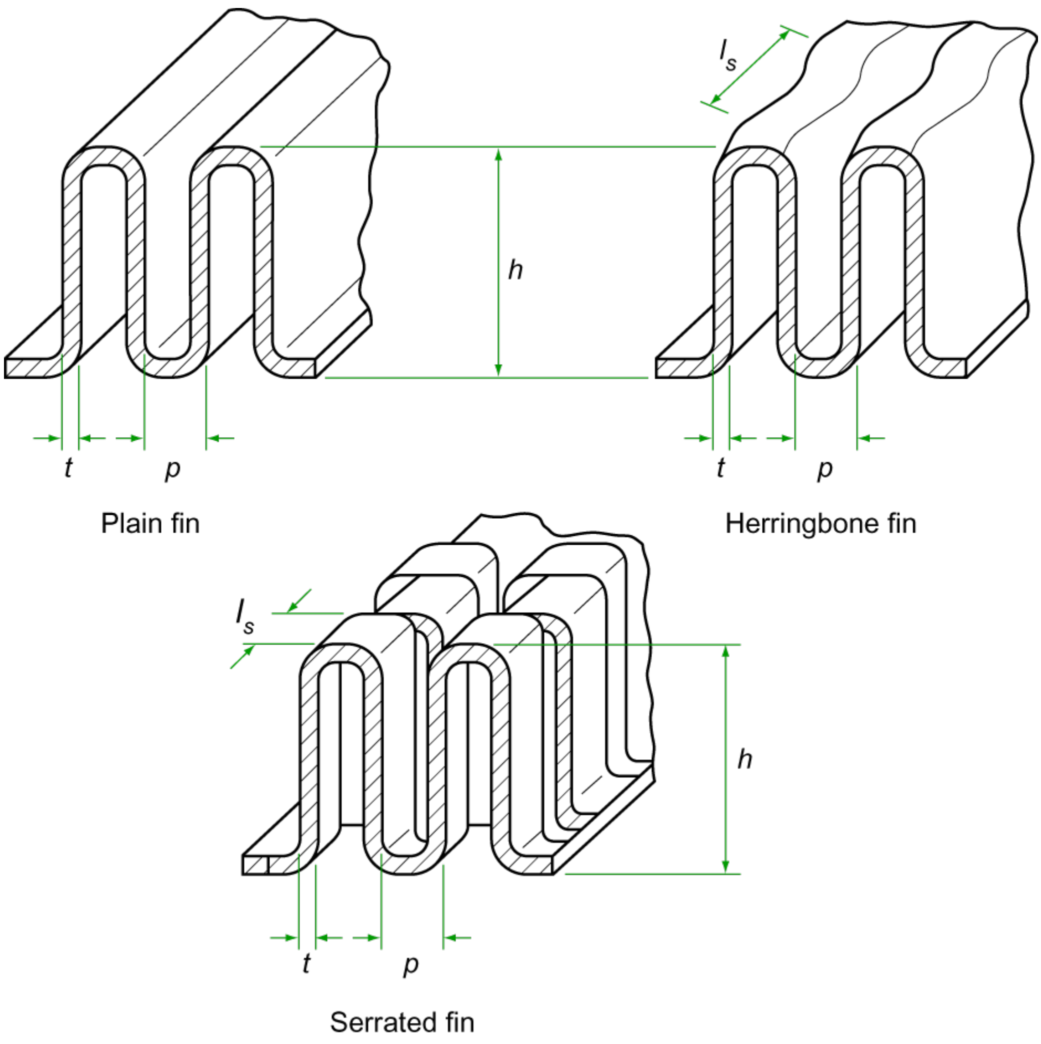


Figure 1-7: Definition of fin dimensions

1.2.7 Distributors

Various distributor types are available for streams entering or leaving a block. The principal distributor types are:

- side distributor (mitred or diagonal)
- end distributor (left, right or central)
- intermediate distributor (special)

Some typical examples are shown in Figure 1-8. Manufacturers may offer other designs.

Special designs of distributor are offered for two-phase streams entering a block (see Section 1.2.8).

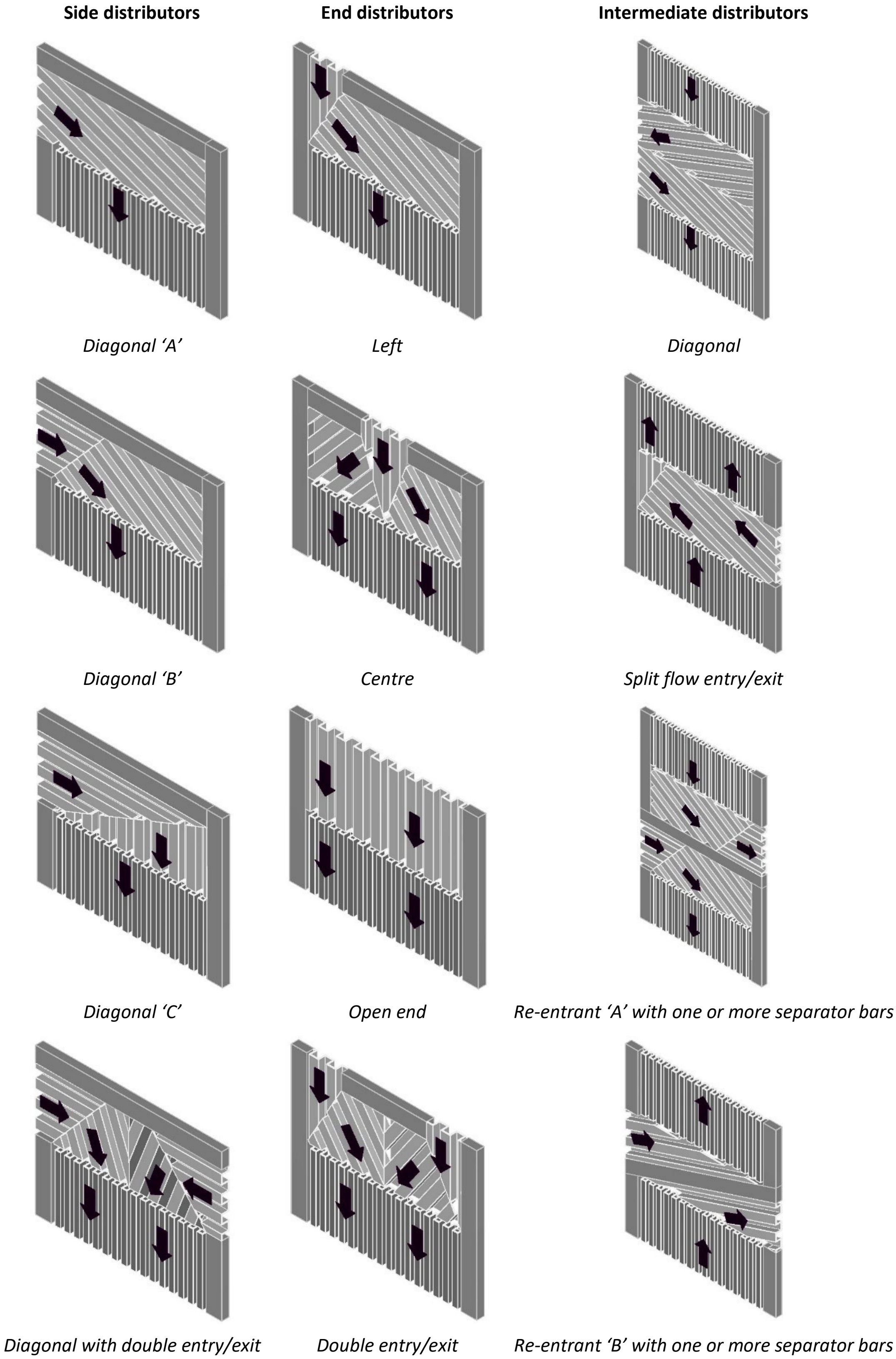


Figure 1-8: Examples of the principal distributor types

1.2.8 Two-phase Distributors

Single-phase distributor types shown in Figure 1-8 can also be used when two-phase streams enter a block. In addition, there are special distributors for two-phase streams entering the block. In these designs, the liquid and the vapour are separated in a vessel upstream of the brazed aluminium plate-fin heat exchanger and then introduced through separate distributors. Some examples are

- perforated tube or bar distributor (Figure 1-9)
- slotted parting sheet, split passages type (Figure 1-10)
- slotted parting sheet, adjacent passages type (Figure 1-11)
- sparge pipe in inlet header

Manufacturers may offer other designs without a separation vessel upstream, such as perforated plate in inlet header.

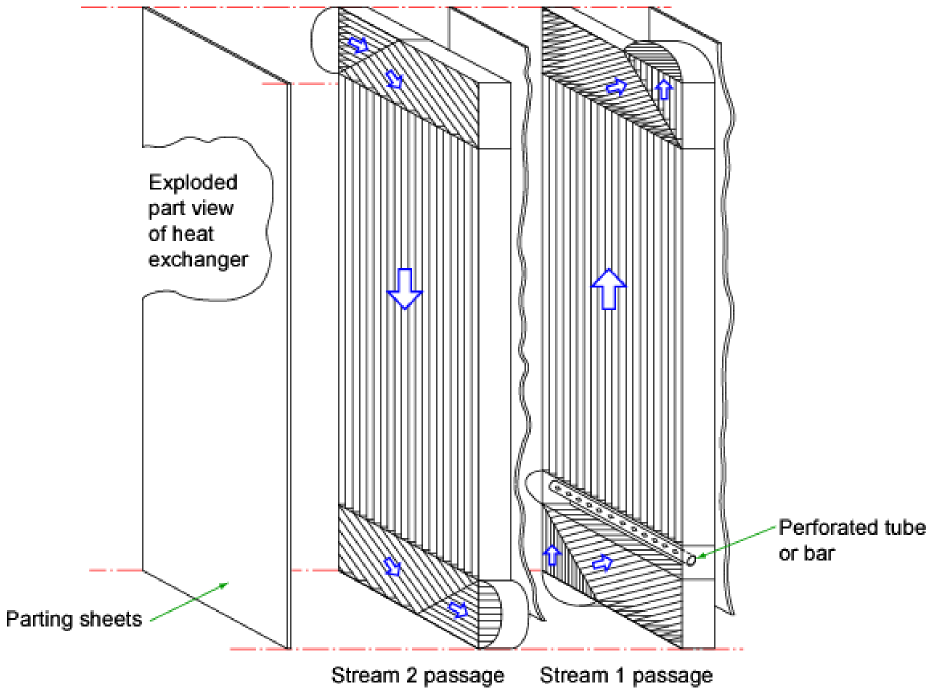


Figure 1-9: Perforated tube or bar distributor

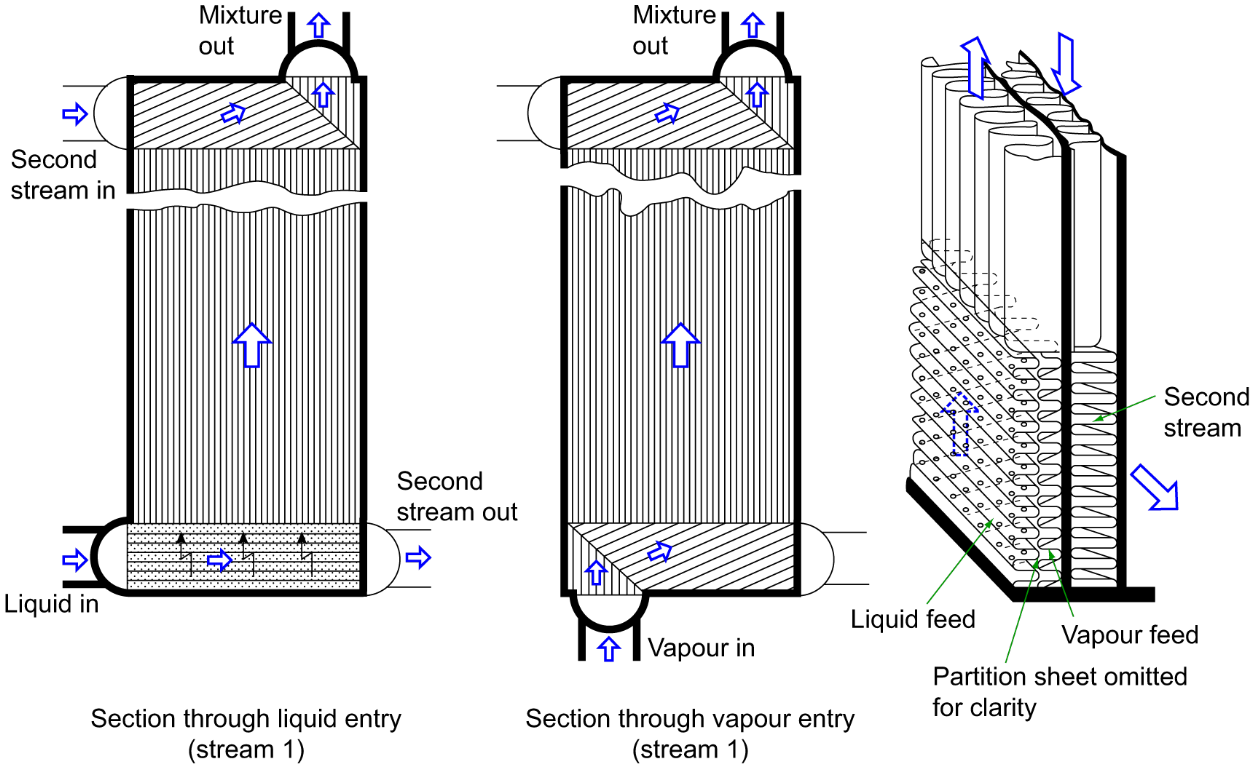


Figure 1-10: Slotted parting sheet, split passages type

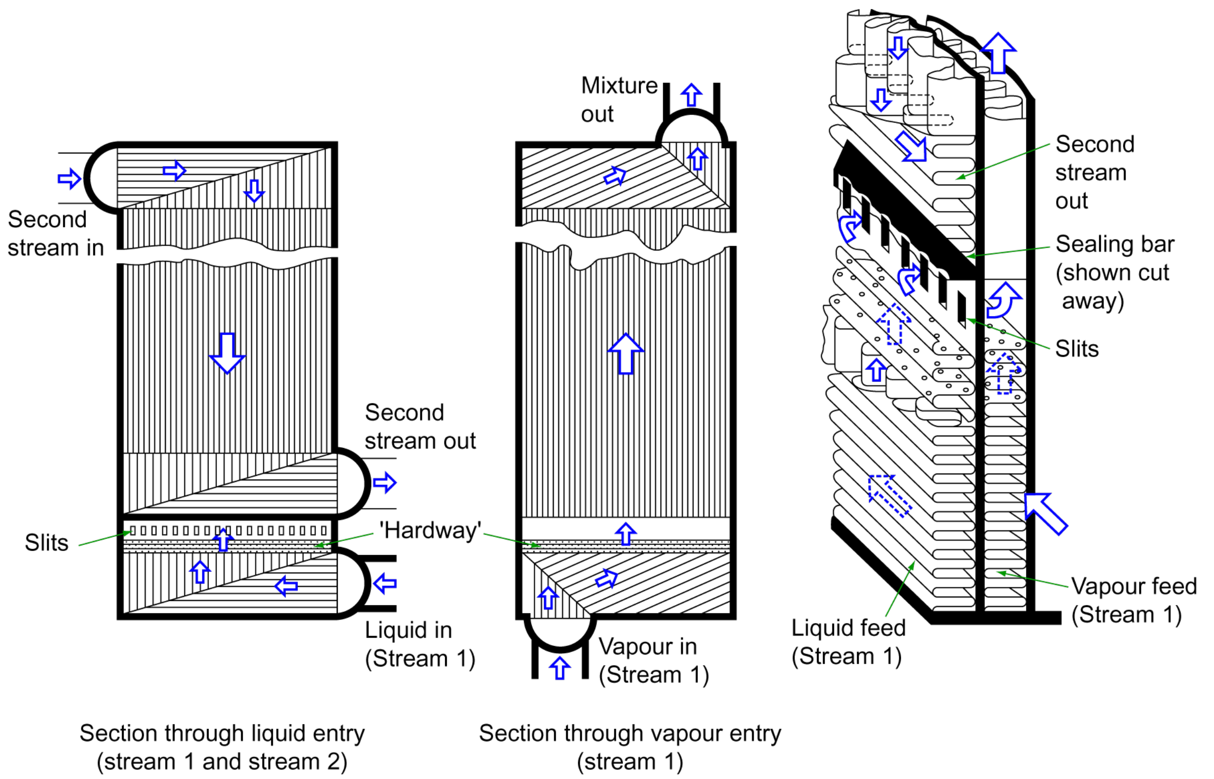


Figure 1-11: Slotted parting sheet, adjacent passages type

1.2.9 Flow Arrangements

Figure 1-12a illustrates the structure of an individual layer. Internal distributor fins (1) conduct the stream from the narrow entry port across the full width of the layer to the heat transfer fins (2). Distribution fins (3) then conduct the stream to the exit port.

Figure 1-12b illustrates the fin arrangement of a layer with an intermediate port (see also Figure 1-8). Various arrangements of the layers for each of the streams provide various possible flow arrangements, as shown in Figure 1-13. Manufacturers may offer further arrangements.

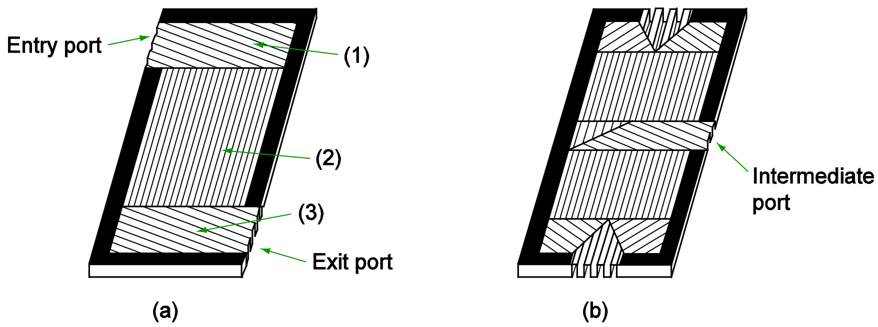


Figure 1-12: Structure of an individual layer

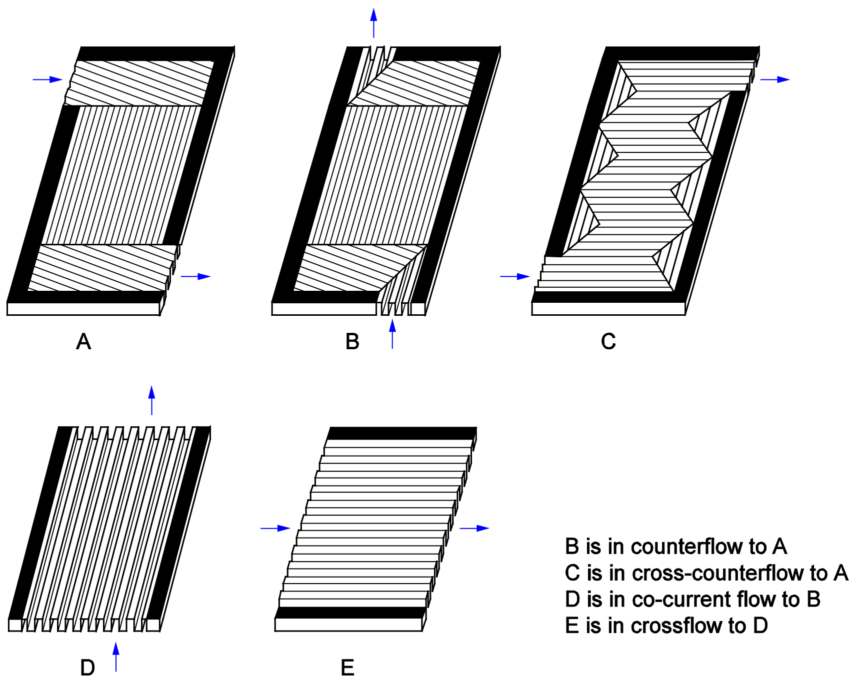


Figure 1-13: Flow arrangements

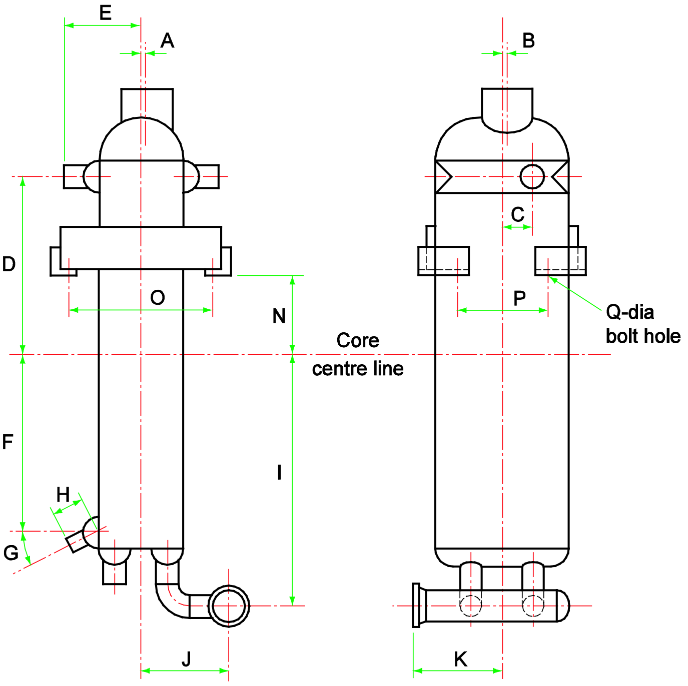
2 TOLERANCES

Standard tolerances for the external dimensions of brazed aluminium plate-fin heat exchangers are shown in Figure 2-1 and Figure 2-2. The core centre line (Figure 2-1) and the base line of supports (Figure 2-2) are used here as datum lines to illustrate these dimensions. However, Manufacturers may use other reference datum lines. The Purchaser and Manufacturer may adopt other tolerance values upon agreement.

Figure 2-3 shows the tolerances for a manifolded assembly of two cores. Here, the base line of the supports may also be used as a datum line as shown in Figure 2-2. Details for flanges are also shown in Figure 2-3.

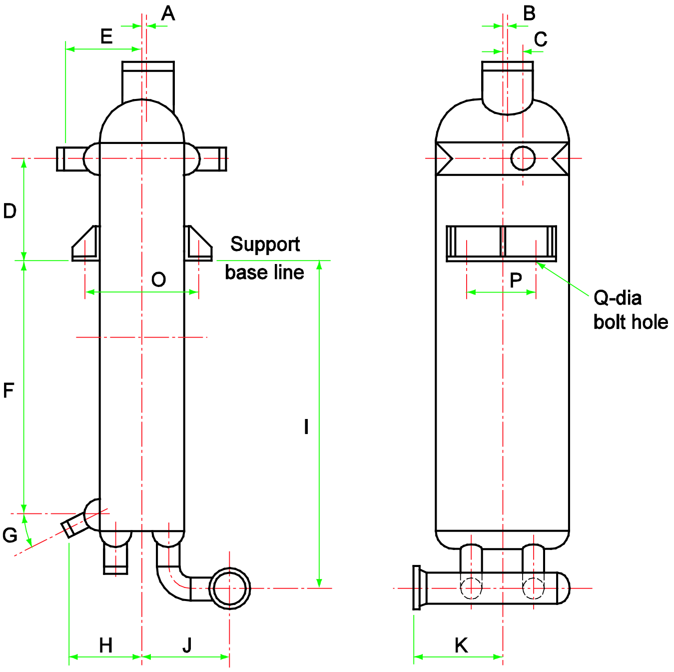
Cap sheets of brazed aluminium plate-fin heat exchangers can be slightly concave and shall be checked for concavity after brazing. Concavity is measured across the width at different positions along the length of the core. Concavity shall not exceed 15 mm.

For spare and replacement exchangers, these tolerances shall also be applied.



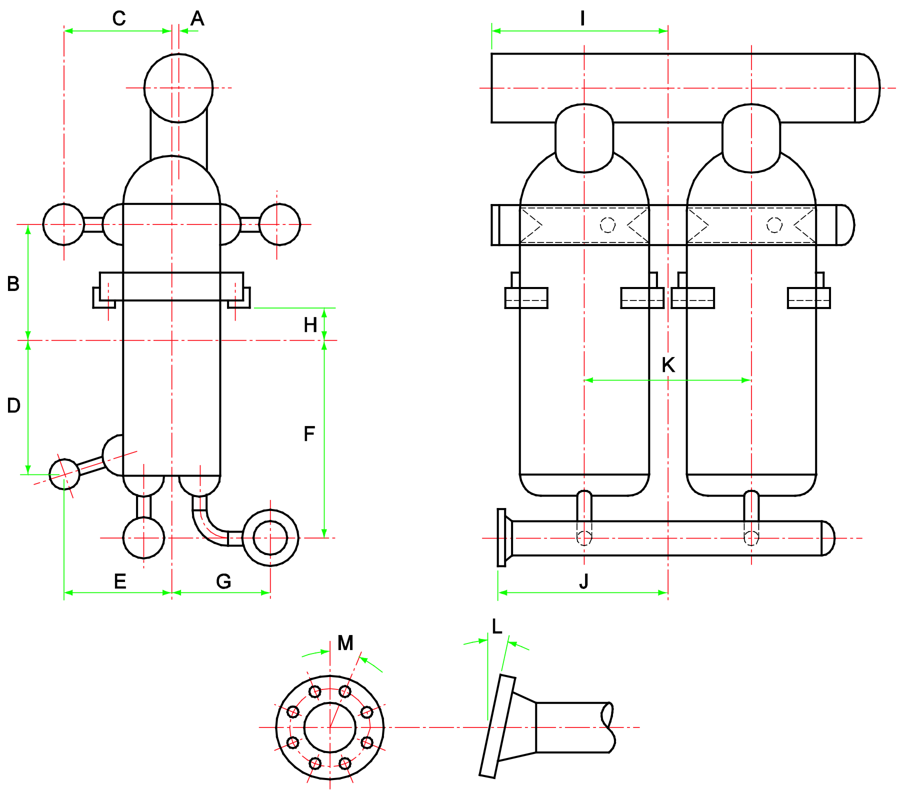
Items	Tolerances	Items	Tolerances
A, B, C, D, E, F, H, I, J, K	± 6 mm for Dim ≤ 1000 mm ± 8 mm for 1000 mm < Dim ≤ 2000 mm ± 10 mm for Dim > 2000 mm	G	± 3°
		N, O, P	± 3 mm
		Q	± 1 mm

Figure 2-1: Important external dimensions of one core using the core centre line



Items	Tolerances	Items	Tolerances
A, B, C, D, E, F, H, I, J, K	± 6 mm for Dim ≤ 1000 mm ± 8 mm for 1000 mm < Dim ≤ 2000 mm ± 10 mm for Dim > 2000 mm	G	± 3°
		O, P	± 3 mm
		Q	± 1 mm

Figure 2-2: Important external dimensions of one core using the support base line



Items	Tolerances	Items	Tolerances
A, B, C, D, E, F, G, I, J, K	± 6 mm for Dim ≤ 1000 mm ± 8 mm for 1000 mm < Dim ≤ 2000 mm ± 10 mm for Dim > 2000 mm	H	± 3 mm
		L	± 1°; max 5 mm at flange periphery
		M	± 1°; max 5 mm at bolt circle

Figure 2-3: Important external dimensions of a manifolded assembly of two cores: General flange details

3 GENERAL DESIGN, FABRICATION, AND CONTRACTUAL INFORMATION

3.1 Shop Operation

The detailed methods of shop operation are left to the discretion of the Manufacturer in conformity with these Standards.

3.2 Design Code

The design of the equipment shall be performed by the Manufacturer in complete compliance with the agreed Code. As a minimum, the latest mandatory edition of the Code in effect at the date of purchase order shall apply. For more information refer to Chapter 5 of these Standards.

3.3 Inspection

3.3.1 Third-party Inspection

Generally, all brazed aluminium plate-fin heat exchangers are subject to an inspection by an independent third-party inspection authority. The authority shall carry out the inspections, witness tests required by the agreed applicable Code, and certify that the heat exchanger is in accordance with Code mandatory requirements.

3.3.2 Manufacturer's Inspection

In some circumstances, in agreement with the Purchaser, the Manufacturer may carry out the inspection and testing of the heat exchanger in the absence of being witnessed by the Code certifying authority. Acceptance criteria shall not be less than that required when performed by a third-party inspection authority in accordance with the Manufacturer's Code registration terms of reference (i.e., Code quality control manual requirements).

3.3.3 Purchaser's Inspection

The Purchaser shall have the right to participate in any inspection and to witness any test that has been contractually agreed. However, Purchaser's inspections and witnesses shall be restricted to non-proprietary operations unless mutual agreement has been made to the contrary. Advance notification of inspections and tests shall be given as agreed upon between the Manufacturer and the Purchaser. Inspection by the Purchaser shall not relieve the Manufacturer of the Manufacturer's responsibilities for compliance with the Code and these Standards.

3.4 Nameplate

3.4.1 Manufacturer's Nameplate

A suitable Manufacturer's nameplate shall be permanently attached to each individual brazed aluminium plate-fin heat exchanger. The location of the nameplate shall allow easy access after installation of the heat exchanger in the plant.

The nameplate shall be of corrosion resistant material, which may be an aluminium alloy or austenitic stainless steel. When specified by the Purchaser, the nameplate shall be attached to a bracket welded to the heat exchanger. If the heat exchanger is to be installed in a permanent enclosure, (e.g., a cold box or vessel), a second identical (duplicate) nameplate shall be fitted to the supplied enclosure by the Manufacturer in a visible location mutually agreed with the Purchaser. Where the enclosure is not provided by the Manufacturer, the duplicate nameplate shall be supplied loose for attachment onto the enclosure by others.

3.4.1.1 Nameplate Structure

Because brazed aluminium plate-fin heat exchangers are capable of accommodating more than two streams, i.e., containing more than two independent pressure chambers, it is common practice that nameplates used by Manufacturers are able to specify data for several streams. Alternatively, several nameplates can be used.

3.4.1.2 Nameplate Data

The nameplate shall, as a minimum, include all data required by the Code. The following data are typical for each stream: the maximum allowable working pressure and the maximum and minimum allowable working temperatures or design metal temperature. Other nameplate information normally given should be the Manufacturer's name and address, the heat exchanger serial number and the year built. In the case of a cold box, a name plate should also be attached outside of the cold box. Nameplates shall be readable from an accessible area.

3.4.1.3 Supplementary Information

The Manufacturer may supply supplementary information pertinent to the identification, operation, or testing of the heat exchanger. This may include information about different design and test pressure conditions or other restrictive conditions applicable to the design and/or operation of the heat exchanger. Such information can be noted on the nameplate or on a supplementary plate attached to the heat exchanger, close to the nameplate location.

3.4.2 Purchaser's Nameplate

Purchaser's nameplate, when used, is to be supplied by the Purchaser and supplement, rather than replace, the Manufacturer's nameplate. The Purchaser shall supply information about size and material of supplementary nameplates to the Manufacturer at the date of order.

3.5 Drawing and Code Data Reports

A general drawings package shall be prepared and submitted by the Manufacturer to the Purchaser and the appointed third-party inspection authority.

3.5.1 Drawings Information

The Manufacturer's drawings package shall contain all the necessary information for approval, and may include the following:

- Overall dimensions and material thicknesses
- Supports and weights
- Specification and identification of materials, and if required by the Code, type of material certificates
- Types of fins used
- Nozzle/flange location, identification numbering, and connection details, - type of fluid for each stream if required by Purchaser or the Code
- Fabrication and test data, extent and location of non-destructive examinations, extent-of-leak testing, test pressures and appropriate weld identification as required by the relevant Code
- Tolerance information
- Centre of gravity for erection

3.5.2 Drawings Approval and Change

The Manufacturer shall submit the outline drawings for the Purchaser's review and approval. An agreed number of hard copies or electronic copies of outline drawings shall be submitted. Other drawings or documents may be furnished as agreed upon by the Purchaser and the Manufacturer.

Review and/or approval of drawings and documents shall be carried out by the Purchaser within a reasonable agreed period.

The Purchaser's approval of drawings shall not release the Manufacturer from his liability and responsibility for compliance with these Standards and applicable Code requirements.

3.5.3 Drawings for Record

After approval of drawings, the Manufacturer shall furnish an agreed number of hard copies or electronic copies of all approved drawings.

3.5.4 Proprietary Rights to Drawings

The drawings and the design indicated are the property of the Manufacturer and are not to be used or reproduced without his permission, except by the Purchaser for his own internal use.

3.5.5 Code Data Reports

After completion of fabrication and inspection of the exchanger(s), the Manufacturer shall furnish an agreed number of copies of the Manufacturer's Data Report.

3.6 Guarantees

Manufacturers are prepared to grant a thermal and hydraulic performance guarantee, and a mechanical guarantee. Details shall be agreed upon by the Purchaser and the Manufacturer.

The following sections give an indication of terms of guarantees which may be expected from the Manufacturer.

3.6.1 Thermal, Hydraulic, and Mechanical Guarantees

The Manufacturer should typically warrant to the Purchaser for a period of twelve (12) months from date of equipment start-up, but not to exceed eighteen (18) months from date of shipment, whichever occurs first, that the equipment provided is free from defects in material and workmanship. Furthermore, the equipment shall meet the thermal and hydraulic performance set forth in the Manufacturer's heat exchanger specifications. No warranty is made against corrosion, erosion, unstable flow or deterioration. If any performance deviations are to be applied in respect to the thermal and hydraulic performance guarantee, the acceptance criteria shall be by agreement between the Purchaser and Manufacturer. The Manufacturer shall not warrant heat exchanger performance and/or mechanical design if the operating process conditions of flows, temperatures, pressures, fluid composition and turndown conditions are more severe than those specified on the Manufacturer's specification sheets, or for pressures or temperatures outside the design range specified on the heat exchanger nameplate, or for damage due to improper installation, operation or storage, or due to external forces applied to the heat exchanger from the connecting piping or support system, which exceed the Manufacturer's specified allowable loads.

The Manufacturer's obligation and liability, at the option of the Manufacturer, is limited to the repair, modification, or replacement of the Manufacturer's equipment if found not to be in compliance with the stated warranty. The Manufacturer's obligations and liabilities are limited to furnishing replacement equipment, ex-works, not conforming to this warranty.

3.6.2 Consequential Damage

In no event shall the Manufacturer be held liable for any costs for gaining access, installing, lost product, lost production, lost profits or any incidental or consequential damages of any nature.

3.6.3 Corrosion

After delivery the Manufacturer assumes no responsibility for any defect of the equipment due to corrosion, except where the Manufacturer has expressively accepted the conditions and/or substances which have caused such corrosion.

3.7 Preparation of Brazed Aluminium Plate-Fin Heat Exchangers for Shipment

3.7.1 General

Brazed aluminium plate-fin heat exchangers shall be suitably protected to prevent damage during shipment. If there are no special instructions from the Purchaser, the following terms shall apply.

3.7.2 Cleaning

Internal and external surfaces shall be free from oil and grease and free from any loose scale or other foreign material.

3.7.3 Drying

The Manufacturer shall ensure that all pressure chambers are sufficiently dry before shipment. Each pressure chamber of the brazed aluminium plate-fin heat exchanger shall be dried to a dew point lower than 0 °C (32 °F).

3.7.4 Flange Protection

All exposed machined contact surfaces shall be protected against mechanical damage by suitable blind covers.

3.7.5 Dummy Passages/Inactive Areas

Openings in dummy passages or inactive heat exchanger areas shall be suitably protected to avoid ingress of water and dust. See Section 4.7.

3.7.6 Pressurising

To avoid ingress of any moisture or dust during transport, brazed aluminium plate-fin heat exchangers shall be shipped with flanges and nozzles hermetically sealed and all pressure chambers pressurised with dry, oil-free nitrogen gas or air to between 0.2 and 1.2 bar g.

All connections shall carry warning labels stating that the heat exchanger is under pressure.

WARNING: REMOVAL OF END CLOSURES IS ONLY ALLOWED AFTER REDUCING THE PRESSURE TO ATMOSPHERIC IN THE RELEVANT CHAMBER BY THE INSTALLED GAUGE/VALVE.

The Purchaser and Manufacturer may agree that pressurising the heat exchanger for transport is not necessary. In that case it may be considered necessary to ship the heat exchanger packed in a plastic foil with a humidity-absorbing agent inserted.

3.8 Scope of Supply

Unless expressively agreed otherwise between the Purchaser and the Manufacturer, the scope of the Manufacturer's supply is as follows:

1. Heat exchanger core(s) complete with headers and nozzles, and, if applicable, manifolded to form an assembly.
2. For stub pipe connections:
   Nozzles are normally seal welded with closures for transport. The Manufacturer shall provide sufficient length of stub pipes to allow easy connection to the plant piping after cutting the closures.
3. For flanged connections:
   The Manufacturer's scope of supply shall end with the face of the flanges, which are normally covered by blind flanges for transport. These blind flanges with pertinent bolts and gaskets are only included to protect the flange faces and to allow moderate pressurising of the individual chambers for transport.

Other options can be agreed between Purchaser and Manufacturer, such as the following:

1. Insulation material, which is not normally included in the Manufacturer's scope of supply.
2. Normally there are no spare parts necessary for the equipment covered by these Standards, other than flange boltings and gaskets where applicable.
3. Steel counterflanges, bolts and gaskets suitable for plant operation are normally not in the Manufacturer's scope of supply but may be specified by the Purchaser.
4. The Manufacturer can supply suitable supports and lifting lugs/devices as described in Sections 3.9.1 and 3.9.2 below.
5. Flanged connections in stainless steel through the use of transition joints.

3.9 General Construction Features

3.9.1 Supports

Generally, all brazed aluminium plate-fin heat exchangers are provided with supports. Any exceptions are to be agreed upon between Purchaser and Manufacturer.

The supports are designed to accommodate the weight of the heat exchanger and its contents, during both operating and test conditions.

For purposes of support design, Manufacturers provide a design margin for external loads from piping, wind and seismic events. This margin is provided by or is available upon request from the Manufacturer. As an alternative, the Purchaser may supply the Manufacturer with the external loadings for the Manufacturer's support design.

Wind and seismic loads shall not be assumed to occur simultaneously.

Support details are described in Chapter 4 of these Standards.

3.9.2 Lifting Devices

Lifting lugs/devices shall be designed using one of the following methods:

1. The Purchaser shall inform the Manufacturer of the way in which he plans to lift and move the heat exchanger,
2. The Manufacturer shall advise the Purchaser of the approved method for lifting and moving the heat exchanger and provide drawings showing the centre of gravity.

If applicable, the Manufacturer can design a suitable device, or provide lifting lugs or equivalent on the heat exchanger, to safely lift and transport the heat exchanger into its installed position.

The Manufacturer shall advise the Purchaser of the proper use of the lifting device and lugs. If these do not exist, it is possible, with prior approval from the Manufacturer, to lift the heat exchanger with belts or ropes, if attention is fully paid to suitable protection of the heat exchanger corners.

3.10 Nonconformity Rectification

3.10.1 Introduction

Rectification work on a brazed plate-fin heat exchanger block is necessary if a nonconformity occurs during the manufacturing process. This section describes procedures and Purchaser notifications to resolve a nonconformity.

The Manufacturer judges the severity of the nonconformity and reviews contractual requirements in determining the involvement of the Purchaser in deciding disposition.

Unless there are contractual requirements to the contrary, the following procedures are followed in performing nonconformity rectification.

3.10.2 Procedures and Documentation

All rectification work shall be carried out according to approved procedures based on sound engineering principles. All rectification shall conform to Code requirements, assure leak integrity and not affect the structural integrity of the heat exchanger. The Manufacturer shall fully maintain the agreed mechanical guarantee.

A nonconformity record document shall be completed by the Manufacturer and be available for review by the Purchaser, on request.

3.10.3 Side-Bar-to-Sheet Joint Leak Rectification

A brazed aluminium plate-fin heat exchanger typically contains more than 1600 linear meters of brazed, side-bar joints. Based on a very controlled brazing process the overall structural and leak integrity of the side-bar to parting sheet joint is excellent. However, due to the large number of potential leak sites, small leaks in the side-bar-to-sheet joint can sometimes occur.

Rectification is by seal welding using a proven procedure developed for this purpose.

External leaks at the brazed connection between side bars and parting sheets may be seal welded without notification of the Purchaser.

3.10.4 Blocking of Layers

A typical brazed aluminium plate-fin heat exchanger contains more than a half million linear meters of braze joints between fins and parting sheets, produced by the brazing process in excellent quality. However, due to manufacturing variations over the large number of fin-to-sheet braze sites, a small area of unacceptable brazing may occasionally occur within a layer and is detected by the pressure test.

Rectification is typically to block the layer using a proven procedure developed for this purpose. The Manufacturer shall perform calculations to estimate the influence of the blocked layer(s) on thermal performance, pressure drop and structural integrity for any exchanger designed by the Manufacturer. In almost all cases the Manufacturer's design margin allows for blocked layers. Blocking of an active layer requires notification to the Purchaser only if the contractual performance is no longer met.

3.10.5 Other Rectification Work

The Purchaser shall be notified and involved in any disposition decision if the rectification has impact on the performance guarantee or installation of the unit.

4 INSTALLATION, OPERATION, AND MAINTENANCE

4.1 General

It is normal for brazed aluminium plate-fin heat exchangers to be installed in the vertical orientation, warm end up. There are, however, some exceptions to this such as reversing heat exchangers for air separation or cross-flow sub-coolers. Heat exchangers are normally supplied with all the necessary supports to facilitate site installation.

Generally, mounting supports are, if possible, positioned in the warm half of the block. This reduces movement at the juncture of the bracket to the support beam during start-up and shut-down cycles.

In addition to the main supports, there may be a need for an additional sliding guide to restrict movement from the vertical plane. Several aspects need to be considered to determine if such a device is necessary, for example,

- physical dimensions of the heat exchanger
- weight of the heat exchanger
- site conditions (earthquake, wind, pipe load etc.)
- relative position of the main support plane to the centre of gravity of the heat exchanger

Upon request, the Manufacturer shall provide values for the allowable forces and moments which may be applied at the junctions of the header to nozzle of the heat exchanger. The Purchaser should then ensure that these values are not exceeded for all connecting pipe runs (refer to Section 5.12.2.4 and Table 5-1 for typical values).

If the connecting pipe joints between the heat exchanger and the Purchaser's pipework are to be made by means of welding, then this shall be done using the relevant qualified weld procedures and welders. The weld filler materials used shall be those approved for welding the materials to be joined. Details of the nozzle material are normally marked on the heat exchanger nozzles as well as being stated on the relevant drawing. The Manufacturer should be asked if the installer has any doubts on this matter.

4.2 Lifting and Handling

Extreme care should be taken in the lifting and handling of brazed aluminium plate-fin heat exchangers. The Manufacturer shall ensure that a heat exchanger is provided with the means of lifting, e.g., lifting lugs, lifting attachments, specified nozzles, etc. The Manufacturer shall provide detailed instructions for lifting and handling; these instructions shall be specific to each heat exchanger and shall be strictly followed. If there are any doubts about lifting and handling of any heat exchanger, the Manufacturer should be consulted.

WARNING: FAILURE TO FOLLOW THE MANUFACTURER'S LIFTING AND HANDLING INSTRUCTIONS FOR A SPECIFIC HEAT EXCHANGER COULD SERIOUSLY DAMAGE THE EXCHANGER.

4.3 Support Beams

The beams onto which the heat exchanger is to be mounted, unless within the scope of supply, are the responsibility of the Purchaser. In the selection of these beams, in addition to the dead weight imposed by the heat exchanger, loads generated by applied external forces and moments should also be considered. It is a common practice in this evaluation to assume that the allowable forces and moments given in the Manufacturer's design documents are not applied simultaneously. The mating faces of the support beam should be flat and be aligned so that, when installed, the deviation of the unit to the true vertical is a maximum of ½° or 15 mm measured over the block length. The alignment may be achieved with the aid of metal shims, but this is not preferred. For light applications the support beams may be manufactured from aluminium, however, it is normal for the beams to be made from the appropriate steel alloy.

4.4 Support Insulation

To prevent heat leakage, thermal insulation is required between the mating faces of the supports. The heat-break insulating material (for example Micarta*) shall be capable of load bearing and allow movement on the support. The thickness of heat-break shall be selected by the Manufacturer depending on the operating conditions. If the heat exchanger is mounted in a cold box, a proportion of the required total thickness of heat-break may be fitted at the junction of the support beam and cold box, in addition to the mating face between the heat exchanger support and the support beam.

4.5 Sliding Guide Frame

For applications where externally applied forces due to wind, earthquake, and pipe loads are large enough to cause lateral movement, it is necessary to limit horizontal movement of the heat exchanger.

The function of the sliding guide is to limit the horizontal movement. Details of typical support systems and guides are shown in Figure 4-1, Figure 4-2 and Figure 4-3. The guide frame should be separated from the heat exchanger with layers of heat-break material. To compensate for core contraction in service, the break shall be tightly packed and fixed to the guide frame with stainless steel screws to prevent it from becoming detached. An additional wear plate (scuff plate) may be fitted to the heat exchanger for protection. The interface members of the frame should be made from the appropriate steel alloy and braced back to the main structural members of the cold box or support frame to give stability.

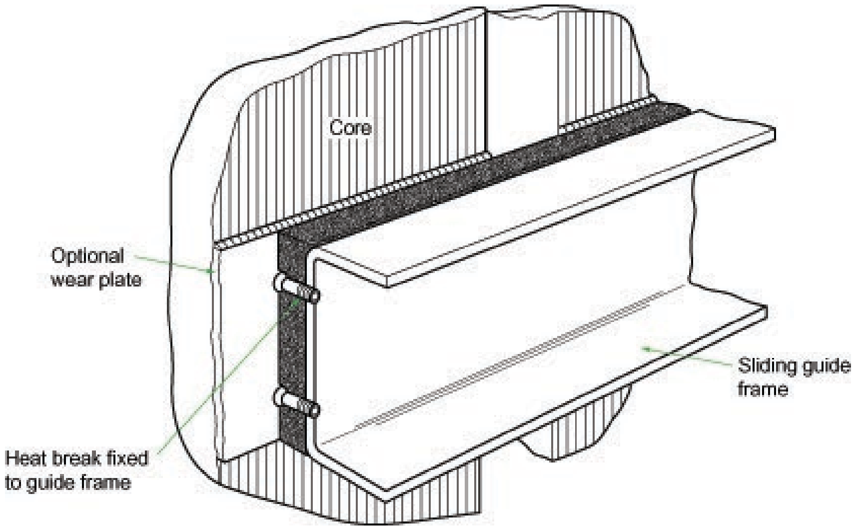


Figure 4-1: Typical sliding guide frame

Figure 4-2 shows a typical assembly of three heat exchangers with a Shear Plate Support Arrangement. The heat exchangers are supported at the upper warm end and guided at the lower cold end.

* Trade name

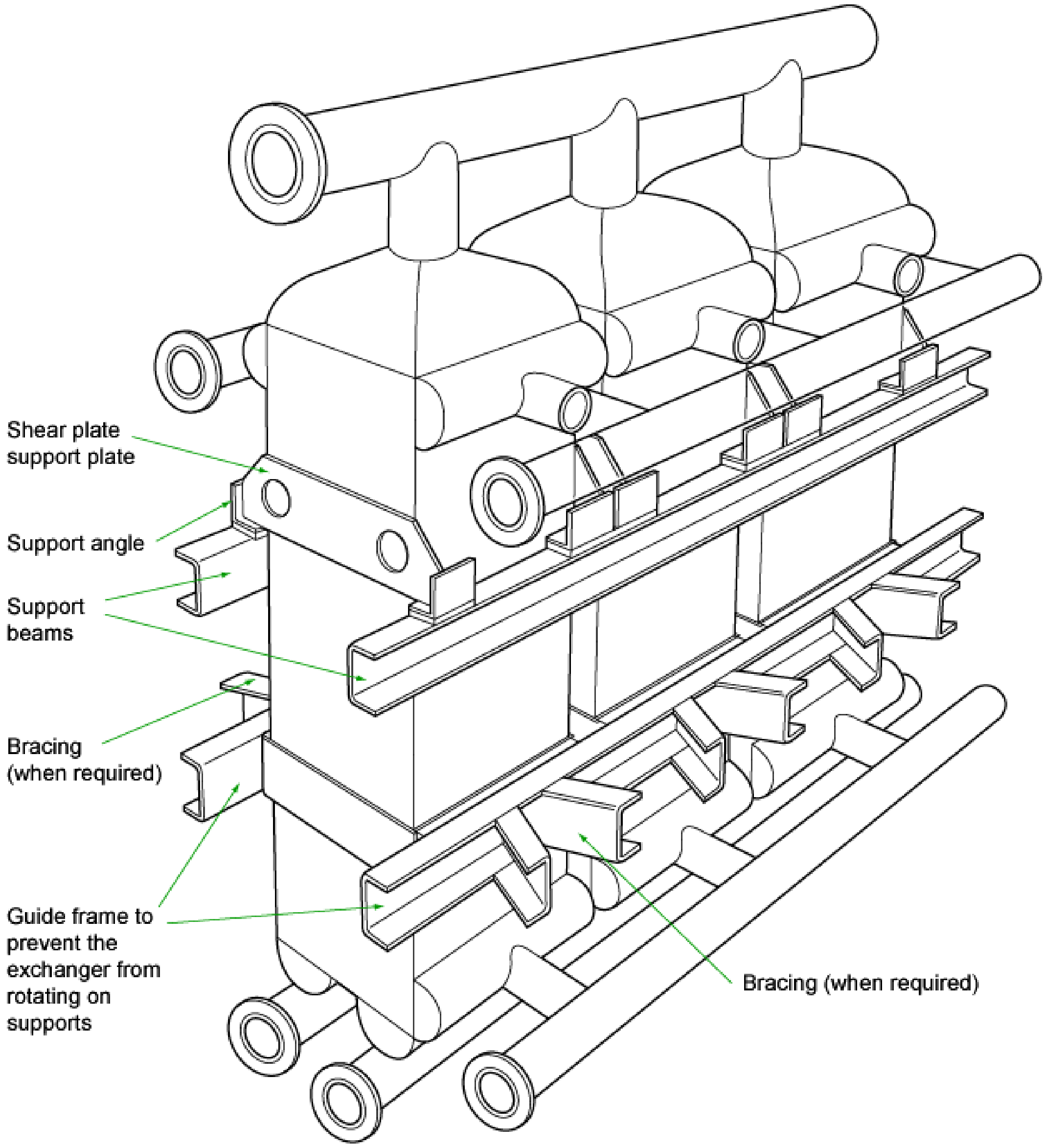


Figure 4-2: Typical heat exchanger assembly of three cores showing shear plate supports

Figure 4-3 shows a typical assembly of three heat exchangers with an Angle Bracket Support Arrangement. The heat exchangers are supported from angle brackets welded onto the side-bar faces.

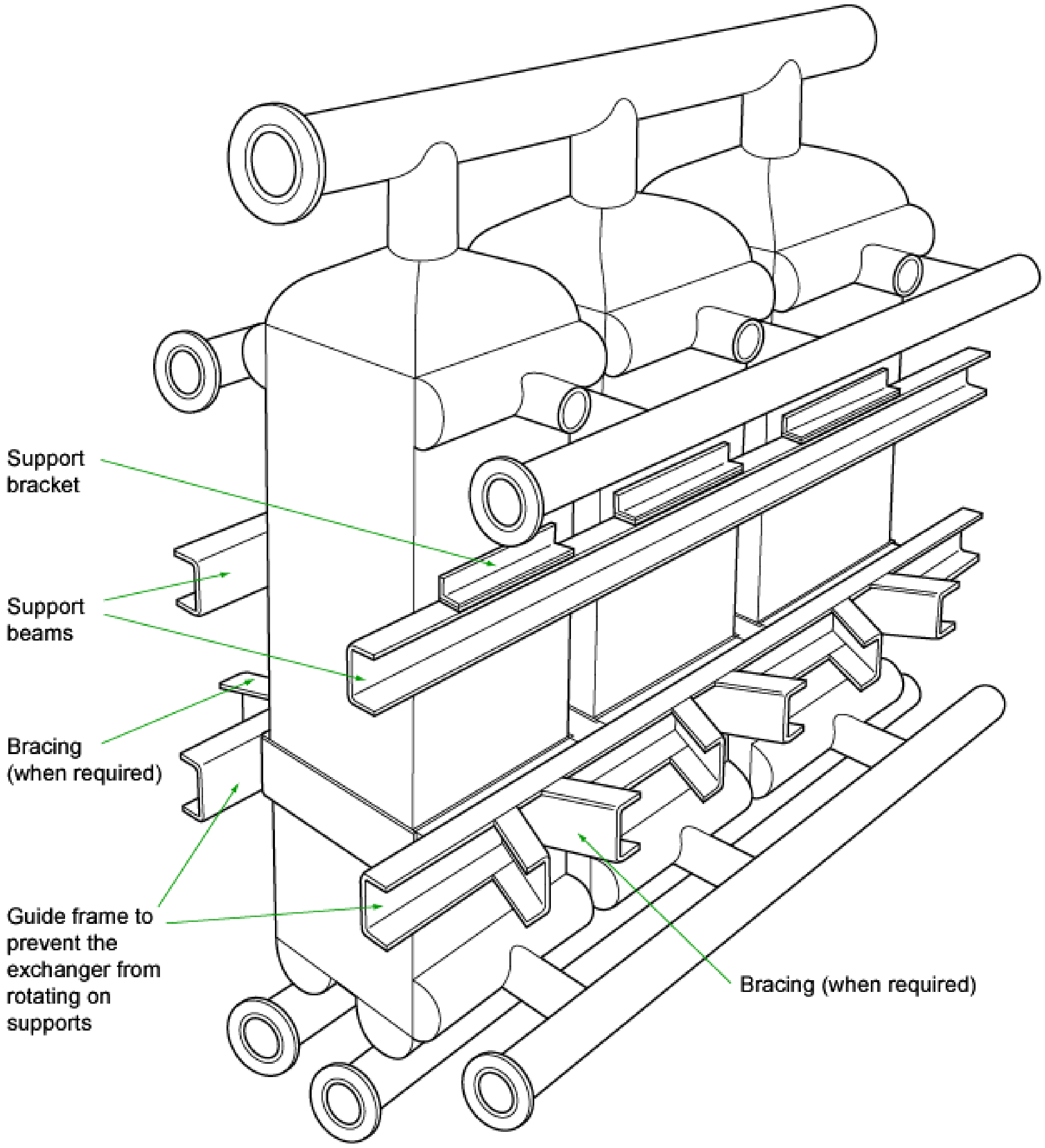


Figure 4-3: Typical heat exchanger assembly of three cores showing angle bracket supports

4.6 Fixing (Mounting) Bolts

There are several methods employed for locating heat exchangers on supports; a mounting bolt system is typical.

The object of the mounting bolt is to keep the heat exchanger in contact with the support beam. It is NOT to fix the heat exchanger rigidly, because a rigid attachment can lead to very high moments and forces being applied at the supports when the heat exchanger contracts or expands during start-up and shutdown.

Manufacturers recommend a mounting assembly that allows differential contraction of the supports on the heat exchanger and the support beams.

A typical method of using mounting bolts is shown in Figure 4-5. With this method a stainless-steel tube some 0.5 mm longer than the combined thickness of the support components is used to prevent rigid locking when the mounting bolt nut is tightened.

Alternatively, the mounting bolt nuts can be installed "finger-tight" to prevent rigid locking and the stainless-steel sleeve omitted. However, the bolts should be installed "head-up" to avoid detachment should the nuts work loose and it is recommended that lock nuts or thread locks be used.

Typically, four mounting bolts per heat exchanger are employed, one at each corner. These should be manufactured from stainless steel, although for light applications an appropriate aluminium alloy may be used.

During start-up, the heat exchanger can contract or expand on the mounting beam in both horizontal directions.

WARNING: THE NECESSARY AMOUNT OF CLEARANCE SHALL BE ALLOWED BETWEEN THE INSERT TUBE (OR BOLT ONLY WHEN USED) AND THE SUPPORT COMPONENTS.

The necessary clearance to take account of the expected thermal movement may be calculated from the following expression for both horizontal directions:

$$X = \alpha_\ell \Delta T_R s \tag{4.1}$$

where

X	=	Required clearance	mm
α_ℓ	=	Coefficient of linear expansion at the average temperature between ambient and operating temperatures (from Figure 4-4)	m/m K
ΔT_R	=	Temperature range at support; difference between operating temperature and ambient temperature	K
s	=	Distance between the extreme bolts in a given plane	mm

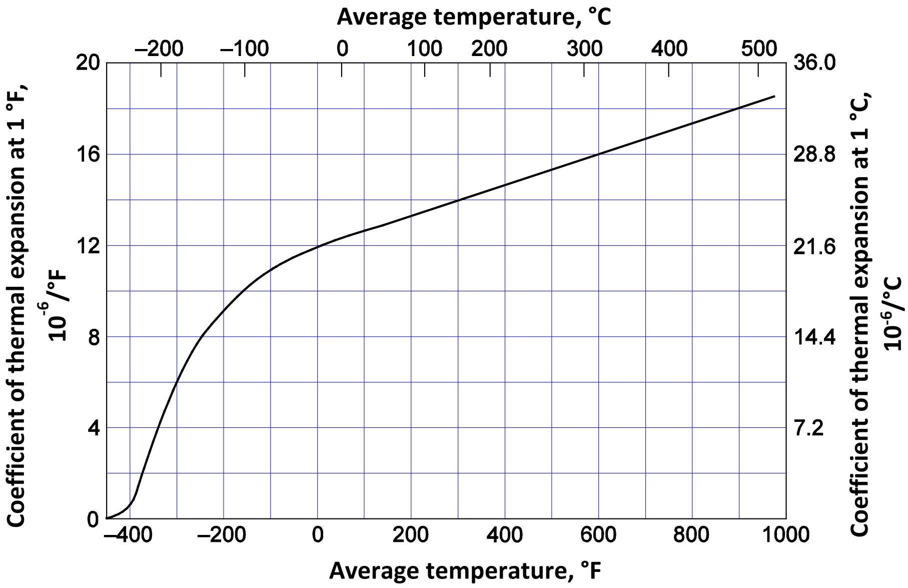


Figure 4-4: Coefficient of thermal expansion of aluminium

For estimated clearances of up to 4 mm, the clearance may be achieved by drilling the mounting hole in the support bracket with an oversize. For larger clearances, the support bracket shall be slotted in the direction of the movement. For larger two-directional movements, the clearance is maintained by slotting both the support bracket and the support beam, with the slots being positioned at 90 degrees to each other.

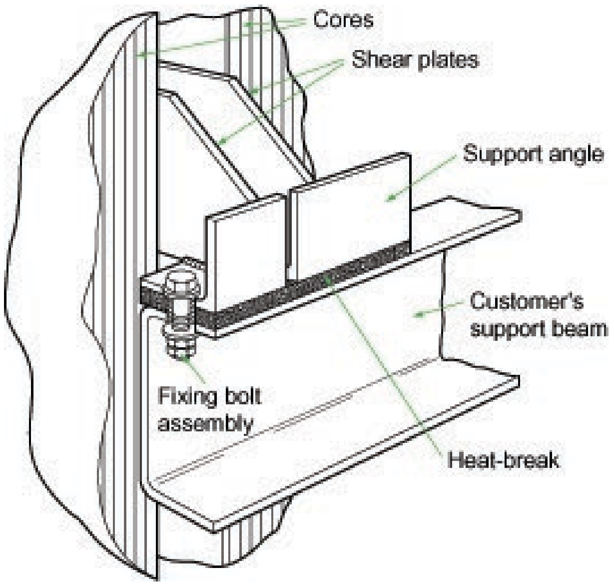


Figure 4-5: Typical shear plate bolt assembly

4.7 Venting of Dummy/Inactive Areas

Many brazed aluminium plate-fin heat exchangers contain void or inactive areas. Typical examples of these inactive areas are

- inactive or dummy layers on the top or bottom of the heat exchanger stack
- the space formed between two streams in the same layer having adjacent side headers
- the dead corner of an end distributor employing the slant bar drainability feature
- the modular space formed by welding together two heat exchanger blocks
- other special features

Unpressurized dummy/inactive layers are provided with vent openings. Manufacturers take precautions to dry and seal the inactive areas prior to shipment.

WARNING: FOLLOWING INSTALLATION AND PRIOR TO SITE TESTING AND OPERATION, IT MUST BE ASCERTAINED THAT THE VENTS OF THE INACTIVE AREAS ARE NOT LEFT PERMANENTLY SEALED. VENTS SHALL BE EITHER FITTED WITH REMOVABLE COVERS, OR CONNECTED TO VENT LINES, OR LEFT OPEN (DEPENDING ON EQUIPMENT CONFIGURATION) AS INSTRUCTED BY THE MANUFACTURER.

4.8 Field Testing

4.8.1 Non-destructive Testing

Manufacturers recommend the following non-destructive testing of the connecting pipework welds to be carried out to maintain an acceptable quality level.

1. A visual inspection of all connecting pipework root welds.
2. A liquid penetrant examination of all connecting pipework and cap welds.
3. A radiographic examination of a minimum of 10% of all closing butt welds. Representative samples of each welder's work should be examined.

The test procedures and acceptance criteria shall be in accordance with the governing construction Code.

Note: Dye penetration examination of the brazed surfaces should not be performed as this type of testing is not appropriate, not required and the results are misleading. This is due to

1. The configuration of the side-bar and parting sheet surface and
2. The braze engagement length (bar width) which greatly exceeds the minimum braze engagement length required by the governing construction Codes.

For Clarification Purposes: Due to the geometry of the side-bar and parting sheet surface, false indications of braze defects occur when subjected to liquid penetration examination. Due to the width of the bars used, the braze engagement length far exceeds the minimum required by the construction Codes and thus the adequacy of the brazed joint does not depend on the brazed joint extending fully to the outside surface of the brazed aluminium plate-fin heat exchanger.

4.8.2 Proof Pressure Testing

Most governing jurisdictions require a pressure test of the piping system after a heat exchanger is installed. The test shall meet the requirements of the relevant Code Authority with the following additional considerations fully taken into account.

It is highly recommended that a pressure test is NOT conducted with water as the test medium. Water removal after the heat exchanger is installed is difficult and residual water trapped within the heat exchanger can freeze during unit operation causing serious damage and lead to premature failure of the heat exchanger.

WARNING: SPECIAL PRECAUTIONS SHALL BE TAKEN IF THE HEAT EXCHANGER IS TO BE PNEUMATICALLY TESTED. PNEUMATIC TESTING CAN BE HIGHLY DANGEROUS IF NOT CARRIED OUT BY FOLLOWING THE RELEVANT LOCALLY APPROVED PROCEDURES.

Each stream of the heat exchanger system shall be tested individually, with the other streams not pressurised. Oxygen-free nitrogen of dewpoint -40 °C or better should be used as the test medium.

WARNING: THE TEST PRESSURE SHALL BE IN ACCORDANCE WITH THE LOCAL CODE REGULATIONS BUT SHOULD NOT UNDER ANY CIRCUMSTANCES EXCEED THE PROOF TEST PRESSURE OF THE HEAT EXCHANGER.

On completion of the proof test for the heat exchangers tested at values above the design pressure, each stream is firstly to be de-pressurised down to its design or operating pressure and a soap bubble leak test carried out on all pipe weld connections. For heat exchangers proof tested at the design pressure or lower, the leak test is to be carried out at the operating pressure.

The evaluation of the proof pressure test shall take into account any variation in ambient temperature from commencement of the pressure hold period to its completion. For the test pressure to have been satisfactorily held over this period, the final pressure can be calculated from

$$\text{Final pressure reading} = \text{Initial pressure reading} \times \frac{\text{Final ambient absolute temperature}}{\text{Initial ambient absolute temperature}}$$

The pressures used in this calculation are absolute, not gauge.

4.9 Insulation

On completion of all field testing the heat exchanger shall be insulated.

For heat exchangers mounted within a cold box the minimum insulated distance, in mm, between the heat exchanger and cold box wall is taken from Figure 4-6. The void space between the heat exchanger and wall shall be packed with insulant. This may be either expanded perlite or rockwool. For perlite a density in the range of 50 to 70 kg/m^3 is normally used. When packing with rockwool care shall be taken to avoid damage to the heat exchanger's connections. Prior to start-up, a continuous dry oxygen-free nitrogen purge is to be connected to the cold box.

For stanchion (pedestal) or frame mounted heat exchangers, the minimum thickness of insulation is also to be taken from Figure 4-6. The insulation used for this type of heat exchanger is usually of the spray-on polyurethane foam type. After application of the insulant, the heat exchanger shall be sealed with a weatherproof jacket.

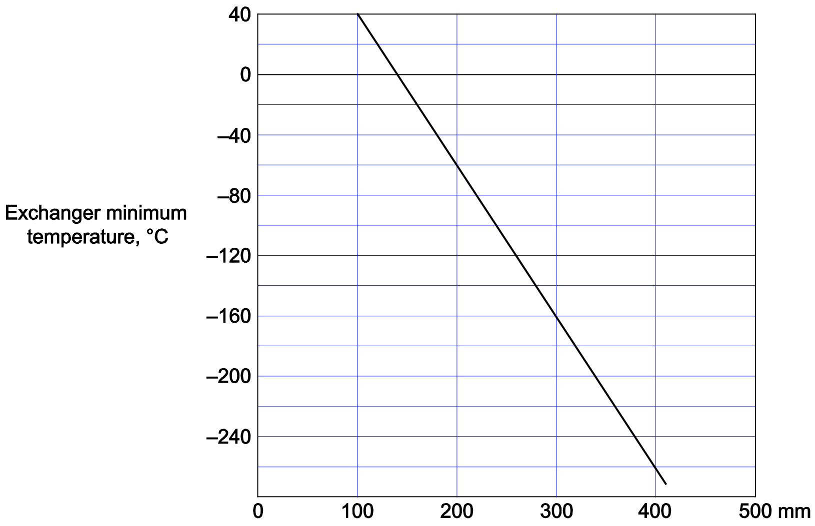


Figure 4-6: Recommended minimum insulation thickness (mm)

4.10 Operation

4.10.1 Start-up

WARNING: TO ENSURE SAFE OPERATION EACH STREAM OF THE HEAT EXCHANGER SHALL BE PROTECTED WITH A PRESSURE RELIEF DEVICE. IT IS THE RESPONSIBILITY OF THE USER TO PROVIDE AND ENSURE PROPER INSTALLATION OF THE PRESSURE RELIEF DEVICES. THE RELIEF PRESSURES SHALL BE SET NO HIGHER THAN THE MAXIMUM ALLOWABLE WORKING PRESSURE OF THE STREAMS, CONSIDERING BOTH THE HEAT EXCHANGER AND THE CONNECTING PIPING.

Prior to start-up, internals of the connecting pipework and vessels system shall be thoroughly cleansed of all particulate matter such as rust, scale, grit or sand. The system should then be purged using oxygen-free nitrogen or other suitable purge gas (dewpoint of -40 °C or less). The objective of this purge is to remove any residual moisture, the presence of which could result in freeze damage to the heat exchanger during operation. Duration of the purge should range from 4 hours to several days depending on size, complexity and physical state of the heat exchanger system. The purge exit should be monitored until consistent readings of dewpoint (approaching that of the inlet purge gas dewpoint) are obtained.

Cool-down of the heat exchanger is only permitted using gas (i.e., no liquid phase present). Cool-down should be carefully controlled to avoid thermal shocking of the heat exchanger and pipework. A rate of 5 °C per minute maximum change of stream temperature is normally recommended to allow for gradual dimensional adjustments but the Manufacturer should be consulted if this rate is likely to be exceeded. With the agreement of the Manufacturer, rates in excess of 5 °C have been approved for certain heat exchanger applications. Furthermore, the cool-down rate shall not exceed 60 °C in an hour. The cooling gas should be introduced to all streams simultaneously to prevent local thermal stresses developing. The gas when introduced to the system should not have a temperature difference greater than 30 °C relative to the local metal temperature.

A record of all relevant data should be kept for each individual start-up (see Section 4.12). This shall be required in the event of problems developing later in the life of the heat exchanger.

Fatigue induced by changing conditions is discussed in Section 8.1.2.

Temperature monitoring devices can be added when requested by the customer.

WARNING: THE MAXIMUM OPERATING PRESSURE FOR THE DESIGN OR WORKING TEMPERATURE SHOWN ON THE HEAT EXCHANGER'S NAMEPLATE AND THE MANUFACTURER'S DRAWINGS SHALL NOT BE EXCEEDED.

4.10.2 Normal Operation

If all the recommended procedures have been followed, then the heat exchanger can give many years of trouble-free service.

Some industrial pollutants, notably mercury, sulphur dioxide, chlorine, nitrogen oxides, etc., can be extremely harmful and corrosive to aluminium. It is therefore advisable that reference be made to Chapter 8 on Recommended Good Practice on operation with potentially corrosive streams.

To prevent particulate matter from entering the heat exchanger, the heat exchanger system shall be operated with mesh filters at the stream inlets. A valved bypass system should be considered to permit cleaning of filters without having to shut down. As a minimum, filtration should remove particles larger than 177 microns (80 Mesh Tyler Standard).

All process fluids entering the heat exchanger should be in steady flow state. Pulsing or surging from pumps or compressors shall be avoided, and the Manufacturer should be consulted about permissible limits.

To prevent over-pressurisation, it is the User's responsibility to install sufficient and suitable pressure-relieving devices into each stream. The relief setting of the devices shall not be greater than the stated maximum allowable working pressure. Relief settings and relief capacities shall comply with the relevant governing Code and there is no need to account for interstream leakage due to the nature of the construction of brazed aluminium plate-fin heat exchangers.

4.10.3 Shut-down

The cautions applicable to start-up in Section 4.10.1 also apply to shut-down. In particular, to prevent thermal shocking, warm-up should be accomplished slowly at no more than the recommended rate of 5 °C per maximum change of stream temperature. Furthermore, the warm-up rate shall not exceed 60 °C in an hour. As with start-up the Manufacturer should be consulted if this rate is likely to be exceeded.

4.10.4 Warming Up

Warming up shall be performed with gas and only after all liquid has been drained from the heat exchanger.

On attainment of ambient temperature the heat exchanger is to be purged with oxygen-free nitrogen supplied by a Vendor with a dewpoint of -40 °C or lower. When a sterile internal atmosphere has been achieved then the heat exchanger should be blanked off using blind flanges. If the shut-down is to continue for any length of time, the streams should be pressurised with dry nitrogen to a pressure of 0.2 to 1.2 bar g.

4.11 Off-design Scenarios

4.11.1 Unplanned Two-phase Inlet Conditions

Two-phase inlet conditions which have not been considered in the design may cause significant maldistribution of heat duty. In particular, when errant liquid is boiled in the exchanger it can cause severe temperature differences and fluctuations between adjacent parts of the exchanger. Basically, un-planned two-phase conditions may occur due to

- flashing of liquids (e.g., lack of sub-cooling, low pressure conditions),
- liquid carry over from upstream equipment,
- exceeding the capability of a two-phase inlet device,
- excursion from specified process conditions.

Good practice

Operators should take care that operating conditions (through data available on a Distributed Control System (DCS), for example see 4.13.1.3) are consistent with the cases provided for the design. Process modelling may allow to assess the process envelope and check cases which shall have a risk as mentioned above.

4.11.2 Cold Restart After Stop of Flows

The longitudinal wall temperature profile, which is determined by the process streams during operation, will slowly equalize by heat conduction after shut-down. If the offline period was of extended duration (e.g., due to a prolonged trip), and streams are re-started with almost the original temperatures, large temperature differences may occur between the fluid and the local wall at the stream inlet position. This is because the wall temperature has changed in the meantime. This can lead to rapid wall temperature changes at the stream inlet position and generate thermal stress, as illustrated in Figure 4-7.

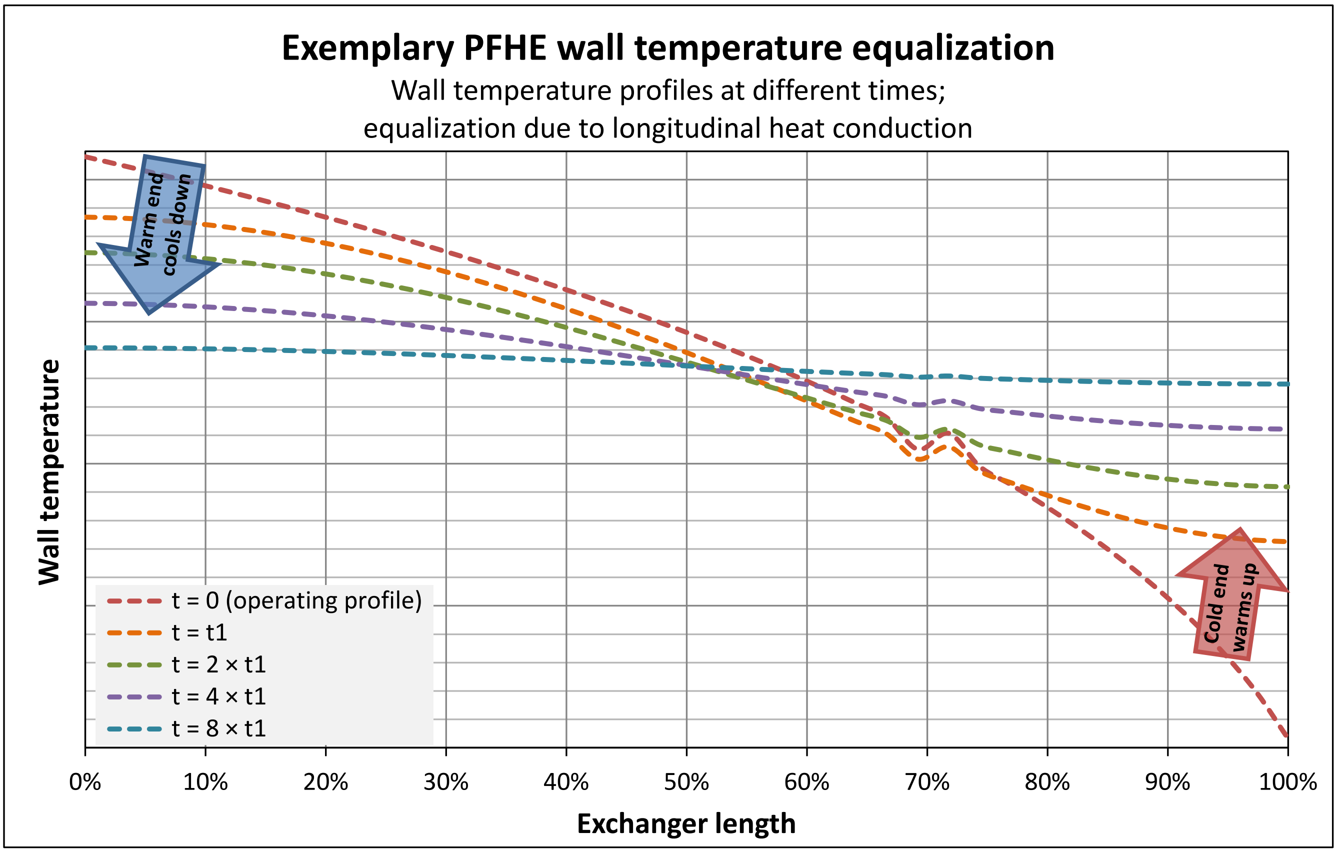


Figure 4-7: Brazed aluminium plate-fin heat exchanger wall temperature equalization (example) after shut-down (assumes all streams are gas phase)

The wall temperature after shut-down might also be influenced by the specific boundary conditions. For example, if the single streams are not stopped simultaneously or if boiling liquid inventory remains in the exchanger.

Good practice

Temperature measurement devices can be installed in or on end bars to measure end bar metal temperatures. On restart, flow is introduced slowly ("crack the valve") until the stream temperature and the end bar temperature are within ALPEMA temperature difference guidelines. The flow is then increased while maintaining ALPEMA recommended temperature rates of change.

4.11.3 Deriming / Warm-up

The Operator should be aware that uneven warm-up may cause large pipe stresses, and warming up from the cold end causes large temperature differences where the warm-up gas is introduced. Thermal stress may be also generated.

Good practice

If uneven warm-up cannot be avoided, this case should be considered in the piping design.

4.11.4 Emergency Depressurization

Auto-refrigeration may lead to large temperature changes if liquids are not de-inventoried before depressurization. Nevertheless, depressurization through the top of the exchanger having not de-inventoried liquid can carry over very cold liquid up to warm end piping, which may not be designed for such cold temperatures.

Good practice

- Shut off flow from the top of the exchanger and vent through the bottom of the exchanger. This will establish parallel (co-current) flow and minimize heat transfer during blowdown.
- Minimize the distance between the blowdown valves and the brazed aluminium plate-fin heat exchanger to minimize the stream volume transferred through the exchanger during blowdown.

4.11.5 Turbomachinery Trips and Control Switch Functions

Safety related switch functions or trips of rotating machines may cause the failure of single streams. This may lead to heat imbalances, and hence rapid wall temperature changes that cause excessive thermal stresses.

Good practice

Plant control systems should be designed to maintain a heat balance in the heat exchanger. Dynamic process modelling is recommended to assess the suitability of the plant control system.

4.11.6 Local Temperature Difference Between Stream and Wall (Streams with Intermediate Inlet Position)

The inlet header position will be determined from the fluid inlet temperature and the metal wall temperature profile, according to the design conditions (see Figure 4-8). Excessive temperature differences between the stream and the local wall may occur due to the following:

- The exchanger temperature profile deviating from the expected profile (e.g., off-design condition, start-up, deviating heat-balancing).
- The stream inlet temperature deviating from the expected value.

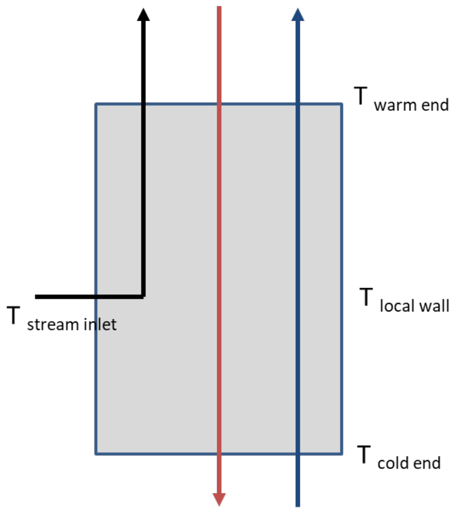


Figure 4-8: Brazed aluminium plate-fin heat exchanger with intermediate inlet position

Good practice

Temperature measurement devices can be installed to measure side-bar metal temperatures. When the stream is introduced into the heat exchanger, flow is introduced slowly ("crack the valve") until the stream temperature and the side-bar temperature are within ALPEMA temperature difference guidelines (see Chapter 8). The flow is then increased while maintaining ALPEMA recommended temperature rates of change (see Section 4.10).

4.12 Instrumentation and Monitoring of Operations

Instrumentation is recommended to monitor the thermal history over the lifetime of brazed aluminium plate-fin heat exchangers. Monitoring is necessary to detect plugging, fouling, thermal stress conditions, instabilities or unexpected controller behaviour. Data delivered by the instrumentation may be used for on-line assessment and further analysis, and include the following:

- Knowledge regarding the fluids' composition is required, but usually this information can be derived from other sources. However, on-line measurements may be helpful for specific streams to increase reliability of simulations.
- Fluid measurements of inlet / outlet temperature, pressure and flow rates. These are usually required for thermal / hydraulic simulation.
- Metal temperature measurements at feasible positions. These may be useful for multiple purposes, for example: calculation of the rate of change of local wall temperatures; determination of the local temperature difference between local wall and fluid before starting the flow; detection of deviations from the design wall temperature profile, including alarming. Metal temperatures allow more accurate thermal / hydraulic simulations, hence more reliable assessments.
- Pressure drop measurements. These are typically used to detect plugging.
- For boiling streams a dedicated differential pressure instrument should be used to monitor pressure oscillations, which indicate unstable flow.

Measurements should be taken and recorded at a maximum of 1-minute intervals.

For hydrocarbon applications, gas detectors should be installed in the vicinity to alert if any external leaks exist. For cold boxes, gas detectors should be installed close to the purge outlet.

Sockets for metal temperature measurements are not in a basic scope of supply. Temperature monitoring devices can be added when requested by the Purchaser. The quantity, type, and location of such devices shall be agreed between the Supplier and the Purchaser.

In general, the quality of measurements may be compromised under specific conditions. For example, fluid temperature measurements may not deliver reliable information when there is no flow or in two-phase conditions (especially for single component streams). Flow rate measurements may not be reliable at low flows or at deviating pressure / temperature.

4.13 Maintenance

Problems with brazed aluminium plate-fin heat exchangers are rare. However, if problems develop, advice should be sought from the Manufacturer. Rectification of problems should not be attempted without prior consultation with the Manufacturer.

The integrity of a brazed aluminium plate-fin heat exchanger can be assessed in part by performing various inspections. Some inspections can be performed while the unit is in operation, while others require the unit to be out of service. The Owners and Operators should establish an inspection schedule based on service, operating conditions, Code requirements, local regulation, and Manufacturers' recommendations.

Inspections should also be performed if damage is suspected, or if there is concern about the integrity of the brazed aluminium plate-fin heat exchanger, or lack of knowledge of the operating history of the heat exchanger.

Extra care is required when investigating problems associated with heat exchangers installed in cold boxes (see Section 9.2.14).

The nitrogen purge for the cold box shall be disconnected.

WARNING: COLD NITROGEN GAS CAN ACCUMULATE AT GROUND LEVEL AND CAUSE FATAL RESULTS THROUGH ASPHYXIATION.

4.13.1 During Operation

4.13.1.1 Visual Inspection

The exterior of the unit should be inspected in operation for anomalies such as frost spots, dripping fluids, and gas clouds. This may indicate external leaking is occurring, or there may be insulation problems. This inspection is recommended every shift.

4.13.1.2 Fluid Composition Testing

The fluid composition of the streams, both upstream and downstream of the unit should be monitored. Changes in composition from upstream to downstream of the brazed aluminium plate-fin heat exchanger may indicate internal cross pass leaking is occurring. In a closed-loop system, gradual changes in composition over time may also indicate cross pass leaking.

4.13.1.3 Operating Data Monitoring

Operating data include stream flow rates, inlet and outlet temperatures and pressures for all streams, often recorded at one-minute intervals or less. This data contains information about how the unit has been controlled, events that may have caused high thermal gradients, and how well it is performing. If there are any issues or problems with a brazed aluminium plate-fin heat exchanger, including plugging, fouling, or high thermal stress, there will most likely be indications of the issues in the operating data.

A log of all plant operational events which might affect the heat exchanger should be kept. The log should record details of all operations, especially start-ups, shut-downs, and any plant malfunctions, power outage, together with any available routine operational data. See Section 8.1 for guidance on thermal stresses.

Operators should continuously monitor the real-time temperatures, pressures, and flowrates. After operational events watch for indications of:

- high local stream temperature differences
- high rate of change of stream temperature
- high pressure drops
- abnormal fluctuations in pressure, temperature, and flowrate

Additionally, the operating data may indicate whether general operation or specific procedures should be improved in order to minimize thermal stresses and extend the service life of the brazed aluminium plate-fin heat exchanger. Periodic inspection of operating data is the easiest way to assess the history of a brazed aluminium plate-fin heat exchanger and can be a valuable tool in improving current operations and procedures.

In examining operating data, the Operator should check for the following conditions:

1. Pressure above nameplate maximum value specific for that stream. This can indicate operational problems, or problems with pressure relief systems.
2. Temperature above nameplate maximum value specific for that stream. This can affect the mechanical integrity.
3. High rates of change of stream or metal temperatures. This can indicate thermal shock in extreme cases, or thermal fatigue for regularly occurring temperature cycles.

4. High local stream-to-stream temperature differences. This can indicate thermal shock, or higher susceptibility to thermal fatigue.
5. Rapidly or cyclically changing flow rates. This can cause thermal cycling, leading to thermal fatigue.
6. High pressure drop. This can indicate plugging, hydrate or ice formation, or fouling.

For a well-controlled process, operating data should be reviewed typically on a monthly basis. For new operations, review the operating data continuously until the operations are understood or until known process control issues are under control.

4.13.2 While Out of Operation

The following recommendations are made for maintenance considerations.

4.13.2.1 External Inspection

Visually examine the exterior of the exchanger for bulges in the cap sheets or other anomalies such as cracks. Bulges in the cap sheets may indicate ice formation has occurred. Cracks likely indicate thermal fatigue or thermal shock damage. If such anomalies are discovered during visual inspection, the Manufacturer should be notified for further investigation before the exchanger is put back in service.

Perform dye penetrant test on all external welds. See Section 4.8.1.

4.13.2.2 Internal Inspection

Insert a borescope into the header and inspect the fins. If visible debris or particulates are found this could indicate plugging. If discolouration or residue is seen this may indicate corrosion or fouling. If visible cracks in fins or internal header welds exist this may indicate thermal fatigue or thermal shock damage.

4.13.2.3 Leak Detection

See Section 4.14 for information on leak detection procedures, and Section 4.13.1.3 for monitoring operating data.

4.14 Leak Detection

4.14.1 Introduction

External leaks will be evident by the appearance of localised freeze spots or vapour clouds on the outer casing of the insulation. In the case of heat exchangers installed in a cold box, an increase in or contamination of the purge gas flowing out of the cold box purge valve will be evident. The smell or sound of the escaping fluid may also be discernible.

For heat exchangers installed within cold boxes, external leaks may quickly increase over time through perlite swirling, leading to erosion of aluminium surfaces.

Internal leaks can manifest themselves in the reduction of product purity and, if the leak is of sufficient magnitude, a redistribution of flow levels will occur between affected streams.

When a leak is suspected, it should be investigated fully and immediately and the Manufacturer's repair procedure be put in action as soon as practicable. In the case of leakage where harmful substances are present, immediate action is necessary.

WARNING: FAILURE TO RECTIFY LEAKING UNITS MAY RESULT IN PERSONNEL INJURY AND/OR SERIOUS DAMAGE TO THE UNIT AND COMPROMISE THE SAFETY OF THE PLANT.

To establish the existence of a leak the following procedures may be applied.

4.14.2 Site Leak Detection Pressure Test

Prior to any site work taking place, all streams shall be purged with either dry nitrogen or dry air. A gas analysis should be performed to ensure that any harmful gas residues are completely removed from the system. In the case of work being carried out within a cold box, the oxygen concentration within the cold box has to be controlled continually with good venting of the box maintained through openings on the top and bottom.

Each of the unit streams is isolated using blind blanking flanges fitted with suitable pressure gauges. In turn and on an individual basis, each stream is pressurised; initially the test pressure should be set at a maximum of, say, 5 bar g since the majority of leaks will be detected at low pressure. Further tests at higher pressures may be necessary depending on the type of leak but the test pressure should not exceed the operating pressure of the stream being tested. The test should be carried out with either dry nitrogen or dry air. The pressure is held for a period of ten to twelve hours. If after this time the pressure level has decayed, and the amount of the decay was not caused by temperature changes, the presence of a leak has been established. The holding time will depend on the sensitivity of the pressure gauges and the volume of the streams on test. A coinciding rise in pressure of any of the isolated streams is indicative of an internal interstream leak. All streams shall be fitted with pressure relief systems to prevent over-pressurisation.

A check is to be carried out to ensure that no mechanical joints are leaking.

If no subsequent rise in adjacent stream pressure is evident then the leak is to the external.

Throughout this procedure the safety aspects covered in Section 4.8.2 (Proof Pressure Testing) of this Standard **SHALL** be rigidly adhered to.

WARNING: PRECAUTIONS SHALL BE TAKEN TO ENSURE THAT LEAKS HAVE NOT CREATED A COMBUSTIBLE SITUATION OR DISPLACED OXYGEN IN ENCLOSED SPACES.

WARNING: SPECIAL PRECAUTIONS SHALL BE TAKEN IF THE HEAT EXCHANGER IS TO BE PNEUMATICALLY TESTED. PNEUMATIC TESTING CAN BE HIGHLY DANGEROUS IF NOT CARRIED OUT BY FOLLOWING THE RELEVANT LOCALLY APPROVED PROCEDURES.

All pressures recorded shall be adjusted in accordance with the method described in Section 4.8.2, to compensate for ambient temperature differences over the duration of the test.

To locate external leaks the soap bubble test described in Section 4.8.2 of this Standard is repeated.

4.14.3 Site Helium Leak Detection Test

The helium leak test may be used to locate leaks that are difficult to find, both internal and external. However, while helium leak testing is a valid testing procedure, especially when dealing with high purity applications, carrying out such tests and interpreting the results on site can be impractical. Only highly trained personnel using specialised equipment shall therefore carry out such helium tests.

4.15 Repair of Leaks

Field repairs to brazed aluminum plate-fin heat exchangers should only be conducted and documented by companies maintaining valid pressure design code qualifications, for example, a valid "R" (Repair) stamp issued by the National Board Inspection Code (NBIC), with competent welders qualified to a code such as Section IX of the ASME Code. In addition, familiarity with thermal/mechanical/hydraulic design of the heat exchanger is required.

To achieve maximum life expectancy of the heat exchanger, root cause analyses of leakage/failure events should be conducted. This may involve metallurgical analysis and/or review of past operating history. If thermal fatigue is suspected as causing the leaks and the repair under consideration is the second repair due to thermal fatigue, it is recommended the customer replace the heat exchanger within a reasonable time frame.

Actual field repair procedures (external/internal leaks) are dependent on the type of leakage identified. Internal leaks result in seal welding/blocking of process layers/passages and the resultant reduction of heat transfer area. Thermal and hydraulic reviews should be conducted before the process of finalizing the repair procedure. Since a thermal and hydraulic review uses proprietary Supplier information, the original equipment Manufacturer should be involved in the evaluation. Any blocked layers shall be vented.

4.15.1 Repair of Leaks to the Brazed Aluminium Plate-Fin Heat Exchanger

Should the need arise to locate and repair a leak associated with the brazed aluminium plate-fin heat exchanger then this should be discussed with the Manufacturer.

Dependent on the type of leak, the location of the leak, accessibility in the plant, together with other criteria, such as climatic conditions at site, a recommendation can be given by the Manufacturer on how, where and when a repair should be carried out. The repair of internal and external leaks involving work on the brazed structure of the heat exchanger requires specialised knowledge and repair techniques. These repairs shall not be attempted without prior consultation with the Manufacturer. It is strongly recommended that any such repairs be carried out only by the Manufacturer or a recognised specialist repair team who have at their disposal all necessary backup and equipment to effect such repairs.

As previously detailed under Section 3.10, repair or rectification can be carried out during the manufacture of the heat exchanger; this is recognised to be a normal and acceptable practise in the industry. These same repair-welding techniques can be employed to repair a heat exchanger in the field, and when carried out by an approved specialist repair team the mechanical integrity of the heat exchanger should not be affected. Within the contractual mechanical guarantee period any repair or modification of the brazed aluminium plate-fin heat exchanger not authorized by the original Manufacturer shall typically lead the mechanical guarantee to become immediately void.

4.16 Storage

The following are general recommendations for interim and long-term storage of brazed aluminium plate-fin heat exchangers. The Purchaser shall refer to the Manufacturer's Installation, Operation and Maintenance (IOM) manual for specific instructions.

An indoor storage area away from any main work area is recommended. Indoor storage is required for exchangers having open layers or nozzles not covered with welded or bolted covers. Any ingress of moisture into these open layers should be avoided. In all storage areas, the following additional recommendations should be followed.

The heat exchangers are shipped in protective crates, wooden boxes, or on wood or steel skids and should be stored in the original shipping package. Stacking of exchangers is not advised without prior approval from the Manufacturer.

The storage area should provide level, uniform support with good drainage. When the heat exchanger is removed from its crate or packaging, it should be laid on wooden sleepers in a horizontal position on the outside cap sheet face of the exchanger.

The storage area should not be located where the heat exchanger is subjected to fluids or atmospheres which are corrosive to aluminium or subjected to vibration.

Avoid a location where other work activity or falling objects will be in the vicinity of the stored heat exchanger. External denting of the heat exchanger can damage the internal structure of the heat exchanger and cause leakage.

Avoid a location which is subject to large fluctuations in temperature, especially below 0 °C (32 °F), or high humidity when the exchanger is not sealed and weather-proofed, as this can cause condensed water to accumulate in the exchanger and freeze when the exchanger is placed in storage or operation. Water freezing inside the heat exchanger can damage its internal structure.

Heat exchangers shall be properly covered and sealed in such a manner that dirt, sand, water, or foreign materials cannot enter open nozzles, ports, or through any other access into the heat exchanger. Periodically, the heat exchanger should be checked to ensure that the transport pressure is maintained as per Section 3.7.6.

For heat exchangers which are not shipped with transport pressure and do not have welded shipping covers on the nozzles, all nozzle openings on the heat exchanger should be covered and sealed while the unit is in a dry condition.

5 MECHANICAL STANDARDS

5.1 Scope

These Standards apply to all vacuum-brazed aluminium plate-fin heat exchangers.

In theory, there is no limit to the size of a brazed aluminium plate-fin heat exchanger core since the internal pressure forces are resisted by the internal structure (fins, sheets). Thus, effectively the size of a single brazed core is limited by the size of the vacuum furnace of the Manufacturer.

However, the wall thickness and diameter of the headers, nozzles and piping connections for a given internal pressure will limit the practical size of the heat exchanger core.

5.2 Definition of a Brazed Aluminium Plate-Fin Heat Exchanger

A brazed aluminium plate-fin heat exchanger is an arrangement of a succession of layers, each being designed for heat exchange duties, flow characteristics and pressure, specific to given process conditions.

All the layers carrying the same stream are connected together by headers (inlet, outlet, intermediate) directly attached by welding onto the brazed core, as illustrated in Figure 1-2. Certain layers may be open to the surrounding space within a vessel, i.e., without any headers attached to the core (typical arrangement for condensers / reboilers, or see Section 9.1).

5.3 Codes for Construction

The design, construction and testing of brazed aluminium plate-fin heat exchangers are governed by the existing national rules applying to pressure vessels.

The design of a heat exchanger is the result of the mechanical strength analysis of:

- the brazed aluminium plate-fin structure under pressure
- the influence of headers on the plate-fin structure
- the header/nozzle assembly

Specific details regarding the design of the individual components are given in Section 5.15.

Brazed aluminium plate-fin heat exchangers are commonly designed under the provisions of the existing Codes, typically:

ASME VIll, Div. 1
European PED and related Codes (e.g., AD 2000, CODAP,
Dutch Pressure Vessel Code, Swedish Pressure Vessel Code,
Raccolta, etc.)
Japanese HPGS Law
AS 1210.

5.4 Typical Materials of Construction

Typical materials for use on the construction of brazed aluminium plate-fin heat exchangers are

Core matrix (fins, plates, side bars)	3003 aluminium alloy
Headers/nozzles	5083 aluminium alloy

For a more comprehensive set of materials, refer to Chapter 6.

5.5 Design Pressures

A brazed aluminium plate-fin heat exchanger is a pressure vessel consisting of more than one independent pressure chamber, operating at the same or different pressures. It shall be designed to withstand the most severe condition of coincident pressures expected in service.

Design pressures for each individual stream shall be specified by the Purchaser. Pressure parts shall be designed for full vacuum if specified by the Purchaser.

A brazed aluminium plate-fin heat exchanger is built from many different categories of components (including fins, side bars, parting sheets, plates, headers and nozzles), assembled into a single unit by brazing and welding. Due to this complexity, it is not possible to compute the Maximum Allowable Working Pressure (MAWP) of each stream during detailed design on the basis of the individual MAWP of each component. Therefore, for each stream, the MAWP is set equal to Design Pressure.

The Purchaser shall also indicate the design pressure of the environment around the heat exchanger, in case it is to be installed inside a pressure vessel. In this case, the heat exchanger shall be designed to withstand the internal pressure forces independently from the external compression forces.

The Purchaser shall also state if the heat exchanger is to be vacuum insulated (i.e., installed in a vacuum vessel) and the Purchaser shall determine the design pressures of streams accordingly.

5.6 Testing

5.6.1 Pressure Test

The brazed aluminium plate-fin heat exchanger shall be pressure tested in accordance with the applicable design Code. This may be carried out by either of the following methods:

5.6.1.1 Hydrostatic Test

The heat exchanger is hydrostatically tested with water. Each individual chamber is to be pressurised up to its test pressure with water.

The minimum hydrostatic test pressure at room temperature shall be 1.3 times the design pressure, except where Code or Purchaser requirements rule otherwise.

5.6.1.2 Pneumatic Test

The heat exchanger is subjected to a pneumatic test, where each individual chamber is pressurised up to its test pressure.

WARNING: A PNEUMATIC TEST MAY ONLY BE PERFORMED PROVIDED THE RULES OF SAFETY FOR SUCH PNEUMATIC TESTING ARE ADHERED TO.

The pneumatic test pressure shall be in accordance with Code and Purchaser requirements.

5.6.2 Leak Test

In order to ascertain the absence of a leak from one chamber towards any other chamber or into the atmosphere, a leak test is necessary. The extent-of-leak testing as well as the allowable leakage rates have to be agreed upon between Purchaser and Manufacturer.

The leak test may be carried out by either, or a combination, of the methods listed below.

5.6.2.1 Air Test

All chambers shall be tested for external and interstream (chamber to chamber) leakage. The test pressure, applied to one chamber only, is typically the design pressure for each individual chamber.

WARNING: AN AIR TEST MAY ONLY BE PERFORMED AFTER AND IF THE PRESSURE TEST DESCRIBED UNDER SECTION 5.6.1 HAS BEEN CARRIED OUT.

5.6.2.2 Helium Test

The following test methods may be used:

External leak test: helium vacuum test under non-metallic cover for checking the leak tightness of the exchanger to the atmosphere.

All chambers of the heat exchanger are evacuated and directly connected with the gas detector. The heat exchanger is sealed within a non-metallic cover and the space between cover and exchanger is filled with helium.

Standard allowable leak rate is 1×10^{-3} mbar litre/s (at pressure difference = 1 bar).

Interstream leak test: helium test for checking the leak tightness between those chambers selected by the Purchaser and the Manufacturer.

Helium is successively admitted to the test chamber. The other chambers are evacuated and connected to the gas detector. Starting with the chamber with the highest operating pressure each chamber is tested for leaks into the other chambers.

The standard allowable leak rate is 1×10^{-2} mbar litre/s (at pressure difference = design pressure) or 1×10^{-3} mbar litre/s (at pressure difference = 1 bar).

A helium test may only be performed after and if the pressure test described under Section 5.6.1 has been carried out.

5.7 Metal Temperature Limitations

5.7.1 Metal Temperature Limitations

The metal temperature limitations for the typical materials used are those prescribed by the Codes. See Section 6.1 for further information.

5.7.2 Design Temperature

All aluminium alloys have advantageous behaviour at low temperatures, i.e., the values of rupture strength and yield strength increase as temperature decreases. Therefore, only the maximum design temperatures are of importance for aluminium alloys.

Higher temperatures may be allowed for short periods and reduced pressures (e.g., for deriming purposes); Manufacturers should be consulted for details.

Temperature monitoring devices can be added when requested by the customer.

5.8 Permissible Temperature Differences Between Adjacent Streams

Due to the nature of brazed aluminium plate-fin heat exchangers which are produced by brazing, all internal components are metallurgically bonded to each other. The simultaneous presence of streams at different temperatures will produce contraction/expansion of the parts subjected to temperature, leading to thermal internal stresses.

The physical properties of process fluids may lead to temperature differences between streams internally within a heat exchanger that may be higher than the temperature differences measured at the corresponding inlet or outlet nozzles, e.g., if there are streams with phase change. This is illustrated in Figure 5-1.

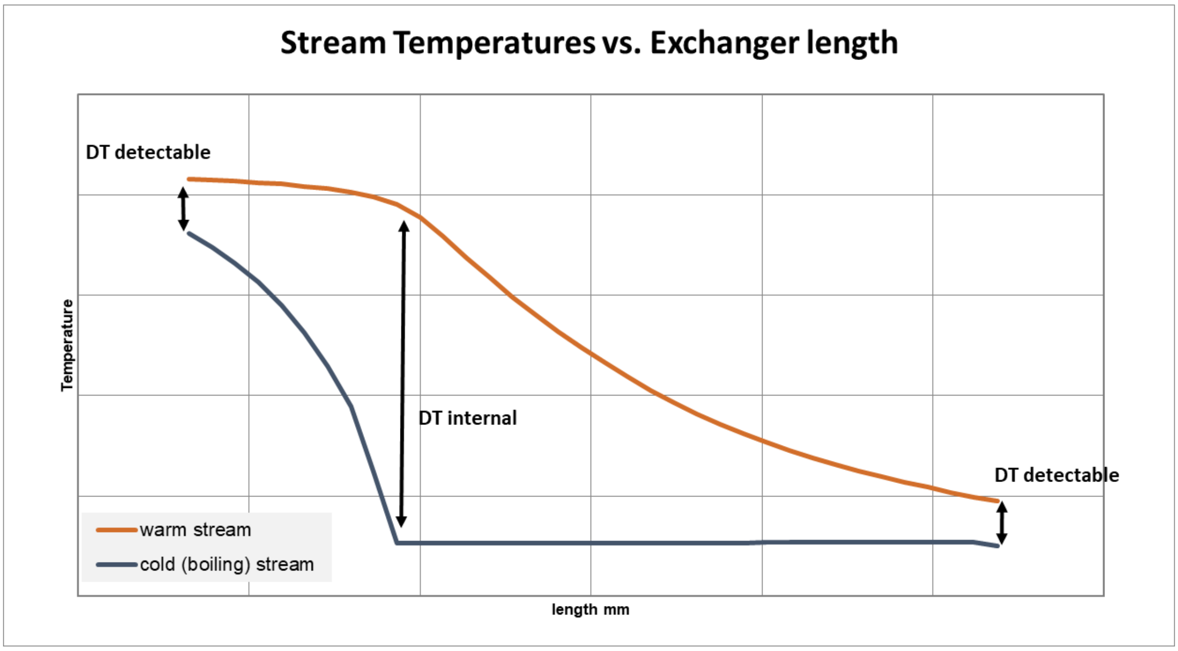


Figure 5-1: Example stream temperature profiles of an exchanger having largest temperature difference internally (boiling / superheating service)

Suitable temperature monitoring devices can be added to monitor this specific temperature difference when requested by the customer. When in doubt, the specific heat release curves provide details.

The thermal stresses developed shall remain within the acceptable limits for the material used. It is generally accepted that, for a typical geometry of a brazed aluminium plate-fin heat exchanger under steady-state conditions, the maximum permissible temperature difference between streams is approximately 50 °C.

However, in more severe cases such as two-phase flows, transient and/or cyclic conditions, this temperature difference should be lower, typically 20 – 30 °C.

For details, reference should be made to Section 8.1, regarding thermal shock/ temperature differences.

There are two methods for reducing the temperature difference:

- either modify the process conditions (modifying the flow rate, installing a bypass line, etc.) to reduce the imposed temperature difference to acceptable limits, or,
- when applicable, design the heat exchanger to reduce the stresses, generally by absorbing the imposed temperature difference in a modified structure.

5.9 Corrosion and Erosion Allowances

Brazed aluminium plate-fin heat exchangers are designed for operation with non-corrosive fluids. No allowance is made in material thickness for corrosion or erosion.

5.10 Service Limitations

The service limitations of brazed aluminium plate-fin heat exchangers are described in Section 1.1.3.

5.11 Typical Range of Sizes

5.11.1 Parting Sheets

Standard parting sheet thicknesses typically vary between 0.8 mm and 2.0 mm and are selected by the Manufacturer mainly according to the design pressures.

Parting sheets are normally clad on both sides with a brazing alloy. However, unclad parting sheets are available, where brazing is performed using brazing foils.

5.11.2 Cap Sheets

Standard cap sheet thicknesses are typically 5 and 6 mm. However, thicknesses from 2 mm to 10 mm are also used for special applications.

5.11.3 Side Bars

Side-bar heights are the same as the fin heights.

Side-bar width is selected by the Manufacturer according to the design pressure and typically varies between 10 mm and 25 mm. Manufacturers use different shapes of side bars for manufacturing reasons.

5.11.4 Fins

Typically, fin height, thickness, and density vary within the following range:

Fin height	2.0 mm to 12 mm
Fin thickness	0.15 mm to 0.7 mm
Fin pitch	1.0 mm to 4.5 mm (25 fpi to 6 fpi)

Not every Manufacturer will use the whole range of these dimensions. In addition, as a result of the restrictions of manufacturing tools, not every combination of fin dimensions can be produced, e.g., large thickness with many fpi may be excluded.

5.12 Headers and Nozzles

5.12.1 Headers

Headers are fabricated from half cylinders with welded end caps. Some typical configurations are shown in Figure 1-4. Headers are usually made either from standard pipe sizes or formed from plates.

At elevated design pressures, the resulting required wall thickness of large diameter headers may exceed a reasonable or producible value. In that case, the use of multiple headers of smaller diameter is common practice.

The use of reinforcing pads around nozzles, common on conventional cylindrical pressure vessels, has limited application to headers as a result of the restricted distance from the block surface to the nozzle.

5.12.2 Nozzles

5.12.2.1 Nozzle Construction

Nozzles are normally welded into the cylindrical part of the headers. Radial nozzles are considered as standard; other installations may be used (see Figure 1-5).

Nozzles are typically selected from commercial seamless standard pipes. Large and special sized nozzles may be made from welded pipes or formed plates.

If the nozzle diameter is to be limited and acceptable flow velocities are to be maintained, several nozzles may be welded into one common header.

5.12.2.2 Flow Velocities in Nozzles

Nozzle sizes are determined primarily on the basis of pressure drop and flow distribution.

5.12.2.3 Nozzle Installation

Nozzles shall be installed to ensure venting and draining of the individual pressure chambers, as far as possible.

Additional vent or drain connections to the header or connecting pipe (min. ¾-in. NPS) may be required.

5.12.2.4 Nozzle Loadings

The associated piping can impose forces and moments on the heat exchanger nozzles. Resultant forces and moments are usually calculated by resolving the forces and moments along and about the three reference axes shown in Figure 5-2.

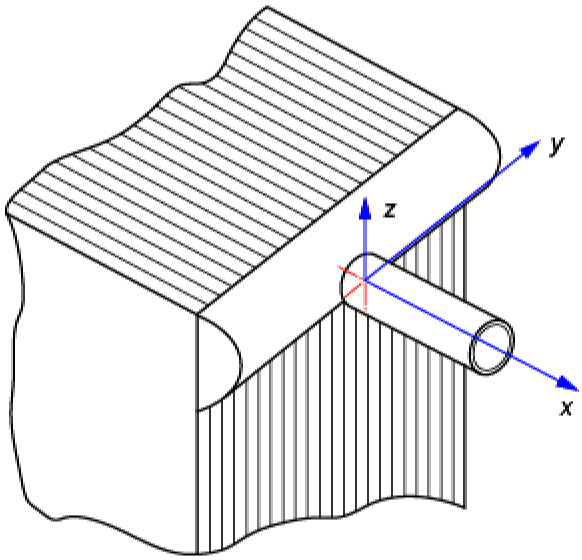


Figure 5-2: Positions of the three reference axes

These resultant forces, F_r, and moments, M_r, are defined by

$$M_r = \sqrt{M_x^2 + M_y^2 + M_z^2} \tag{5.1}$$

$$F_r = \sqrt{F_x^2 + F_y^2 + F_z^2} \tag{5.2}$$

Some typical maximum allowable values of moments ($M_{\max}$) and forces ($F_{\max}$) can be taken from Table 5-1.

Manufacturers shall specify maximum allowable forces and moments. Where simultaneous application of forces and moments occurs, the following rule can be applied as a guideline, where $M_{\max}$ and $F_{\max}$ are those in Table 5-1:

$$\left(M_r / M_{\max}\right) + \left(F_r / F_{\max}\right) \leq 1 \tag{5.3}$$

Table 5-1: Typical resultant forces and moments allowable at nozzle-to-header intersection

Nozzle size, in.	M_{max} , Nm Resultant moment, M_r	F_{max} , N Resultant force, F_r
2	120	810
3	330	1500
4	660	2660
6	1530	3600
8	2160	5540
10	2700	6740
12	3300	9000
14	3900	10800
16	4640	12900
18	5400	15000
20	6000	16500
24	7200	20600

Nozzle loadings at nozzle-to-header intersection must not be confused with allowable loadings at flanged connection.

Note that Welding Research Council (WRC) Bulletins 297 and 537 are not suitable for calculation of the allowable nozzle loadings of brazed aluminium plate-fin heat exchangers as the header/nozzle diameter ratios are generally out of WRC range.

5.12.3 Aluminium Flanged Connections

5.12.3.1 Principles

Weld-neck type flanges with raised-face (RF) sealing surfaces are selected. Other types and facings can be provided. For compatibility with the mating flanges, dimensions, drilling details and facings shall comply with the applicable flange Standards, e.g., ASME B16.5 for NPS $\leq$ DN600 (24 in.), ASME B16.47, series A or B, for NPS > DN600 (24 in.), or DIN/EN, with the possible exception of the flange ring thickness which may have to be increased depending on the service conditions. When non-standard flange ring thicknesses are required, the ring thickness shall be specified to allow procurement of suitable studs.

Bolt holes shall straddle natural centre lines.

The Manufacturer shall provide recommended bolt torque values on request.

5.12.3.2 Design

Aluminium flanges shall be provided in accordance with ASME B31.3 Appendix L to B16.5 geometry or, alternatively, the Manufacturer shall determine the geometry by designing the flanges according to the provisions of the applicable pressure vessel Code, e.g., ASME VIII/1, App. 2, or AD Merkblätter B7/B8. Flange design is dependent on the bolting and gasket. The Purchaser shall specify type and material properties of bolts and gaskets that shall be used in service. If such information is not available, the Manufacturer shall advise of the assumptions made in the flange design (allowable stress, number and diameter of bolts, deformation stress and seating factor of gasket). It is the Purchaser's responsibility to verify that these assumptions meet all operating conditions.

5.12.4 Transition Joints

Bi-metallic transition joints are available from several Suppliers, using different design and manufacturing techniques (e.g., hot pressing, explosion bonding, cold-pressure welding, friction-stir welding). Qualification of transition joints shall be supported by adequate reliability tests such as leak tests, thermal shock tests, etc. Strength design for transition joints shall meet applicable piping Code regulations, e.g., ASME B31.3. Manufacturers shall consider the maximum allowable external loads provided by the Supplier of the transition joints.

Temperature limits have normally to be considered during welding to avoid impacting the bonded aluminium-steel connection. Care should also be taken to stay within these limits during installation.

Temperature limits have also to be considered when the heat exchanger is dried after hydrostatic testing. When a value of temperature limit is lower than the drying temperature, transition joints shall be installed on nozzles after hydrostatic testing and drying are complete.

5.13 Effect of Production Process on Materials

As a result of the manufacturing process of brazing, the core of the heat exchangers will be in the fully annealed condition, referred to as the 'O' temper.

5.14 Arrangement of Layers

In principle there are no limitations or restrictions on permissible layer arrangements.

Usually the ratio of "warm" to "cold" passages varies between 1:1 (single banking) and 1:2 (double banking) or vice versa. Details are described in Section 7.2.2.

5.15 Brazed Aluminium Plate-Fin Heat Exchanger as a Pressure Vessel

Brazed aluminium plate-fin heat exchangers are vessels, for which the mechanical design procedure is governed by a pressure vessel Code. However, as a result of unique construction features, certain items need specific attention.

5.15.1 Fins

As well as their thermal performance, fins have to be selected by the Manufacturer for their ability to resist the pressure loadings as structural members.

These loads essentially produce tensile stresses in the fins. The maximum allowable design pressure for the individual fins shall be determined either by calculation or by burst test methods. The same fin corrugation may be acceptable up to different design pressures, depending on the requirements of the applicable Codes.

5.15.1.1 Calculation Procedure

Pressure vessel Codes generally do not contain formulae for the fins in brazed aluminium plate-fin heat exchangers.

The calculation methods used by Manufacturers have been approved by the applicable Code Authority. The stresses thus calculated are compared with the maximum allowable stress of the Code.

5.15.1.2 Burst Test Method

A representative brazed sample is pressurised up to bursting. By applying a safety factor to the measured burst pressure, the allowable maximum design pressure for the particular fin type is derived depending on the applicable Code, including adjustments for tolerances in material properties.

5.15.2 Parting Sheets

The parting sheet thickness shall be sufficient to resist the tensile stresses from the pressure forces acting on the side bars. These stresses depend on pressure loadings on the adjacent sides of the parting sheet and on the height of the adjacent side bars.

5.15.3 Side Bars

The side-bar height is the same as the fin height. Side-bar width is chosen to take account of pressure loading from the header attachment and allow a reasonable mass for welding.

5.15.4 Cap Sheets

Due to their thickness, cap sheets are lightly stressed structural members of the heat exchanger. Their main task is to protect the core against physical damage and provide a base to allow welding of supports and other attachments.

5.15.5 Headers and Nozzles

It is common practice to determine wall thicknesses and reinforcement of openings from the requirements of the applicable Code.

Manufacturers give consideration to the weld efficiency factors for the attachment of headers to the core, as required by the applicable Code.

5.16 Special Features

The Purchaser should indicate within the specification of the equipment any special conditions which may include vibration, seismic loading, thermal cycling, pressure cycling or vacuum or external pressure conditions or layers where there shall be a “no flow” condition during some operating modes. The Manufacturer shall then take into account these effects in his design.

6 MATERIALS

6.1 Typical Materials of Construction

The heat exchangers covered by these Standards are constructed from aluminium alloys. Materials are selected for their braze-ability, weld-ability, and other characteristics. Typical materials used for construction and their maximum applicable design temperatures, are shown in Table 6-1 and Table 6-2. Figure 1-2 details the components.

Table 6-1: Typical materials used in the construction of brazed aluminium plate-fin heat exchangers and their maximum applicable design temperature (Celsius)

CODES	ASME		European Standard (EN)		Japanese Industrial Standard (JIS)	
COMPONENTS	**Alloy No.**	**Max. Applicable Design Temperature [*1]**	**Alloy No.**	**Max. Applicable Design Temperature [*2]**	**Alloy No.**	**Max. Applicable Design Temperature [*3]**
Heat Transfer Fin	SB-209 3003 3004	 204 °C 204 °C	 EN-AW-3003	 150 °C[*5]	H4000 A3003P A3004P	 200 °C 200 °C
Distributor Fin	SB-209 3003 3004	 204 °C 204 °C	 EN-AW-3003	 150 °C[*5]	H4000 A3003P A3004P	 200 °C 200 °C
Side Bar	SB-221 3003	 204 °C	 EN-AW-3003	 150 °C[*5]	H4100 A3003S	 200 °C
Centre Bar	SB-221 3003	 204 °C	 EN-AW-3003	 150 °C[*5]	H4100 A3003S	 200 °C
Parting Sheet [*4]	SB-209 3003	 204 °C	 EN-AW-3003	 150 °C[*5]	H4000 A3003P	 200 °C
Cap Sheet	SB-209 3003	 204 °C	 EN-AW-3003	 150 °C[*5]	H4000 A3003P	 200 °C
Header	SB-209, 221 & 241 3003 5052 5083 5454 6061	 204 °C 204 °C 65 °C 204 °C 204 °C	 EN-AW-5754 EN-AW-5083 EN-AW-5454	 100 °C[*6] 100 °C[*6,7] 100 °C[*6]	H4000 A3003P A5052P A5083P A5454P A6061P	 200 °C 200 °C 65 °C 200 °C 200 °C
Nozzle	SB-209, 221 & 241 3003 5052 5083 5086 5454 6061 SB-221 & SB-241 6063	 204 °C 204 °C 65 °C 65 °C 204 °C 204 °C 204 °C	 EN-AW-5754 EN-AW-5083 EN-AW-5454	 100 °C[*6] 100 °C[*6,7] 100 °C[*6]	H4080 & H4000 A3003TID&TE A3003P A5052TID&TE A5052P A5083TID&TE A5083P — A5086P A5454TE A5454P A6061 TD&TE A6061P A6063TID&TE	 200 °C 200 °C 65 °C 65 °C 200 °C 200 °C 200 °C
Flange	SB-247 5083 6061	 65 °C 204 °C	 EN-AW-5083 EN-AW-5754	 100 °C[*6,7] 100 °C[*6]	H4140 A5083FD A6061FD	 65 °C 200 °C
Support	SB-209 & 221 5052 5083 6061 6063 —	 204 °C 65 °C 204 °C 204 °C —	 EN-AW-5754 EN-AW-5083 EN-AC-43000	 100 °C[*6] 100 °C[*6,7] N/A	H4000 A5052P A5083P A6061P	 200 °C 65 °C 200 °C

Remarks
*1 : Maximum applicable temperature is as per ASME Sec. VIII, Div.1, where the official unit is British (degree F).
*2 : Maximum applicable temperature is as per EN-13445, where the official unit is Metric (degree C).
*3 : Maximum applicable temperature is as per Japanese High Pressure Gas Safety Law, where the official unit is Metric (degree C).
*4 : They may be typically clad with brazing material.
*5 : Valid for wall thickness < 50 mm. For temperatures > 150 °C the material would be in the creep range.
*6 : For temperatures > 100 °C the material would be in the creep range.
*7 : For temperatures > 80 °C only non-corrosive service is allowed.

Table 6-2: Typical materials used in the construction of brazed aluminium plate-fin heat exchangers and their maximum applicable design temperature (Fahrenheit)

CODES	ASME		European Standard (EN)		Japanese Industrial Standard (JIS)	
COMPONENTS	**Alloy No.**	**Max. Applicable Design Temperature [*1]**	**Alloy No.**	**Max. Applicable Design Temperature [*2]**	**Alloy No.**	**Max. Applicable Design Temperature [*3]**
Heat Transfer Fin	SB-209				H4000	
	3003	400 °F	EN-AW-3003	302° F[*5]	A3003P	392 °F
	3004	400 °F			A3004P	392 °F
Distributor Fin	SB-209				H4000	
	3003	400 °F	EN-AW-3003	302° F[*5]	A3003P	392 °F
	3004	400 °F			A3004P	392 °F
Side Bar	SB-221				H4100	
	3003	400 °F	EN-AW-3003	302° F[*5]	A3003S	392 °F
Centre Bar	SB-221				H4100	
	3003	400 °F	EN-AW-3003	302° F[*5]	A3003S	392 °F
Parting Sheet [*4]	SB-209				H4000	
	3003	400 °F	EN-AW-3003	302° F[*5]	A3003P	392 °F
Cap Sheet	SB-209				H4000	
	3003	400 °F	EN-AW-3003	302° F[*5]	A3003P	392 °F
Header	SB-209, 221 & 241				H4000	
	3003	400 °F			A3003P	392 °F
	5052	400 °F	EN-AW-5754	212 °F[*6]	A5052P	392 °F
	5083	150 °F	EN-AW-5083	212 °F[*6,7]	A5083P	150 °F
	5454	400 °F	EN-AW-5454	212 °F[*6]	A5454T	392 °F
	6061	400 °F			A6061P	392 °F
Nozzle	SB-209, 221 & 241				H4080 & H4000	
	3003	400 °F			A3003TID&TE A3003P	392 °F
	5052	400 °F	EN-AW-5754	212 °F[*6]	A5052TID&TE A5052P	392 °F
	5083	150 °F	EN-AW-5083	212 °F[*6,7]	A5083TID&TE A5083P	150 °F
	5086	150 °F			— A5086P	150 °F
	5454	400 °F	EN-AW-5454	212 °F[*6]	A5454TE A5454P	392 °F
	6061	400 °F			A6061 TD&TE A6061P	392 °F
	SB-221 & SB-241				A6063TD&TE	392 °F
	6063	400 °F				
Flange	SB-247				H4140	
	5083	150 °F	EN-AW-5083	212 °F[*6,7]	A5083FD	150 °F
	6061	400 °F			A6061FD	392 °F
			EN-AW-5754	212 °F[*6]		
Support	SB-209 & 221				H4000	
	5052	400 °F	EN-AW-5754	212 °F[*6]	A5052P	392 °F
	5083	150 °F	EN-AW-5083	212 °F[*6,7]	A5083P	150 °F
	6061	400 °F			A6061P	392 °F
	6063	400 °F				
	—	—	EN-AC-43000	N/A		

Remarks *1 : Maximum applicable temperature is as per ASME Sec. VIII, Div.1, where the official unit is British (degree F).
*2 : Maximum applicable temperature is as per EN-13445, where the official unit is Metric (degree C).
*3 : Maximum applicable temperature is as per Japanese High Pressure Gas Safety Law, where the official unit is Metric (degree C).
*4 : They may be typically clad with brazing material.
*5 : Valid for wall thickness < 1.97 in. For temperatures > 302 °F the material would be in the creep range.
*6 : For temperatures > 212 °F the material would be in the creep range.
*7 : For temperatures > 176 °F only non-corrosive service in allowed.

7 THERMAL AND HYDRAULIC DESIGN

7.1 Introduction

The brazed aluminium plate-fin heat exchanger has special features and advantages which make it quite different from other types of heat exchangers:

1. A very large heat transfer surface area can be made available per unit volume of heat exchanger. This surface area is composed of primary and secondary (finned) surfaces. Even taking into account the fin efficiency of the secondary surface, the effective surface area per unit volume can be typically five times greater than that of a shell-and-tube heat exchanger.
2. A range of fin types is available. The fin type is selected to suit the characteristics of a stream. For example, serrated, wavy and perforated fins (Figure 1-6) are particularly suitable for gas streams.
3. One heat exchanger can incorporate several streams and heat can be exchanged simultaneously amongst several streams in a multi-stream heat exchanger. Suitable headers and distributors also permit streams to enter and leave the heat exchanger at intermediate points along its length as well as at the ends.

7.2 Features of a Brazed Aluminium Plate-Fin Heat Exchanger

7.2.1 Primary and Secondary Heat Transfer Surfaces and Thermal Length

Heat is transferred from or into a stream within a finned passage. The primary heat transfer surface within the heat exchanger consists of the bare parting sheet and the fin base directly brazed to the parting sheet (Figure 7-1).

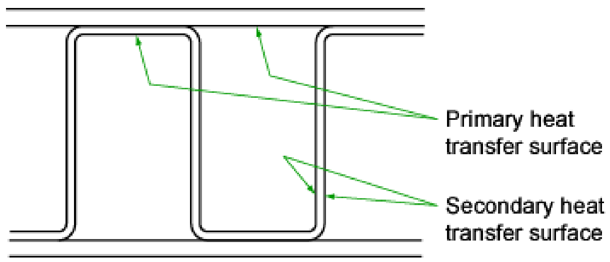


Figure 7-1: Cross-sectional view of fin and parting sheet

The secondary heat transfer surface is provided by the fins. The effectiveness of the secondary surface to transfer heat is given by the fin efficiency.

Per unit area of each layer,

- the primary surface is given by $2\ (1 - nt)$
- the secondary surface is given by $2n\ (h - t)$

where

n	=	Fin density, i.e., number of fins per unit length	m^{-1}
t	=	Fin thickness	m
h	=	Fin height	m

The thermal length of a single pass of a brazed aluminium plate-fin heat exchanger is typically defined as the effective length of the finned region between, but not including, the distributors.

7.2.2 Single and Multiple Banking

There are two types of layer arrangement for a stream: single banking and multiple banking (typically double banking, as in Figure 7-2).

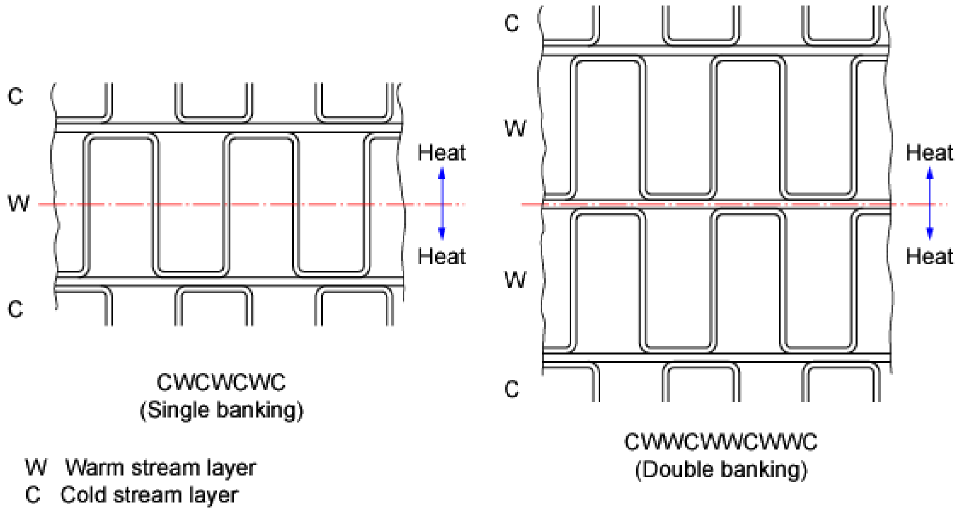


Figure 7-2: Single and double banking

Single banking is the simplest arrangement where each warm stream layer (W) is adjacent to a cold stream layer (C). The thermal efficiency of this fin arrangement is given in Section 7.4.5.

Double banking is also illustrated in Figure 7-2. Here, two layers of equal height are provided for a warm stream with a large flow rate within the allowable pressure drop. More than two layers can also be used. The thermal efficiency of double banking is also given in Section 7.4.5.

In double banking, the parting sheet between the two layers becomes a secondary surface and the fin efficiency is reduced because of the increased length of the heat path along the fins.

7.2.3 Multi-stream Brazed Aluminium Plate-fin Heat Exchangers

The brazed aluminium plate-fin heat exchanger is capable of accommodating many streams within its structure and heat can be exchanged among several streams simultaneously.

A multi-stream brazed aluminium plate-fin heat exchanger, with streams also entering and leaving at intermediate positions between its ends, can accommodate over twelve different streams. The selection and design of the layer arrangement, the fins of the layer and the effective length of each stream is of crucial importance.

7.3 Thermal Design Procedure

The design procedure for a brazed aluminium plate-fin heat exchanger is different, in many respects, from a traditional two-stream exchanger such as a shell-and-tube exchanger. The main differences are:

1. In most cases, more than two streams can be handled.
2. The secondary surface area provided by the fins is a large portion of the total heat transfer area.
3. There are a variety of fin types available for giving the best heat transfer and pressure drop characteristics for each individual stream (see Section 7.6.1).
4. An efficient sequence of layers is required where each layer is the flow channel for a given stream with the appropriate choice of fin. This is known as the layer stacking arrangement and is discussed in Section 7.6.3.

5. There is a strong inter-relationship between the mechanical and thermal-hydraulic design because key elements in the thermal design, such as fin density, height, and thickness, are governed by the mechanical design.
6. Optimising a design involves working with a large number of variables, and this is best handled using a specially developed software combined with expert knowledge from an experienced designer.
7. The designer requires much more information to cover the many streams and the greater detail often required for each stream. Figure 7-3 and Figure 7-4 give examples of specification sheets which allow for this extra information.

The calculation method given in Section 7.4 is a simple one that effectively converts a multi-stream heat transfer process into a two-stream one. The first step is to generate the temperature-enthalpy plot (T - Q curve) for all the cold streams and all the warm streams. Plotting these curves on the same chart is very revealing in showing where close temperature approaches (temperature pinches) arise, which require special care in design. An overall heat transfer coefficient is also calculated which combines the individual heat transfer coefficients for all the streams.

It is stressed that this calculation method is an approximation which can provide good solutions for simpler heat transfer processes. More rigorous calculation methods are available, which take into account the detailed variations from stream to stream, including the temperature differences between individual parting sheets. An experienced designer should therefore be consulted at an early stage in detailed design.

7.4 Thermal Relations

7.4.1 Basic Heat Transfer Relation

The required surface area of a brazed aluminium plate-fin heat exchanger can be obtained from

$$UA_r = \frac{Q}{MTD} \qquad (7.1)$$

where

U	=	Overall heat transfer coefficient between streams	W/m^2 K
A_r	=	Required overall effective heat transfer surface	m^2
Q	=	Heat to be transferred	W
MTD	=	Mean temperature difference between composite or combined streams	K

1	**BRAZED ALUMINIUM PLATE-FIN HEAT EXCHANGER SPECIFICATION**									
2	Customer				Project			Location		
3	Item number				Service			Date		Revision
4	Stream i.d. / fluid name			Unit	A/	B/	C/	D/	E/	F/
5	Flow rate	Total		kg/s						
6		Vap./Liq.	In	kg/s	/	/	/	/	/	/
7		Vap./Liq.	Out	kg/s	/	/	/	/	/	/
8	Molecular weight	Vap.	In/Out	-	/	/	/	/	/	/
9		Liq.	In/Out	-	/	/	/	/	/	/
10	Density	Vap.	In/Out	kg/m^3	/	/	/	/	/	/
11		Liq.	In/Out	kg/m^3	/	/	/	/	/	/
12	Viscosity	Vap.	In/Out	cP	/	/	/	/	/	/
13		Liq.	In/Out	cP	/	/	/	/	/	/
14	Specific heat	Vap.	In/Out	J/kg K	/	/	/	/	/	/
15		Liq.	In/Out	J/kg K	/	/	/	/	/	/
16	Thermal conductivity	Vap.	In/Out	W/m K	/	/	/	/	/	/
17		Liq.	In/Out	W/m K	/	/	/	/	/	/
18	Temperature	In/Out		K	/	/	/	/	/	/
19	Operating pressure	In		kPa						
20	Allowable frictional pressure drop			kPa						
21	Heat load			MW						
22	Corrected MTD			K						
23	Fouling resistance			m^2 K/W						
24	Design pressure / test pressure			kPa	/	/	/	/	/	/
25	Design temperatures max / min.			kPa	/					
26	Number of cores and assemblies			-	In parallel	In series	Number of cores/assembly		Number of assemblies	
27	Core size			mm	Width	Height	Length			
28	Flow pattern			-	Counter	Cross-counter		Cross	Parallel	
30	Approx. weights			kg	Core empty	Core operating		Assembly empty	Assembly operating	
31	Number of layers			-						
32	Fin heat transfer surface/core			m^2						
33	Fin free flow cross sectional area			m^2						
34	Core opening size		In/Out	mm	/	/	/	/	/	/
35	Nozzle number × size		In/Out	mm	/	/	/	/	/	/
36	Manifold pipe size		In/Out	mm	/	/	/	/	/	/
37	Calculated frictional pressure drop			kPa						
38	Code and/or regulation:									
39	Notes									
40										
41										
42										

User supplied data

Manufacturer supplied data

Data sheet (rev.3/2023) supplied by ALPEMA

www.alpema.org/alpemaspecsheet.zip

Figure 7-3: Typical specification sheet

1	**BRAZED ALUMINIUM PLATE-FIN HEAT EXCHANGER SPECIFICATION**														
2	Customer				Project					Location					
3	Item number				Service					Date				Revision	
4	Stream i.d.		Fluid name							Reference pressure				kPa	
5	Composition	Component													
6		Mole (%)													
7	Stream physical property data at Reference Pressure	Temp.	Heat load or specific enthalpy	Mass quality	Vapour Physical Properties					Liquid Physical Properties					
					MW	Density	Heat capacity	Viscosity	Thermal cond.	MW	Density	Heat capacity	Viscosity	Thermal cond.	Surface tension
8		K	MW	-	-	kg/m^3	kJ/kg K	cP	W/m K	-	kg/m^3	kJ/kg K	cP	W/m K	N/m
9	1														
10	2														
11	3														
12	4														
13	5														
14	6														
15	7														
16	8														
17	9														
18	10														
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30	22														
37	23														
38	24														
39	Notes:														
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41															
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43															
44															

All data on this sheet supplied by User

Data sheet (rev.3/2023) supplied by ALPEMA
www.alpema.org/alpemaspecsheet.zip

Figure 7-4: Typical stream specification sheet (one per pressure level for each stream)

7.4.2 MTD and UA_r

The MTD can be obtained by calculating the logarithmic mean temperature difference ($LMTD$) in each section where both warm and cold composite T - Q curves are linear. An example composite curve is shown in Figure 7-5.

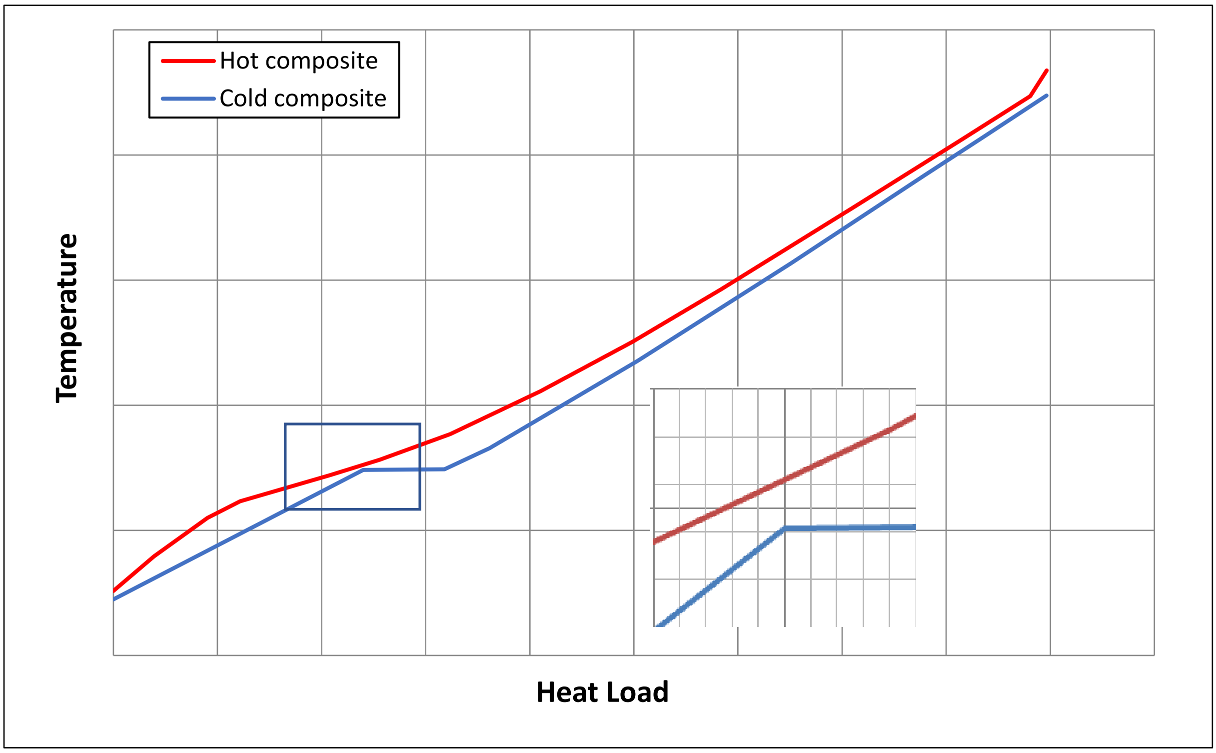


Figure 7-5: Example composite curve

Equation (5.1) becomes

$$UA_r = \sum \frac{Q_i}{LMTD_i} \tag{7.2}$$

where

$$LMTD_i = \frac{\Delta T_{i+1} - \Delta T_i}{\ln\left(\Delta T_{i+1} / \Delta T_i\right)} \tag{7.3}$$

while ΔT_i and ΔT_{i+1} are the temperature differences between warm and cold streams at each end of Section i (K).

This $LMTD$ can be used for counter-flow or parallel-flow.

For cross-flow and cross-counter-flow, however, the $LMTD$ shall be corrected. Details are given in Reference (1).

For a multi-stream brazed aluminium plate-fin heat exchanger, the MTD shall be obtained from the two composite temperature-enthalpy curves for the combined warm and combined cold streams. Further information can be found in References (1) to (3).

7.4.3 Overall Effective Heat Transfer Surface of Exchanger

The overall effective heat transfer surface can be estimated from Equation (5.4). The thermal resistance of the parting sheet between the two streams can usually be ignored primarily because it is made from thin aluminium sheet.

$$\frac{1}{UA_d} = \frac{1}{\sum(\alpha_0 A)_{wi}} + \frac{1}{\sum(\alpha_0 A)_{ci}} \tag{7.4}$$

where

α_0	=	Effective heat transfer coefficient of a stream	W/m² K
A	=	Effective heat transfer surface of a passage or layers of a stream	m²
A_d	=	Designed (or estimated) overall effective heat transfer surface	m²
wi, ci	=	Warm or cold stream *i* (subscripts)	K

7.4.4 Effective Heat Transfer Coefficient of Each Stream

The heat transfer coefficient of each stream can be estimated from Equation (5.5).

$$\alpha = \frac{jG_m C_p}{\mathrm{Pr}^{2/3}} \tag{7.5}$$

where

α	=	Heat transfer coefficient of a stream	W/m² K
j	=	Colburn factor for a finned passage	—
G_m	=	Mass flux of a stream	kg/m² s
C_p	=	Specific heat capacity of a stream at constant pressure	J/kg K
Pr	=	Prandtl number of a stream ($C_p\,\mu/k$)	—
k	=	Thermal conductivity of a stream	W/m K
μ	=	Dynamic viscosity of a stream	N s/m²

The effective heat transfer coefficient of each stream, α_0, can be estimated from Equation (5.6) which takes the fouling resistance into account.

$$\frac{1}{\alpha_0} = \frac{1}{\alpha} + r \tag{7.6}$$

where

r	=	Fouling resistance of a stream	m² K/W

Equation (5.5) can be used for single-phase streams, i.e., all vapour or all liquid flow. For two-phase condensing or vaporizing flows, various equations are available for predicting the two-phase heat transfer coefficient; given for example in Reference (3). Suppliers, however, can use calculation methods based on experience with two-phase streams.

The Colburn factor, j, is highly dependent on the type of fin, its nominal geometry, and details of manufacture, as well as the Reynolds number of the stream. Information about the Colburn factor j can also be obtained from Reference (2).

Heat transfer coefficients of each stream shall be calculated locally where the thermodynamic and/or physical properties of the stream change rapidly, for example, at a phase change or in the super-critical state. For these conditions, a step-by-step calculation along the stream shall be necessary.

7.4.5 Heat Transfer Surface of Each Passage

The effective heat transfer surface area for a passage, A, is obtained from the following expression:

$$A = A_1 + \eta_1 \phi A_2 \tag{7.7}$$

where η_1 can be estimated from the following expression for single banking:

$$\eta_1 = \frac{\tanh(\beta/2)}{\beta/2} \tag{7.8}$$

Instead for double banking, A is obtained from the following expressions:

$$A = \frac{1}{2}A_1 + \eta_2\left(\frac{1}{2}A_1 + \phi A_2\right) \tag{7.9}$$

$$\eta_2 = \left(\frac{1}{\beta+\gamma}\right)\left(\frac{B-1}{B+1}\right) \tag{7.10}$$

where

$$\beta = h\left(\frac{2\alpha_o}{\lambda_m t}\right)^{0.5} \tag{7.11}$$

$$\gamma = \frac{\beta}{2h}\left(\frac{1}{n} - t\right) \tag{7.12}$$

$$B = \left(\frac{1+\gamma}{1-\gamma}\right)e^{2\beta} \tag{7.13}$$

A_1	=	Primary heat transfer surface of a stream (Figure 7-1)	m^2
A_2	=	Secondary heat transfer surface of a stream (Figure 7-1)	m^2
η_1	=	Passage fin efficiency for single banking	—
η_2	=	Passage fin efficiency for doublebanking	—
h	=	Passage fin height	m
t	=	Passage fin thickness	m
n	=	Passage fin density	m^{-1}
α_o	=	Effective heat transfer coefficient of a stream	W/m^2 K
ϕ	=	Unperforated fraction [1 – (Percentage perforation)/100]	—
λ_m	=	Thermal conductivity of fin material (aluminium)	W/m K

7.4.6 Rough Estimation of the Core Volume and Weight

To obtain a quick indication of the heat exchanger volume required for a certain duty, the following simple relation may be used:

$$V = \frac{Q / MTD}{C} \tag{7.14}$$

where

V	=	Required volume of heat exchanger or heat exchangers (without headers)	m^3
Q	=	Overall heat duty	W
MTD	=	Mean temperature difference between streams	K
C	=	Coefficient: 100,000 for hydrocarbon application 50,000 for air separation application	W/m^3 K

The values of 100,000 and 50,000 represent the product, UA_d, assuming an overall heat transfer coefficient of 200 W/m^2 K and 100 W/m^2 K, respectively, and a mean geometric heat transfer surface density of 500 m^2/m^3.

The weight of a complete heat exchanger may be assumed to be 1000 kg per unit core volume (m^3). This value varies in practice between 650 and 1500 kg/m^3.

7.5 Hydraulic Relations

The Purchaser usually specifies the allowable pressure loss for each stream, within the Manufacturer's scope of supply.

In the hydraulic design of the heat exchanger, the fin type and passages are chosen to meet this pressure loss requirement. In order to ensure uniform flow distribution of a stream among its passages, the components of the pressure drop are evaluated. Uniform distribution of a stream over the width of a layer is provided by good design of the distributors.

7.5.1 Components of Pressure Loss

The individual pressure losses within a heat exchanger typically consist of (See Figure 7-6):

1. Expansion loss into the inlet header
2. Contraction loss at the entry to the core
3. Loss across the inlet distributor
4. Loss across the heat transfer length
5. Loss across the outlet distributor
6. Expansion loss into the outlet header
7. Contraction loss into the outlet nozzle
8. Gravitational loss (or gain)

Additional pressure losses in piping and/or manifolding outside the Manufacturer's scope of supply are to be accounted for by the Purchaser. The Purchaser usually specifies the allowable nozzle to nozzle pressure drop.

General methods for predicting these pressure losses are given in Reference (3). Manufacturers make use of their experience to select the most appropriate method of estimating the losses given in Items 1 to 7 above.

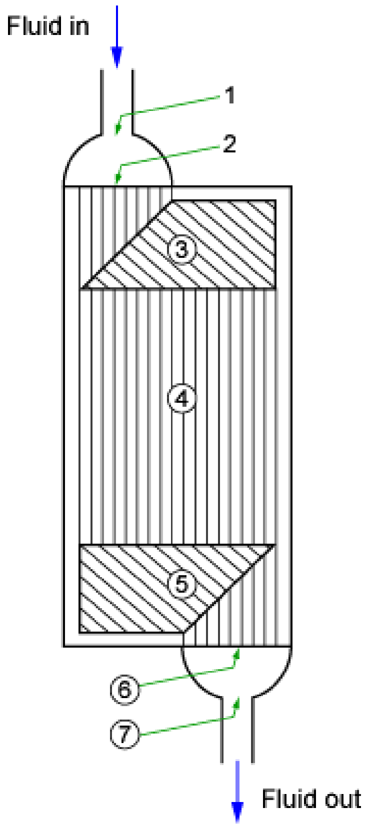


Figure 7-6: Pressure loss components

7.5.2 Single-Phase Pressure Loss

The frictional pressure loss across a plate-fin passage and at any associated entry, exit and turning losses, can be expressed by

$$\Delta P = 4\, f \left(\frac{l_p}{d_h}\right)\left(\frac{G_m{}^2}{2\rho}\right) + K\left(\frac{G_m{}^2}{2\rho}\right) \tag{7.15}$$

where

f	=	Fanning friction factor	—
l_p	=	Passage length	m
d_h	=	Hydraulic diameter of passage	m
G_m	=	Mass velocity (mass flux) of stream	kg/m^2 s
ρ	=	Density of a stream	kg/m^3
K	=	Expansion, contraction, or turning loss coefficient	—
ΔP	=	Overall pressure drop	Pa

Note: By convention, the upstream mass flux is used for estimating expansion losses and the downstream for contractions.

7.5.3 Two-Phase Pressure Loss

In brazed aluminium plate-fin heat exchangers with two-phase streams where fluid quality and physical properties are changing, it is necessary to divide the heat exchanger into suitable increments of length in order to assess the overall pressure gradient simultaneously with the thermal design calculations.

The pressure gradient in a two-phase flow can be divided into three components:

- Frictional component
- Static head component
- Accelerational component

Each Manufacturer uses suitable design correlations for estimating these components from experience. General estimating methods are given in Reference (3).

7.6 General Considerations in the Thermal and Hydraulic Design

7.6.1 Design Margin

There are no standard margins for heat transfer (or thermal duty), surface area and pressure drop. Overdesign factors are stream and/or application dependent and may vary from 0 and 10%, unless other project specific values are specified. Because clean fluids are processed, fouling factors are not used unless specified by the customer. See Section 8.2.1.

Distributor areas are normally not included in calculation of the heat transfer surface area.

7.6.2 Choice of Fin Geometry

Each fin shall conduct the required amount of heat and also withstand the design pressure at the design temperature as a structural component.

Fin geometry is therefore selected to meet both requirements. Details of the required structural performance are given in Chapter 5. Details of the fin's required thermal performance are given earlier in Section 7.4.4.

The choice of fin will also influence the most economical design of an exchanger for a specific application. Table 7-1 provides general information on common applications for each type of fin (see Figure 1-6).

Table 7-1: Common applications for each type of fin

Corrugation	Description	Application	Features	
			Relative pressure drop	**Relative heat transfer**
Plain	Straight	For general use	Lowest	Lowest
Perforated	Straight with small holes	Most frequently used for any purpose. Sometimes used for the "hardway" finning	Low	Low
Serrated	Straight, offset half a pitch - usually about every 3-4 mm	Frequently used, especially for low pressure gas streams in air separation plants	Highest	Highest
Herringbone or long-lanced serrated	Smooth but in waves of about 10 mm pitch (can be perforated), or serrated with long serration pitch	Often used for gas streams with low allowable pressure drop	High	High

7.6.3 Layer Stacking Arrangement

With multi-stream heat exchangers, the choice of the stacking sequence or layer-pattern shall take into account the local heat balance among streams and any local non-linearity of the Enthalpy-Temperature Curves of each stream. A thermally well-balanced stacking arrangement would result in a nearly uniform metal temperature at any cross section of the heat exchanger, thus allowing the detailed design to proceed with the assumption of a common wall temperature.

The deviations from a uniform metal temperature can be evaluated by using a more detailed (layer-by-layer) analysis, taking into account heat being transferred by metal conduction between non-adjacent layers.

7.6.4 Two-Phase Distribution

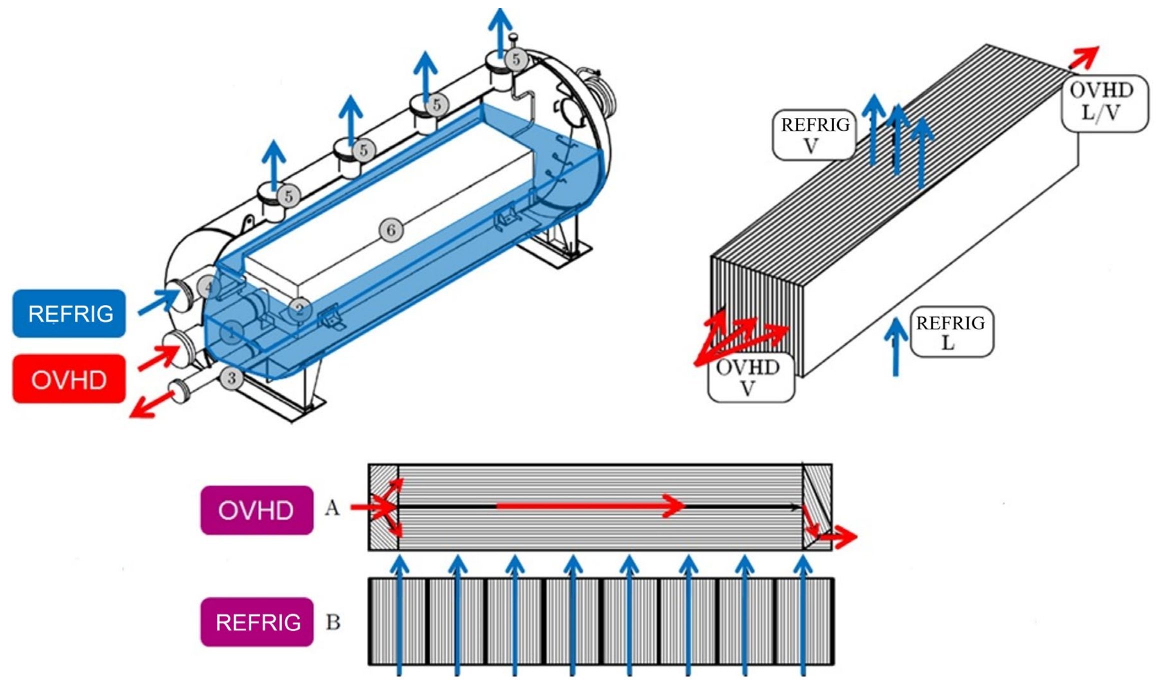
When a stream at the exchanger inlet is made up of a mixture of vapour and liquid, it can be necessary to separate the two phases first and then to distribute them in separate distribution systems that mix the two phases in a controlled way within the exchanger. Re-mixing of the two phases can be done in the header or in the passages depending on the distribution system being used. Details of typical systems are shown in Section 1.2.8.

7.6.5 Thermosyphons

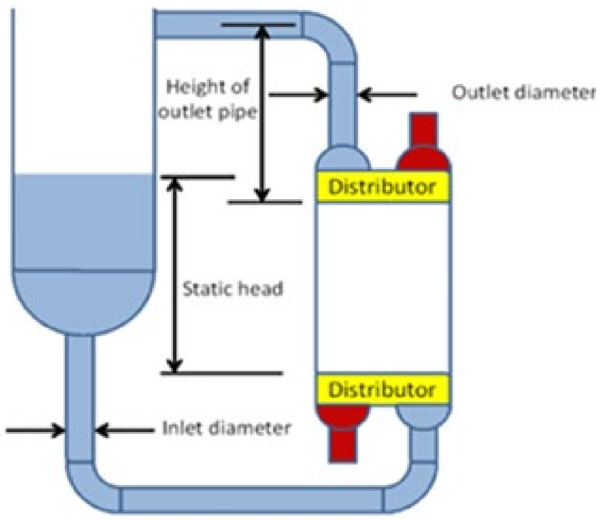
There are two main types of thermosyphons: internal and external.

One example of an internal thermosyphon is block-in-shell (Figure 7-7a). Internal thermosyphons are also installed in distillation columns. The cold side refrigerant is typically an isothermal boiling fluid. Operations must adhere to the Manufacturer's recommended liquid levels.

External thermosyphon reboilers are widely used in cryogenic gas plants, for example, and sometimes refer to their position towards the column to which they are linked, i.e., side reboiler, bottom reboiler. The exchanger, the liquid downcomer pipe, and the liquid-vapor return pipe are referred to as the thermosyphon loop. In this case, it is considered as external thermosyphons. It is similar when linked to a thermosyphon column or drum. See Figure 7-7b for example.



a. Internal thermosyphon



b. External thermosiphon with recirculation

Figure 7-7: Example internal and external thermosyphons

External thermosyphon reboiler piping design is critical to proper operation and is outside the scope of brazed aluminium heat exchanger Manufacturers. Some key attributes are that the heat exchanger should be as close to the column or drum as possible with as simple a piping arrangement as possible, the downcomer pipe should be self-venting, a control valve should be installed on the downcomer piping just before the brazed aluminium plate-fin heat exchanger, and a differential pressure gauge should be installed across the boiling pass to detect unstable flow.

7.6.6 Manifold Assemblies

For multiple core assemblies, the so-called U-piping configuration for a stream is preferred to Z-piping for improved flow distribution. Figure 7-8 illustrates this point.

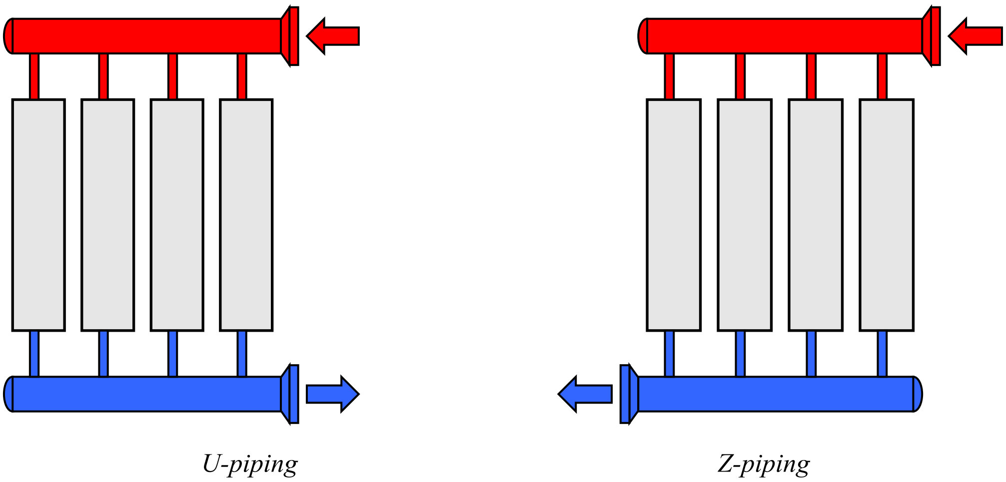


Figure 7-8: Manifold assemblies

It is recommended, at least for the main streams, to locate the inlet and outlet connections on the same face of the unit for improved thermal balance.

8 RECOMMENDED GOOD PRACTICE

8.1 Thermal and Mechanical Stresses Within Brazed Aluminium Plate-Fin Heat Exchangers

8.1.1 Introduction

As with any pressurised heat exchanger, stresses in each component of a brazed aluminium plate-fin heat exchanger shall be maintained within allowable code limits. Pressure loads, externally applied loads (e.g., piping forces and moments), and thermally induced loads produce stresses which shall be maintained within permissible limits to prevent component damage or failure.

Manufacturers design each brazed aluminium plate-fin heat exchanger for the intended design pressure loads; Users are provided with details of allowable external loads that may be exerted on the exchanger. A margin above the stresses created by these loads is made available by the Manufacturer for thermally induced loads which may occur in service. In this section, the mechanism by which thermal stresses are induced is explained.

Recommendations are given for the measures to be taken in the operation of brazed aluminium plate-fin heat exchangers so that the overall combined stresses remain within allowable limits during standard and non-standard operating conditions.

Fatigue induced by changing conditions is discussed in Section 8.1.2.

8.1.2 Failure Mechanism

The components of a brazed aluminium plate-fin heat exchanger are relatively close and rigidly connected in all directions to each other. As a result, conditions which generate large local metal temperature differences in and between the components of its structure will cause significant thermal stress in these components.

Local metal temperature differences result from the components, or portions of the components, warming or cooling at different rates in response to a thermal input (change). These differences produce a transient differential expansion or contraction within or between the components; mechanical restraint to these thermally induced structural movements results in thermal stress in the components. If the local metal temperature differences are large, the combined thermally induced stresses and other stresses from imposed loads can exceed the yield stress and possibly the ultimate stress of the material.

Temperature differences between adjacent parts of a heat exchanger, having the potential to produce significant thermal stresses, can arise from:

1. Continuously unsteady operating conditions: for example, large flow fluctuations; unstable flow in boiling channels; inadequate plant control systems. For vaporizing passages, a differential pressure measurement across the boiling pass has good sensitivity to detect unstable flow.
2. Transient operating conditions: for example, start-up; shut-down; plant upsets; deriming; cool-down and warm-up; etc.
3. Steady-state stream temperature differences (see Section 5.8), causing high wall temperature gradients and/or large temperature differences between the brazed structure and the attached header. Thermal stress may arise even if the transient into such condition is slow.
4. Plugging or other causes for flow maldistribution.

An example of the creation of thermal stress is illustrated by the quick opening of a valve. If this action allows a significant quantity of cold fluid with a high thermal capacity to enter a warm heat exchanger, then those parts of the heat exchanger which can lose heat rapidly will contract quickest. The fins in the region of the inlet port would thus contract more quickly than the side bars on either side of the port; tensile thermal stress would be created within the fins and compressive stress in the side bars. These stresses will diminish as temperature differences decrease and thermal equilibrium is restored. Thermally induced failures can also occur in other components of a heat exchanger apart from the fins. The next most susceptible component is the parting sheet.

Continuously unsteady operating conditions, as described above, can generate cyclic stresses exceeding the yield strength, and failure by fatigue may result.

During transient operating conditions, if the combined stresses exceed the ultimate tensile strength of the material, components may fail.

8.1.3 Criteria for Life Assessment

Many factors influence the lifetime of a brazed aluminium plate-fin heat exchanger. Some potential factors that lead to a gradual degradation of the material are: corrosion, erosion and fatigue. Corrosion and erosion can lead to loss of containment. Fatigue can result in leaking.

Visual inspection and non-destructive examination can provide insights on current conditions but not provide insight on the life of the exchanger.

Alternatively, theoretical methods may allow Manufacturers to offer assessments regarding specific damage causes, i.e., pressure and thermal fatigue.

An operating condition or set of conditions supplied by the customer is the basis for fatigue assessment, which is basically performed by means of stress / stress change analysis and cumulative fatigue analysis. The Purchaser should be aware of the unavoidably approximate nature of any fatigue calculation.

The stress analysis typically considers the following:

- metal wall temperature distribution, which basically determines the thermal stress distribution,
- internal pressures, which determine the pressure stresses,
- pipe stress analysis, which should include cases which can generate stress for the attached components (header-nozzle connection, bi-metallic junctions, flanges). This can include no-flow streams during cool-down / warm-up. See Section 9.2.12.

Metal wall temperature distributions can be evaluated by means of thermal-hydraulic simulations, based on the available measured historical data or based on assumptions. Different scenarios might cause different stress levels or stress maxima at different locations of the exchanger. Therefore, such simulations might be required to assess several scenarios.

The availability, the reliability, and the validity of the measured or simulated data impact the accuracy of the assessment.

8.1.4 Recommendations

The lifetime of a brazed aluminium plate-fin heat exchanger is comparable to that of other heat exchange equipment types. Well operated brazed aluminium plate-fin heat exchangers installed over 50 years ago are still in operation. Situations that can reduce brazed aluminium plate-fin heat exchangers' lifetime include operation outside recommended operation conditions, even occasionally, and repairs.

The most typical causes of damage in brazed aluminium plate-fin heat exchangers result from thermal stresses, and the good mitigation practices (and bad practices), drawn from long experience, are summarized in Chapters 4, 5 and 8.

Some of the examples given as 'bad practice' are unavoidable due to upset conditions in long-term plant operation, and these are not entirely unacceptable.

Brazed aluminium plate-fin heat exchangers have been installed and operated successfully for many years in various plants where upset conditions are unavoidable.

However, since thermal fatigue damage accumulates, it is realistic to extend the service life by reducing thermal stresses as much as possible, although they cannot be reduced to zero.

Therefore, in order to tolerate uncontrollable upset conditions as much as possible, and to reduce the possibility of component damage or failure during the operational conditions described above, the following recommendations are made:

1. Limit the pressure and external loads to those allowed (stated) by the Manufacturer.
2. As with any heat exchanger, bring the brazed aluminium plate-fin exchanger to or from operating or derime conditions slowly to avoid excessive thermal stress. This is of particular importance when introducing a liquid or two-phase stream due to the heat capacity of the stream and its ability to transfer heat rapidly to or from the components. Recommended rates for start-up and shut-down, cool-down, warm-up, deriming, etc. are presented in Chapter 4.
3. Limit the temperature differences between adjacent streams at any point in the heat exchanger to those recommended in Chapter 5 or by the Manufacturer. Temperature differences recommended in Chapter 5 are general to all brazed aluminium plate-fin heat exchangers. Other recommendations may be provided by the Manufacturer when supplying a heat exchanger for a particular application.
4. Exercise particular care in applications where a liquid is totally vaporised within the heat exchanger. Boiling to total dryness can produce large temperature differences and also induce flow instabilities. The Manufacturer's recommended allowable temperature differences for these applications shall be strictly adhered to. Also, the design of the process plant shall ensure that stable flow occurs.
5. Design and operate the plant equipment and piping connected to the heat exchanger to prevent flow excursion and instabilities (for example, intermittent slugging of liquid to the exchanger). This is particularly important with boiling streams.
6. Limit cyclic or frequently repeated temperature fluctuations of any stream to ±1 °C per minute. Continuous flow fluctuations should be avoided because they can generate cyclic thermal stresses and sometimes become the cause of damage. Continuous flow fluctuations could occur due to reasons such as inadequate plant control systems or flow instabilities and can be monitored by suitable flow instrumentation. When such flow instruments are not available, other indicators (e.g., pressure drops, metal temperatures, valve openings or liquid levels in thermosyphons) could be utilized to identify flow fluctuations. If any stream temperature fluctuations are a concern, appropriate modelling should be used to explore control methods that minimize thermal cycling.

Safety against over pressure for each process streams shall be installed by the Purchaser. Devices should follow recommended practices such as API 521 for pressure safety valves.

8.1.5 Summary

Brazed aluminium plate-fin heat exchangers are robust exchangers which are very tolerant of large steady-state stream-to-stream temperature differences. Being relatively compact and rigid structures, brazed aluminium plate-fin heat exchangers are susceptible to damage if subjected to transient or continuously unsteady operating conditions which produce excessive thermal stressing. Excessive thermal stressing can be avoided by following the precautions (recommendations) outlined above, thus helping to ensure long life of the heat exchanger.

8.2 Fouling and Plugging of Brazed Aluminium Plate-Fin Heat Exchangers

8.2.1 Fouling

Fouling is generally not encountered for processes in which brazed aluminium plate-fin heat exchangers are traditionally used: air separation; hydrocarbon separation and liquefaction of gases.

In the case where a degradation of thermal performance is observed with little or no change in pressure drop of the product, fouling may be suspected.

Fouling can impact purity of product (product quality).

8.2.1.1 Prevention

Before deciding on the use of brazed aluminium plate-fin heat exchangers, the fluid conditions have to be examined for the presence of solids, foreign particles forming deposits during operation of the heat exchanger, especially in low temperature regions.

It is important also to consider fouling that may arise from contaminants in the process fluids. A typical example is the use of seal oil with refrigerant streams, which could deposit as a solid film on the fin surfaces and reduce the thermal performance of the heat exchanger.

Although mesh filters are recommended to remove particulate matter (see Section 4.10.2), mesh filters are not suitable to protect brazed aluminium plate-fin heat exchangers when the stream is not clean in operation (e.g., contains contaminants such as compressor oil, heavy components, wax, molecular sieve powder or water). The Purchaser should implement suitable pre-treatment systems combined with filtration upstream of the brazed aluminium plate-fin heat exchanger. Good practice should be to install filtration on streams at any location where there is a possibility of contaminating the process fluid.

WARNING: GASES CONTAINING TRACES OF NO_x SHALL NOT BE USED: NO_x WILL ACCUMULATE IN THE CRYOGENIC PORTION OF THE EQUIPMENT. EXPERIENCE HAS SHOWN THAT SUCH PRODUCTS CANNOT BE REMOVED FROM THE INTERIOR OF THE EQUIPMENT AND MAY SUDDENLY EXPLODE DURING WARMING UP OF THE PLANT.

8.2.1.2 Remedial Action

If the liquid/solid transformation of the fouling agent is reversible with temperature, changing the operating conditions of the heat exchanger and thus warming up the fouled zone may be sufficient to eliminate the deposits.

In cases where this technique is not effective, solvent cleaning may be used. Brazed aluminium plate-fin heat exchangers can be modified or designed to incorporate a solvent injection system, thus allowing flushing of the contaminated surfaces.

8.2.2 Plugging

Plugging is defined as the obstruction of fin channels inside a brazed aluminium plate-fin heat exchanger as a result of solid particles having entered the unit. Prevention of particulate plugging is discussed below. Also refer to the ALPEMA filtration recommendations of Section 4.10.2.

The effect of plugging on a brazed aluminium plate-fin heat exchanger may be very serious on its thermal performance since, generally, the plugging medium will not be distributed evenly to all passages, or uniformly within the width of the passages and will thus cause severe maldistribution. Simultaneously, the pressure drop of the plugged stream will increase significantly. Uneven plugging may lead also to induced thermal stresses.

An example of plugging is shown in Figure 8-1.

Figure 8-1: Plugged plate-fin channels

There are concerns with brazed aluminium plate-fin heat exchangers in regards to pressure drop readings that reach high levels. As the pressure drop increases, preferential flows to the open areas increase with the potential for erosion if there is any particulate matter entrained in the fluid. Although the risk is very small, erosion can weaken the fin matrix to a state where a pressure release could occur. In addition, with certain areas of the heat exchanger blocked, maldistribution of fluids can contribute to undesirable thermal profiles across the width/length of the exchanger that can result in high localized thermal stresses. These localized thermal stresses can lead to internal/externals leaks in the heat exchanger.

Accumulation of rime in the flow channels is a form of plugging. Importantly, the structure of rime (soft) is different from ice (hard). H_2O and CO_2 are typical rime components, for example in air separation units. Plugging in oxygen service is a particularly severe hazard.

WARNING: IN THE CASE OF EXTREMELY SEVERE PLUGGING, THE SAFETY ASPECTS OF THE PLANT SHALL BE CONSIDERED.

8.2.2.1 Prevention

Plugging of brazed aluminium plate-fin heat exchangers can be prevented by following these recommended actions:

- The Purchaser should develop and implement procedures that maintain the cleanliness / dryness of the cores during storage and installation onsite.
- The end closures of brazed aluminium plate-fin heat exchangers should always be maintained in place until the connection of nozzles or flanges to the plant pipework.
- The cleanliness of the connecting pipes should be checked to make sure that rust, debris, dust, and other contaminants cannot enter the heat exchanger.

Commissioning strainers, usually mesh filters, are recommended at all stream inlets of the brazed aluminium plate-fin heat exchanger to protect it from plugging by any solid particles such as metal chips from pipe welding, rust, debris, dust, and sand particles due to sand blasting.

It is recommended to install a differential pressure drop measurement across the filter, connected to the DCS, with an alarm for excessive pressure drop to avoid breaking the mesh and plugging the exchanger.

As a minimum, filtration should remove particles larger than 177 microns. A mesh size of 80 Tyler is capable of covering most applications. Mesh filter sizes can be recommended by the brazed aluminium plate-fin heat exchanger's Manufacturer depending on fin channels installed for each stream.

Even though plugging risk is higher during commissioning or start-up, it is recommended to retain mesh filters during normal operation in order to avoid accidental plugging.

Filter Manufacturers should be consulted for recommendations on filter types and maintenance considerations.

Manufacturers should be consulted on design techniques for distributors to mitigate the risk of accumulation of hydrates or heavies. For example, this may include distributor re-entrant B and diagonal C (see Figure 1-8). These distributors provide more flushing of undesirable constituents like heavies, or hydrates or water that can freeze and lead to plugging and/or damage.

8.2.2.2 Remedial Action

Should the heat exchanger be significantly plugged for any reason (absence of filter, wrong cleaning procedure of plant pipework, failure of filtering equipment, etc.), the consequences on both the thermal and pressure drop performance will be obvious.

The plugged stream can generally be identified quickly and corrective action be planned to be taken during a shut-down of the plant. This should not be done without the advice of the Manufacturer.

As a general guideline, if the measured pressure drop significantly exceeds the design allowable pressure drop, typically two times or more, it is recommended that cleaning efforts be performed.

It is recommended to identify the source of the pressure drop and take remedial action to remove the blockage. Blockage/high pressure drop could be attributed to hydrates (moisture), particulate matter (desiccant/mole sieve from dryers/pipe scale, etc.), or compressor oils. Hydrate suppression can be mitigated by methanol injection or possibly by derime of the exchanger (complete warm-up of the plant and circulation of warm gas). Particulate matter can be removed by back-blowing. Compressor oils can be removed by solvent flushing.

Deriming refers to the method for getting rime (typically H_2O and CO_2) out of a brazed aluminium plate-fin heat exchanger heat exchanger. It is typically performed by blowing a comparatively warm, low-pressure gas through the channels when the fluid stream shows an adverse increase in pressure drop.

The mechanical methods to remove plugging from a brazed aluminium plate-fin heat exchanger require an air or nitrogen gas discharge from the exchanger via one of the methods below:

- Back-blow the plugged stream, having installed a bursting disc at the inlet and pressurising up to the rupture of the disc (this operation has to be repeated until no particles are observed being discharged) **or**
- Install a special "deplugger" at the outlet of the heat exchanger, made of a volume of air under pressure and a quick-opening valve, to produce a shock wave inside the heat exchanger core.

In the case of severe plugging, a deplugging action may be undertaken on every passage, having connected the "deplugger" successively to each individual passage.

The use of a solvent and gas bubbling uses the bubbles generated inside the liquid which fills the structure, and these provide the mechanical energy to dislodge the particles.

8.2.2.3 Cleaning Fluids

It is essential that clear identification of the fouling/plugging material and selection of the suitable cleaning fluid should take place prior to implementing any cleaning action. The following list summarizes considerations that need to be taken into account for washing brazed aluminium plate-fin heat exchangers. The list is not exhaustive.

- Type of washing fluid

  Examples include naphtha, toluene, iso-propanol, or special fluids (for example, mixtures), depending on type of plant/contamination.

- Washing connections

  It is recommended to install suitable spading, i.e., to avoid flushing other items such as associated machinery.

- Washing direction

  It is recommended to wash in the opposite direction to the operational flow, particularly if plugging by particles is suspected. However, washing in the flow direction can also be effective. The solvent shall circulate for sufficient time.

  It is recommended to install filtering and/or oil separation equipment into the circulation line (i.e., before the washing fluid returns to the heat exchanger)

- Weight considerations

  Heat exchanger support structures shall be checked for the additional weight introduced by washing fluids.

The following washing equipment is usually required:

- Appropriate type and volume of cleaning fluid
- Pump
- Spading elements
- Connecting lines and related parts
- Storage tank(s)
- Possibly filtering and/or oil separation equipment

Depending on the type of plant and the type of contamination, the equipment, fluids and procedures may be different.

8.3 Corrosion

Brazed aluminium plate-fin heat exchangers are satisfactorily used in many processes without experiencing corrosion problems. However, as with any heat exchanger, when corrosion is possible, caution shall be exercised both on the choice of process fluids and the environment to which the brazed aluminium plate-fin heat exchanger is exposed. Purchasers/Operators should contact the Manufacturer to determine the best course of action to avoid corrosion problems.

WARNING: CAUTION SHALL BE EXERCISED BOTH IN THE CHOICE OF PROCESS FLUIDS AND THE ENVIRONMENT TO WHICH HEAT EXCHANGERS ARE EXPOSED WHEN CORROSION IS POSSIBLE.

8.3.1 Process Environments Containing Water

The corrosion processes due primarily to water or which involve water as one of the contributors will stop or be unable to start in those portions of the brazed aluminium plate-fin heat exchangers which are operating below the freezing point of water. This may not be 0 °C due to water purity variations and supercooling phenomena. Above the freezing point, for example during deriming, consideration shall be given to other factors. Water service can be grouped into three categories:

8.3.1.1 Water service in neutral environments

Brazed aluminium plate-fin heat exchangers can be used extensively in the processing of many materials containing water provided the water is and remains relatively neutral in character while within the exchanger (pH of 6 to 8) even in the presence of halides. The compatibility of the aluminium heat exchanger with a process stream containing neutral water can be affected by factors such as the degree of heavy metals contained within the process stream and deposit formation.

For example, aluminium heat exchangers used together with copper and its alloys, or with other heavy metals such as iron, nickel, and lead, should be avoided unless an inhibitor is used to protect the aluminium heat exchanger components. The pitting corrosion resulting from the use of process streams containing heavy metals is usually less severe when the soluble ions of these heavy metals are decreased. Consequently, the presence of heavy metals in acidic or neutral water service process streams in conjunction with brazed aluminium plate-fin heat exchangers will be more detrimental than in alkaline process streams. Austenitic stainless steels are very acceptable for use in combination with brazed aluminium plate-fin heat exchangers in neutral water service process streams.

8.3.1.2 Water service in acidic environments

Aluminium alloys commonly used in heat exchangers are resistant to acidic process streams or local acidic conditions in the 4.5 to 6.0 pH range. However, an inhibitor should be used in this pH range if heavy metals or halides are present in the process stream. Below a pH of 4.5, corrosion can initiate by breakdown of the protective oxide film and by galvanic coupling between components or areas of the aluminium heat exchanger and other more noble metals in the process equipment. Structurally significant corrosion can result from direct chemical conversion of the exposed nascent aluminium after the protective oxide has broken down. As is the case with neutral environments, the formation of deposits can change both the environmental conditions at which corrosion begins and the severity of the attack once the corrosion begins.

8.3.1.3 Water service in alkaline environments

Aluminium heat exchanger alloys have excellent corrosion resistance in mildly alkaline environments (pH of 8 to 9). An alkaline process stream may discolour the surface of the aluminium components, but this darkening of the surface is only superficial and will not affect the structural or operational integrity of the heat exchanger. The use of brazed aluminium plate-fin heat exchangers in more severe alkaline environments (pH > 9) should be done only after a very careful analysis and consideration of the chemical process streams involved. Other factors such as process and impurity concentrations and temperatures within the operating environment to which the equipment will be subjected also need to be given some consideration.

To summarise, the pH value should remain between 4.5 and 8.5 and the presence of halides and heavy metal ions should be avoided.

8.3.2 Process Environments Containing Mercury

In general, mercury will not react with aluminium unless it is allowed to exist in contact with the heat exchanger in its liquid state and there is water present. If these conditions exist within a heat exchanger, then mercury contamination can result in problems. This attack is most severe when coupled with another corrosion process.

Another possible problem resulting from mercury in the process stream affects aluminium alloys that contain a high level of magnesium. A rapid reaction of mercury with a magnesium-based secondary phase within the aluminium can take place in the absence of water. If features are not designed into the equipment to address this problem and conditions are conducive, mercury corrosion cracking can occur and propagate at substantially lower levels of stress than that required if mercury were not present.

Many brazed aluminium plate-fin heat exchangers are able to operate successfully with fluids containing mercury by using precautions that are available. Purchasers can remove mercury from the feed gas with commercially available systems. Operators may also use special shut-down procedures (nitrogen blanketing) to restrict moisture and avoiding, for metallurgical reasons, elevating temperatures above 100 °C for long periods, for example during deriming operations.

Manufacturers can offer several options when mercury service is specified. Design features can eliminate the build-up or pocketing of mercury. Often it is possible to avoid the use of susceptible alloys. When those choices are not possible, precautions are available either to isolate or protect the high-magnesium containing alloys from mercury attack.

In summary:

For any concentration of mercury in gas, mercury guard beds should be considered because the same gas field can sometimes contain large variations in mercury levels over time.

For mercury concentrations greater than 0.1 μg/Nm^3 (12.2 pptv), mercury guard beds shall be used. In addition, the use of mercury tolerant features can be applied to further increase the safety margin against the risk of mercury induced corrosion damages.

8.3.3 Atmospheric or Environmental Corrosion

Brazed aluminium plate-fin heat exchangers will generally not suffer to any structurally appreciable extent from atmospheric corrosion to the external surfaces of the core, considering the internal process streams to be sealed/protected from the atmospheric/environmental conditions. Slight cosmetic corrosion may result if the exchangers are left outside in a humid environment with temperature changes that result in condensation of the humidity on the aluminium surfaces.

Extra precautions should be taken if the exchangers are exposed to an environment containing appreciable quantities of salt spray or salt air, for example, during extended open storage at site locations in coastal areas or during ocean transport. In the case of ocean freight without seaworthy packing; e.g., transport of exchanger batteries, it is recommended that, immediately after arrival on site, all surfaces be washed with water with a chlorine content < 25 ppm. Manufacturers should be contacted regarding the detailed procedures to be used to wash the core. After external washing, all surfaces need to be dried thoroughly.

Since it is difficult to ensure the leak tightness of any heat exchanger insulation system it is important that safety systems which use water to control fire hazards do not expose the heat exchangers to sea, brackish or other forms of salt water. This water could become trapped between the heat exchanger insulation and the heat exchanger metal surfaces resulting in corrosion of the exposed surfaces. Even tap water can result in corrosion under these conditions, and Manufacturers should be contacted regarding procedures to be used to dry the cores.

8.3.4 Other Services

There are many possible service environments for satisfactory operation of brazed aluminium plate-fin heat exchangers. Not all corrosion risks are addressed in this guideline. If there is uncertainty about the fluid and/or process conditions, contact the Manufacturer for specific advice.

8.4 Disposal

Waste management contributes to the protection, preservation and improvement of the quality of the environment and to the prudent and rational utilization of natural resources. According to this principle, it is essential that brazed aluminium plate-fin heat exchangers and peripherals should be properly disposed of, and recycled to the maximum possible extent, after their removal from site. Aluminium or carbon steel parts may be easily recycled, however other components will require special attention.

9 SPECIAL APPLICATIONS AND EXCHANGER PERIPHERALS

9.1 Block-in-Shell Heat Exchangers

9.1.1 General

The term block-in-shell is used to describe a heat exchanger system of one or more brazed aluminium plate-fin heat exchanger blocks installed in a shell made either of aluminium or steel (see Figure 9-1). Enclosing the block in a larger shell iss sometimes advantageous because it avoids the need to have a separate knock-out drum.

The arrangement is similar to a tubular exchanger known as a kettle reboiler. Hence these are sometimes also known as block-in-shell exchangers. The block-in-shell exchanger offers several advantages over the tubular kettle reboiler as is described below.

9.1.2 Features/Advantages

Advantages of a block-in-shell exchanger include extreme compactness, operating efficiency, reliability, and smooth operation. In comparison with the tubular kettle reboiler, the block-in-shell type has the advantages of

- reduced temperature approach and consequent energy saving
- up to ten times greater heat transfer surface per unit volume
- multi-stream capability in a single unit when required
- elimination of leak-prone mechanical joints
- smaller overall size, weight, and footprint
- lower installation costs
- reduced liquid inventory

9.1.3 Arrangement/Construction

Most block-in-shell exchangers only involve two streams and are therefore of a comparatively simple design. This design is shown in Figure 9-1. Here, stream A is a condensing stream and B is a boiling stream operating in a natural circulation mode. Headers and nozzles attached to the block are only required for the condensing stream. All passages occupied by the boiling passages in the block are directly open to the fluid in which the block is immersed. The internal piping, especially in the case of multiple cores in a single vessel, may have a substantial effect on the overall shell tangent to tangent distance.

When the shell is made of steel, material transitions (so-called "transition joints") are needed between the aluminium heat exchanger block and the steel kettle.

Manholes, nozzles for level indication, and support saddles can be provided by the Manufacturer.

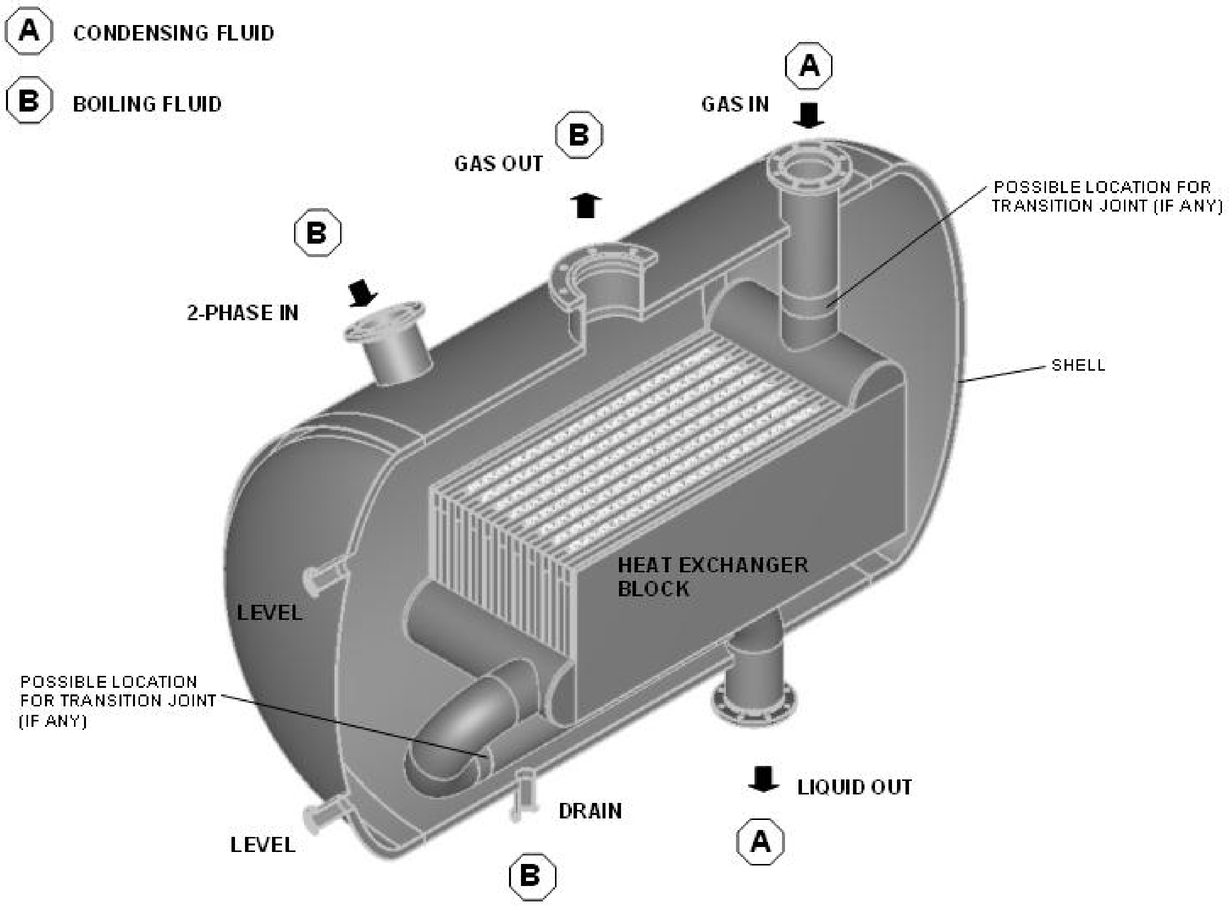


Figure 9-1: Block-in-shell heat exchanger

9.1.4 Thermal and Hydraulic Design

The boiling stream is arranged in vertical up-flow with the condensing stream usually in horizontal flow, giving a cross-flow layout. Circulation of the evaporating stream arises because the density of this two-phase stream within the block is lower than the density of the liquid surrounding the block. This static pressure imbalance gives a driving force to create the up-flow in the boiling channels, i.e., a thermosyphon effect. The actual flow rate achieved is determined by the balance between the driving head and the combined frictional losses and fluid acceleration in the channels. Proprietary software is available for calculating the flow rates and inter-related heat transfer rates in these units.

Sufficient space is allowed above the block to allow the liquid to disengage from the vapour so that a very small amount of liquid is entrained in the outlet vapour leaving the shell. The maximum allowable liquid entrainment from the block-in-shell heat exchanger shall be specified by the Purchaser.

9.1.5 Mechanical Design/Testing

The block is mechanically designed for both internal and external pressure. Prior to installation into the shell, the passages of the condensing stream block are tested in the normal way, and in accordance with applicable Codes, for pressure containment and leaks. Following installation of the block in the shell, the block is tested under external pressure.

9.1.6 Typical Applications

These include

- LNG plants (liquefaction of natural gas against a refrigerant vaporizing in natural circulation)
- Ethylene plants

9.2 Cold Boxes

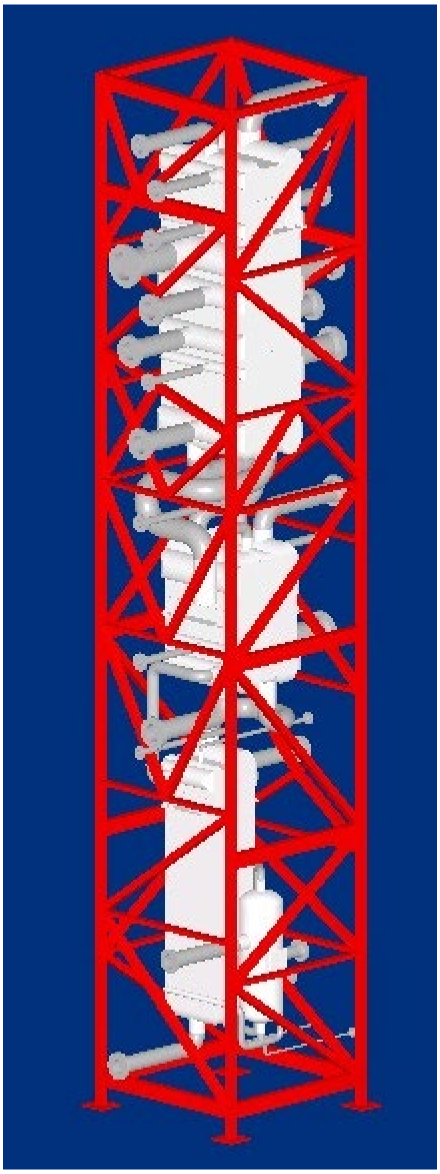


Figure 9-2: Cold box

9.2.1 General

The term **cold box** is used to describe a casing typically of carbon steel which houses cryogenic equipment (see Figure 9-2). The cold box may have a rectangular or circular footprint. The cold box provides structural support, insulation containment, and protection for the internal equipment. A cold box may contain any type of cryogenic equipment, such as brazed aluminium plate-fin heat exchangers, rectification columns, knock-out drums, interconnecting piping, valves, and instrumentation. The exterior is typically painted or coated.

9.2.2 Advantages

Some of the advantages of installing cryogenic equipment into a cold box are:

- It arrives on site nearly "ready to operate" (that is, insulation is normally added on site), reducing cost and the time associated with onsite construction.
- All interconnecting piping between the various components is already complete inside the cold box, thus eliminating flange pairs, piping, and the expense of insulating interconnecting piping.
- No additional support structures are necessary.
- Thermal insulation (perlite) can normally be removed or added more easily than other methods of insulation.
- It provides excellent protection of equipment and insulation against external influences like rain, snow, wind, etc.
- The design facilitates ease of hydrocarbon leak detection through analysis of the exiting purge gas.

9.2.3 Design Considerations

Design considerations include the following:

- Cold box structures shall comply with all project specified environmental and transportation conditions.
- Cold box structures shall be designed for all operational loads including equipment weights, fluid weights, nitrogen purge pressure, and perlite loads.
- Insulation clearances shall be selected to minimize heat leak.
- Maintenance and access requirements shall be defined by the buyer in the project specifications.
- All piping and equipment inside the cold box shall be welded. Flanged joints should not be allowed.

9.2.4 Structure and Panels

A cold box is designed as a welded and self-supporting carbon steel structure with welded carbon steel sheeting. When ambient temperatures are very low, special grades of carbon steel, at least for the structure, shall be considered. All internal equipment is supported by the steel structure of the cold box. Beams directly supporting the internal cryogenic equipment are typically stainless steel. If the expected operating temperatures of the steel supports are above –28.9 °C (–20 °F), carbon steel may be used. The structural supports should be separated from the heat exchanger with layers of heat-break insulating material, if required. The cold box panels are continuously welded on the exterior to provide a seal.

9.2.5 Thermal Insulation

The space between the various cold box components is normally filled with expanded perlite for thermal insulation. Raw Perlite is a type of volcanic stone containing some water. In production, when heated above 1000 °C, the entrapped steam creates "expanded perlite", a sponge type material with a very low density of only 50 – 70 kg/m^3 and very low thermal conductivity, and thus it is perfectly suited for thermal insulation.

Adequate perlite fill and drain connections are included in the cold box casing. Manways can often be used for the same purpose.

Perlite fill and drain shall be performed by a construction company in accordance with procedures provided by the Manufacturer.

Perlite saturated with hydrocarbons after extended use is susceptible to ignition during filling, draining, removal and even storage. A danger of suffocation, ignition and extra forces on cold box internal components potentially caused during filling with and draining perlite should be described in the procedure.

9.2.6 Nitrogen Purge

In order to maintain the original properties of the insulation material and to ensure the cold box atmosphere is free of oxygen, hydrocarbons and humidity, each cold box is equipped with a nitrogen purge system. Dry nitrogen is typically fed into the cold box near the base through perforated tubes which distribute the purge gas equally over the cross section of the cold box. Depending on the height of the cold box, other distributor tubes may be installed at higher elevations. The purge gas normally leaves the cold box at the roof. The nitrogen purge system is externally controlled by valves and instrumentation, such as pressure gauges and flow indicators. Typically, one volume change per day is recommended. Manufacturers can provide recommendations or can supply the complete system.

9.2.7 Wall Penetrations

All pipes entering and leaving the cold box need to penetrate the cold box sheeting. This can be done either with fixed or flexible wall penetrations. Generally, flexible wall penetrations utilising seal boots (e.g., rubber boots) are used. These penetrations keep the cold box airtight while allowing certain movements of the piping in relation to the cold box wall. Fixed penetrations can be provided on pipe penetrations where there is adequate flexibility within the cold box but are generally limited to smaller lines.

9.2.8 Attachments

Cold boxes can be equipped with the following external attachments:

- Roof railings
- Ladders or stairs
- Platforms
- Top davits
- Pipe supports

9.2.9 Safety Devices

Cold boxes are designed for positive internal pressure of nitrogen purge (e.g., 10 mbar-g). In case of leakage from the internal process equipment, the cold box shall be protected against pressurisation over the design limit. Manufacturers provide protection by breather valves, emergency flaps, rupture discs, hydrocarbon detection ports on the breather valve nozzles, or a combination of these devices. It is important to understand that in an unlikely emergency case, a large volume of expanded perlite can escape through such safety devices with consequent danger of suffocation for people beside the cold box. Operators need to take this into account in their plant layout.

9.2.10 Temporary Bracings

In many cases cold boxes are fabricated and transported in a horizontal position. Often temporary bracing is installed to support internal equipment during shipment. After final cold box erection, and prior to any site pressure testing, the temporary bracing shall be removed. Temporary bracing should be colour coded to ensure the correct parts shall be removed once the cold box is erected.

9.2.11 Fire Protection

The cold box steel casing should not be considered as fire protection.

9.2.12 Flanged Connections and Nozzle Loads

Flanged connections to nozzles, valves, etc. should not be made within the cold box casing.

The allowable nozzle load procedure for heat exchangers located in cold boxes and assemblies is the same as listed in Section 5.12.2.4. The scope of work is as follows.

- For pipe runs that attach to the heat exchanger and exit the cold box, the Supplier shall provide the allowable pipe loads and thermal movement at the header to nozzle intersection. The Purchaser shall perform pipe stress analysis on these lines from the plant piping anchor point to the brazed aluminium plate-fin heat exchanger anchor point (which is the header to nozzle intersection). The Supplier shall provide the piping isometrics, with the locations of any internal pipe supports, wall thickness and alloy for pipe runs that are located in the cold box scope of supply to the Purchaser.

- For pipe runs that start and end within the cold box, the Supplier is responsible for the pipe stress analysis.

Note that the flange interface (connector of the cold box to the plant piping) shall not be considered an anchor: that would not be a technically sound assumption since the flange is not the true anchor of the piping system.

Refer to Section 8.1.3 for additional comments on pipe stress analysis, and Section 5.12.2.4 for nozzle loadings at nozzle-to-header intersection. The dimensional tolerances for flanges and nozzles are the same as those listed in Chapter 2.

9.2.13 Shipping, Handling, and Installation

The Manufacturer shall provide the details necessary for shipping, handling, and installation of the cold box.

Templates are generally not required for preparation of civil works and installation of cold boxes in their final position.

9.2.14 Inspection and Maintenance

The nitrogen purge for the cold box shall be disconnected.

WARNING: PERSONNEL SHALL NOT BE ALLOWED INTO THE COLD BOX UNTIL THE UNIFORM LEVEL OF OXYGEN IN THE BOX ATMOSPHERE HAS REACHED A MINIMUM OF 19% BY VOLUME.

A continuous alarm type monitor of the oxygen level in the cold box should be kept. The monitor should be checked at regular intervals of one hour by a second monitor. All monitoring should be at face level of working personnel.

If only part of the box insulation is to be removed to allow access to the heat exchanger, the working cavity so formed shall be secured with scaffolding and planking to prevent the residual insulant from collapsing. See EIGA, 'Doc 146/21 Perlite Management'. If welding is to take place, the cavity should be lined with fireproof material.

WARNING: COLD NITROGEN GAS CAN ACCUMULATE AT GROUND LEVEL AND CAUSE FATAL RESULTS THROUGH ASPHYXIATION.

In addition to Section 4.13, the following are good practice for inspection and maintenance of cold boxes insulated by perlite or other materials. This list is not exhaustive.

- Check for ice formation on the outside of the cold box casing.

  This could indicate either a degradation of thermal insulation, or leaks within the cold box internals, with the inside of the cold box casing exposed to cold process fluid.

- Check for appropriate level and condition of perlite filling:

  The perlite level can be expected to settle over time. Therefore, the level of perlite filling should be checked and recorded on a regular basis, e.g., monthly for the first year and then yearly.

  Ice formation around the upper edge of the cold box casing could indicate a low perlite level.

  Such checks are typically done by looking into the cold box from the top following the onsite safety regulations.

  During plant shut-downs, the perlite conditions (e.g., thermal insulation properties) should be checked. Perlite conditions to be as per Manufacturer's instructions.

- Check for non-spec purge gas composition leaving the cold box:

  The composition of the purge gas leaving the cold box should be checked and recorded on a regular basis, e.g., monthly.

  A composition of the purge gas leaving the cold box deviating from the one entering the cold box (usually N_2) indicates that process fluids from the cold box internals are leaking into the cold box internal atmosphere.

- Check for traces of process fluid in the cold box vicinity:

  The cold box vicinity should be checked regularly, e.g., monthly, for traces of process fluids. Such checks are typically done wearing a gas detector, when walking the cold box area. In addition to checking the cold box at ground level, the cold box side walls and roof should also be checked, depending on accessibility.

- Check the physical condition of flexible wall penetrations:

  Piping wall penetrations, (e.g., neoprene boots), should be checked regularly for possible defects. Defective wall penetrations allow the purge gas to escape and allow humidity to enter the cold box. This jeopardizes thermal insulation properties.

- Check the purge system:

  The purge system shall be operated with flowrates and pressures (usually N_2) as per Manufacturer's instructions. They shall be checked and recorded on a regular basis, e.g., monthly.

- Check the proper function of safety devices, if any (e.g., emergency flaps):

  This shall be checked and recorded on a regular basis, e.g., monthly.

- Check the general appearance of the cold box exterior:

  The cold box exterior should be regularly checked for possible defects.

9.3 Offshore and Marine Installations

Offshore service is a growing application for brazed aluminium plate-fin heat exchangers. The compact and lightweight design make these exchangers ideally suited for offshore applications. Design and approval requirements vary depending on type of application such as whether installed on a fixed platform, a floating platform, or a ship. Offshore and marine installations generally require a marine certified third-party agency to perform a design review and third-party inspection during fabrication. ALPEMA Members should be contacted for experience and capabilities.

NOTATION

		SI	IMPERIAL
A	Effective heat transfer surface of a passage or layer	m^2	ft^2
A_1	Primary heat transfer surface of a stream	m^2	ft^2
A_2	Secondary heat transfer surface of a stream	m^2	ft^2
A_d	Designed or estimated overall effective heat transfer surface	m^2	ft^2
A_r	Required overall effective heat transfer surface	m^2	ft^2
B	Defined by Equation (5.13)	—	—
C	Coefficient, defined by Equation (5.14)	W/m^3 K	Btu/ft^2 °F
C_p	Specific heat	J/kg K	Btu/lb °F
d_h	Hydraulic diameter of passage	m	ft
F	Force	N	lb
f	Fanning friction factor	—	—
G_m	Mass flux/velocity of a stream	kg/m^2 s	lb/ft^2 hr
H	Stacking height of a core	mm	in.
h	Fin height	mm	in.
j	Colburn factor for a finned passage	—	—
K	Expansion/contraction/turning loss coefficient	—	—
L	Core length	mm	in.
l_p	Passage length	mm	in.
l_s	Serration length or distance between crests on herringbone fins	mm	in.
$LMTD$	Logarithmic mean temperature difference	K	°F
M	Moment	Nm	lb ft
MTD	Mean temperature difference	K	°F
n	Fin density	m^{-1}	in.$^{-1}$
p	Fin pitch	mm	in.
Pr	Prandtl number	—	—
Q	Overall heat duty; heat to be transferred	W	Btu
r	Fouling resistance	m^2 K/W	ft^2 °F hr/Btu
s	Distance between the extreme bolts in a given plane	mm	in.
t	Fin thickness	mm	in.
U	Overall heat transfer coefficient between streams	W/m^2 K	Btu/hr ft^2 °F
V	Volume of heat exchanger or exchangers	m^3	ft^3
W	Width of core	mm	in.
X	Required clearance distance	mm	in.

Greek

		SI	IMPERIAL
α	Heat transfer coefficient of a stream	W/m^2 K	Btu/hr ft^2 °F
α_l	Coefficient of linear expansion at average temperature	m/m K	ft/ft °F
α_o	Effective heat transfer coefficient of a stream	W/m^2 K	Btu/hr ft °F
β	Defined by Equation (5.11)	—	—
γ	Defined by Equation (5.12)	—	—
ΔP	Overall pressure drop	N/m^2 (Pa)	lb/in^2
ΔT	Local temperature difference between warm and cold streams	K	°F
ΔT_r	Temperature range at support	K	°F
η_1	Passage fin efficiency for single banking	—	—
η_2	Passage fin efficiency for double banking	—	—
λ_m	Thermal conductivity of fin material	W/m K	Btu/hr ft °F
ρ	Density of stream	kg/m^3	lb/ft^3
ϕ	Unperforated fraction of fin	—	—

Subscripts

c	Cold stream

		SI	IMPERIAL
i	Section		
w	Warm stream		
x, y, z	Direction		

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ALPEMA®

THE STANDARDS OF THE BRAZED ALUMINIUM PLATE-FIN HEAT EXCHANGER MANUFACTURERS' ASSOCIATION

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